# 5 Uncertainty and International Financial Markets

Until now, we have limited our discussion of uncertainty mainly to economies that trade riskless real bonds in an environment of unexpected output, government spending, and productivity shocks. Our earlier framework is a useful one for many purposes, but it has at least two important drawbacks. First, it prevents study of the nature, pricing, and economic role of the increasingly wide array of assets traded in today's international financial markets. Second, it obscures the channels through which prior asset trades influence an economy's reaction to unexpected events.

This chapter is based on the idea that people often have sufficient foresight to make asset trades that protect them, at least partially, against future contingencies affecting their economic well-being. An individual can guard against such risks by buying assets with payoffs that are themselves uncertain, but tend to be unexpectedly high when the individual has unexpected bad economic luck elsewhere. Some types of hedges, such as health, disability, homeowner's, and auto insurance, are familiar. But other types of risky assets, such as currencies, stocks, long-term bonds, and their derivatives, can also play an insurance role. Indeed, the market value of any asset with uncertain returns depends in part on its effectiveness as a means of insurance. Asset pricing in general equilibrium will be one important application of our analysis.

International trade in risky assets can dramatically alter the way an economy's consumption, investment, and current account respond to unanticipated shocks. Consider the simple example of a small endowment economy in which the representative citizen faces uncertainty over the future path of gross domestic product. Suppose further, for the sake of illustration, that the representative individual lays off 100 percent of his output risk in international markets. He might accomplish this purpose either by selling off all local industries to foreign investors in exchange for riskless international bonds, or simply by purchasing an insurance policy that guarantees an income level. Now think about the current-account effect of a temporary rise in the country's gross domestic product. In the bonds-only framework of our earlier chapters, a temporary productivity shock causes a current-account surplus motivated by residents' desire to smooth consumption. But if foreigners have taken on all the country's output risk, a shock to its *GDP* does not affect its *GNP*. The increase in domestic output is matched exactly by a lower net inflow of asset income from abroad. Neither income, consumption, nor the current account changes.

This simple example sidesteps many issues. What price will a country pay to insure its income, and will it choose full insurance at that price? Will countries invest in riskless bonds or buy instead a portfolio of risky foreign assets? The point is clear enough, though. The presence of international markets for risky assets weakens and may sever the link between shocks to a country's output or factor

productivity and shocks to its residents' incomes. Sophisticated international financial markets thus force us to rethink the channels through which macroeconomic shocks impinge on the world economy.

In reality, factors such as moral hazard or imperfect international contract enforcement make it impossible for any country fully to insure itself against all the risks it faces. For much of the analysis in this chapter, however, we suspend skepticism over the scope and efficiency of international financial markets and consider a world with no practical limitations on the kinds of contracts people (and countries) can write with one another. As in the classic models of complete contingent claims markets developed by Kenneth Arrow (1964) and Gerard Debreu (1959), we assume that there is a market for insuring *any* type of risk. While admittedly extreme, this assumption provides an indispensable starting point for clear thinking about the economic effects of risks and markets for risks.

The Arrow-Debreu paradigm of complete markets allows us to think about risk allocation in the same way we think about the allocation of standard commodities at a point in time and over time. Thus, despite more elaborate notation, the maximization problems individuals solve in complete-markets models are no more difficult mathematically than those of our earlier chapters. In addition, standard principles of trade theory apply to the international sharing of risks, allowing us to analyze asset trade in terms of the familiar principle of comparative advantage.

Because it is a straightforward generalization of dynamic macroeconomic analysis under certainty, the Arrow-Debreu setup is easier to analyze than intermediate cases with partially complete markets, one of which we take up toward the end of this chapter. Yet the intuition that a complete-markets benchmark yields also provides critical help in inferring and interpreting the predictions of models with incomplete financial markets.

A final advantage of complete-markets analysis derives from its very strong empirical predictions concerning a host of issues, including global portfolio diversification, asset pricing, and patterns of international consumption growth (to name just a few). In many cases these predictions do not stand up to careful empirical testing, as we shall see, but by looking at the complete-markets model's empirical failures we gain important clues about the kinds of asset-market imperfections most likely to be important in practice.

## 5.1 Trade across Random States of Nature: The Small-Country Case

We introduce trade in risky assets within the simplest possible setting: a small open endowment economy that exists for two periods (labeled 1 and 2) and produces and consumes a single tradable good.[1] To simplify further, we assume for most of this section that on date 2 only two "states of nature" are possible. The two states occur randomly according to a specified probability distribution and differ only in their associated output levels.



### 5.1.1 Uncertainty and Preferences

As usual when population size is constant, we will normalize population to 1 and identify the consumption and endowment of a representative individual with national aggregate consumption and output. The representative individual has known first-period income $Y_1$ and starts out with zero net foreign assets. From the perspective of date 1, however, output on date 2 is uncertain. Either of two states of nature may occur on date 2. In state $s$, which occurs with probability $\pi(s)$, the economy's output equals $Y_2(s)$, $s = 1, 2$.

An individual with uncertain future income cannot predict his future optimal consumption level exactly. In general, the best he can do is predict a range of consumption levels, each contingent on the state of nature that occurs. In analyzing how an individual plans consumption under uncertainty, we therefore focus on a set of desired *contingency plans* for consumption. Exactly which plan is brought into play on a future date will depend on the history of economic outcomes up to then. In terms of our simple two-period, two-state example, we denote by $C_2(s)$, $s = 1, 2$, the two state-contingent consumption plans for date 2.

How does an individual evaluate lifetime utility when future consumption prospects are uncertain? Our usual assumption will be that satisfaction is measured on date 1 by lifetime *expected utility*, that is, by average lifetime utility given the chosen contingency plans for future consumption.[2] Let $C_1$ denote consumption on date 1, which must be chosen before the uncertainty is resolved and thus cannot depend on the state of nature that occurs on date 2. The individual's lifetime expected utility on date 1 is

$$U_1 = \pi(1)\{u(C_1) + \beta u[C_2(1)]\} + \pi(2)\{u(C_1) + \beta u[C_2(2)]\},$$

which, because $\pi(1) + \pi(2) = 1$, implies

$$U_1 = u(C_1) + \pi(1)\beta u[C_2(1)] + \pi(2)\beta u[C_2(2)]. \tag{1}$$

1. For related models of international asset trade, see Cole (1988) and Svensson (1988).

2. We also took this approach in the stochastic model of Chapter 2. The assumption that individuals maximize expected utility can be justified by appealing to the standard Savage (1954) axioms on individual preferences over state-contingent commodities. As with the assumption of intertemporally additive preferences that we discussed in Chapter 1, the assumption of expected utility has the advantages that it is analytically tractable and delivers sharp predictions. However, the framework can be restrictive for some purposes (as we shall illustrate soon) and is subject to some well-known anomalies; see Machina (1987). Section 5.1.8 will briefly consider a richer preference setup.

Notice the tacit assumption that the utility function $u(C)$ in eq. (1) does not depend in any way on the realized state of nature. This need not be the case in general: an individual who unexpectedly falls ill, for example, may well experience a shift in his relative preference for various commodities. We will discuss this possibility in section 5.5.2, but we assume throughout most of this chapter that $u(C)$ is stable across states of nature.

### 5.1.2 Arrow-Debreu Securities and Complete Asset Markets

The asset-market structure that the Arrow-Debreu paradigm posits makes the choice of consumption in different *states* completely analogous to the choice of consumption on different *dates* or, for that matter, to the choice of different consumption goods on a single date. We assume that there is a worldwide market in which people can buy or sell contingent claims. These contingent claims have period 2 payoffs that vary according to the exogenous shocks that actually occur in period 2; that is, their payoffs depend on the state of nature.

Specifically, suppose that on date 1 people can buy or sell securities with the following payoff structure: the owner (seller) of the security receives (pays) 1 unit of output on date 2 if state $s$ occurs then, but receives (pays) nothing in all other states. We call this security the *Arrow-Debreu security* for state of nature $s$ and assume that there is a competitive market in Arrow-Debreu securities for every state $s$.[3]

Of course, we will continue to allow people to borrow and lend, that is, to sell and buy *noncontingent* (or riskless) assets, bonds, that pay $1 + r$ per unit on date 2 regardless of the state of nature, where $r$ is the riskless real rate of interest. If there exist Arrow-Debreu securities for every state, however, the bond market is redundant, in the sense that its elimination would not affect the economy's equilibrium. With only two states, for example, the simultaneous purchase of $1 + r$ state 1 Arrow-Debreu securities and $1 + r$ state 2 Arrow-Debreu securities assures a payoff of $1 + r$ output units next period regardless of the economy's state, just as a bond does. Bonds thus add nothing to the trading opportunities people have once a full set of Arrow-Debreu claims can be traded. This example provides a very simple illustration of how prices for more complicated assets (such as options) can easily be constructed once one knows the primal Arrow-Debreu prices.[4] When we

3. One possible interpretation of the model's initial date 1 is as the date on which securities markets first open. From a multiperiod perspective, however, there is a more sophisticated interpretation: the current and future endowments as of date 1 could be the endogenous result of contingent securities trade *prior* to that date. Our discussion of dynamic consistency in appendix 5D shows that the economy's equilibrium on dates 1 and 2 will be the same (given the same endowments as of date 1) regardless of which interpretation is adopted.

4. By analogy, any $T$-period bond can be viewed as a collection of $T$ pure "discount" bonds, each of which makes a payment in a single period only. (A one-period discount bond makes a payoff after one period, a two-period discount bond makes a single payoff after two periods, etc. In contrast, a standard $T$-period bond makes interest payments in all $T$ periods, and repays principal in the last period.) In the same way, analyzing simple assets that pay off only in a single state of nature allows one to construct the price of any more complex asset.



say an economy has *complete asset markets*, we mean that people can trade an Arrow-Debreu security corresponding to every future state of nature.[5]

It may seem unrealistic to assume that markets in Arrow-Debreu securities exist—no price quotations for such assets are reported in the *Wall Street Journal* or *Financial Times!* Virtually all assets, however, have state-contingent payoffs. Some of these assets, such as stocks and stock options, are traded in organized markets, while others, such as various types of insurance contracts, are not. Later in this chapter (section 5.3 and appendix 5A) we shall see that repeated trading in familiar securities such as stocks can sometimes replicate the allocations that arise when a complete set of Arrow-Debreu securities is traded. Thus, even though Arrow-Debreu securities may seem to be stylized theoretical constructs, thinking about them helps to clarify the economic roles of the more complex securities tracked daily in the financial press.

### 5.1.3 Budget Constraints with Arrow-Debreu Securities

We now turn to analyzing a country's budget constraint under uncertainty and complete asset markets. Let $B_2(s)$ be the representative individual's net purchase of state $s$ Arrow-Debreu securities on date 1. [Thus, $B_2(s)$ is the stock of state $s$ Arrow-Debreu securities the individual holds at the end of date 1 and the start of date 2.] Let $p(s)/(1 + r)$ denote the world price, quoted in terms of date 1 consumption, of one of these securities—that is, of a claim to one output unit to be delivered on date 2 if, and only if, state $s$ occurs.[6] Since this price is determined in a world market, it is exogenously given from the standpoint of the small country.

As usual in an exchange economy, the value of a country's net accumulation of assets on date 1 must equal the difference between its income and consumption:

$$\frac{p(1)}{1+r}B_2(1) + \frac{p(2)}{1+r}B_2(2) = Y_1 - C_1. \tag{2}$$

(We need not explicitly consider purchases of bonds because, as we have seen, bonds are redundant given the two Arrow-Debreu securities available.) When date 2 arrives, the state of nature $s$ is observed, and the country will be able to consume

5. Scholars of Islamic banking have long emphasized that the ban in the Qur'an (holy book) on *riba*, or interest, does not rule out profit-sharing or other arrangements where the lender takes on risk; see Khan and Mirakhor (1987). When there are complete markets for Arrow-Debreu securities, a ban on noncontingent debt contracts alone would not interfere with the efficiency of the economy.

6. Thus $p(s)$ is the price of date 2 consumption *conditional* on state $s$ in terms of *certain* date 2 consumption. We adopt this notation for two reasons: to remind the reader that transactions in Arrow-Debreu securities transfer purchasing power across time as well as states, and to render the resulting budget constraints and Euler equations in a form that is easily compared with their certainty analogs.

**Box 5.1**
Lloyd's of London and the Custom Market for Risks

If the world truly had complete markets, one would be able to insure against virtually any of life's vicissitudes. The typical college student would have access to insurance covering the risk of an unsuccessful career.* Homeowners or prospective home buyers would be able to hedge against changes in real estate values. Taxpayers might purchase insurance against unanticipated tax increases. Indeed, the kinds of risks one would be able to insure in a true complete-markets environment are limited only by one's imagination.

Not surprisingly, there *are* markets for insuring exotic risks—at a price. The leading provider of unusual insurance policies over the past three centuries has been Lloyd's of London.† Lloyd's consists of a group of wealthy individuals—the famous "names"—who accept unlimited liability for the insurance their underwriters provide. Lloyd's origins are in maritime insurance, an area in which it has remained active. The company made a fortune insuring merchant vessels and gold cargoes during the Napoleonic Wars, but paid more than $1 billion in claims during the 1991 Persian Gulf war. In addition to its core shipping and reinsurance businesses, Lloyd's has long stood ready to quote rates for singular contingencies. Lloyd's was a pioneer in nuclear power plant insurance (although its share of the U.S. market had fallen to 7 percent by the time of the 1979 Three Mile Island accident). At Lloyd's, star baseball pitchers can insure their arms, top opera singers can insure their voices, and thoroughbred racehorse owners can insure their prize stallions. Need insurance on a commercial satellite? It's big business at Lloyd's. A store owner can buy riot insurance, and Lloyd's will tailor certain kinds of political risk insurance contracts. Worried about being kidnapped and ransomed? "K&R" insurance, as it is known in the trade, peaked during the terrorism sprees of late of 1970s when world premiums exceeded $75 million, but it is still available.**

Most of the world's insurance business, of course, deals with more mundane matters such as life, fire, and auto insurance. Lloyd's accounts for only a very small fraction of the overall OECD market for standard insurance policies. The table below gives total gross insurance premiums as a percent of GDP for the entire OECD and for the five large countries that account for more than four-fifths of the total.‡

Total Gross Insurance Premiums Paid by Country, 1993

| Country | Total Premiums (as percent of GDP) | Life Insurance Only (as percent of GDP) |
|---|---|---|
| France | 8.6 | 4.7 |
| Germany | 7.9 | 2.8 |
| Japan | 8.8 | 6.5 |
| United Kingdom | 12.9 | 7.6 |
| United States | 10.6 | 4.3 |
| OECD | 8.2 | 4.0 |

*Source:* Organization for Economic Cooperation and Development, *Insurance Statistics Yearbook, 1985–1993* (Paris: OECD, 1995).



**Box 5.1** *(continued)*

It must be emphasized that the figures in the table refer only to the conventional definition of the insurance industry. Therefore, while significant, these figures grossly understate the true overall level of insurance in the world economy according to the much broader concept used in this chapter. Nevertheless, the rarity of exotic contracts of the type Lloyd's of London writes suggests we are making a considerable leap of faith in assuming complete contingent-claims markets. As with other scientific abstractions—for example, the perfect-competition paradigm in economics or the frictionless surface of physics—bold simplification pays off by providing a conceptual framework without which complex real-world situations would be impossible to grasp. As we proceed, we will look closely at the empirical evidence on complete markets.

* In this spirit, several universities (including Yale) have experimented with loan programs in which repayments are indexed to future income.

† Lloyd's has suffered severe financial setbacks in recent years, especially because of uncertainty over settlements on U.S. lawsuits involving asbestos and pollution. But even if Lloyd's the institution does not survive, it is likely that other insurers will fill its place.

** For further reading on Lloyd's, see Hodgson (1984).

‡ On a per capita basis, Switzerland was the most heavily insured country in the OECD in the early 1990s, followed by (in order) Japan, the United States, and the United Kingdom. Although a substantial fraction of all insurance policies is resold through the reinsurance market, foreign holdings of domestic insurance policies are small in most OECD countries (less than 10 percent). This is an example of the home bias phenomenon that we discuss below.

the sum of its endowment and any payments on its state $s$ contingent assets,[7]

$$C_2(s) = Y_2(s) + B_2(s), \qquad s = 1, 2. \tag{3}$$

Using eqs. (3) to eliminate $B_2(1)$ and $B_2(2)$ in the asset-accumulation identity (2), we derive the intertemporal budget constraint for this Arrow-Debreu economy:

$$C_1 + \frac{p(1)C_2(1) + p(2)C_2(2)}{1+r} = Y_1 + \frac{p(1)Y_2(1) + p(2)Y_2(2)}{1+r}. \tag{4}$$

Equation (4) is a slight variation on the usual present-value constraint. It says that the date 1 present value of the country's uncertain consumption stream must equal the date 1 present value of its uncertain income, where contingent quantities are evaluated at world Arrow-Debreu prices. Here international markets allow the country to smooth consumption not only across time but across states of nature. Suppose that the country's output is extremely low in state 1 and extremely

7. In the present setting a person's period 2 income can fall short of required payments on the state-contingent securities issued in period 1 only if this shortfall is planned. Such a plan would be fraudulent, and, as in previous chapters, we continue to assume that people do not plan to violate their intertemporal budget constraints.

high in state 2. Then by going "short" in state 2 securities [choosing $B_2(2) < 0$] and "long" in state 1 securities [$B_2(1) > 0$], the country can smooth consumption across states. Indeed, a consumer could assure himself of a completely *nonrandom* period 2 consumption level by, for example, selling his future state 1 output for $p(1)Y_2(1)/(1+r)$ in bonds and his future state 2 output for $p(2)Y_2(2)/(1+r)$ in bonds. That strategy guarantees the safe date 2 consumption level

$$C_2 = p(1)Y_2(1) + p(2)Y_2(2).$$

But, as we now see, a strategy of full insurance is not necessarily optimal.

### 5.1.4 Optimal Behavior

The country's optimal saving and portfolio allocations maximize expected utility (1) subject to constraint (4). To accomplish that end we use eqs. (2) and (3) to express the consumption levels in eq. (1) as functions of asset choices, by analogy with the procedure followed in earlier chapters. The resulting problem is to maximize over $B_2(1)$ and $B_2(2)$ the unconstrained expected utility

$$U_1 = u\left[Y_1 - \frac{p(1)}{1+r}B_2(1) - \frac{p(2)}{1+r}B_2(2)\right] + \sum_{s=1}^{2}\pi(s)\beta u[Y_2(s) + B_2(s)].$$

The necessary first-order conditions are

$$\frac{p(s)}{1+r}u'(C_1) = \pi(s)\beta u'[C_2(s)], \qquad s = 1, 2. \tag{5}$$

Equation (5) is closely related to the intertemporal Euler equation introduced in Chapter 1, although it pertains to an Arrow-Debreu security rather than a riskless bond. The left-hand side of eq. (5) is the cost, in terms of date 1 marginal utility, of acquiring the Arrow-Debreu security for state $s$. After referring back to eq. (1), you will see that the right-hand side of eq. (5) is the *expected* discounted benefit from having an additional unit of consumption in state $s$ on date 2. As usual, eq. (5) can be rearranged to show that the marginal rate of substitution between $C_1$ and $C_2(s)$ is equal to the two goods' relative price:

$$\frac{\pi(s)\beta u'[C_2(s)]}{u'(C_1)} = \frac{p(s)}{(1+r)}, \qquad s = 1, 2. \tag{6}$$

One can use eqs. (5) to derive the intertemporal Euler conditions for more complex securities that pay off in more than one state of nature. The first such asset that comes to mind is probably a riskless bond, which pays $1+r$ output units on date 2 for every one output unit worth of bonds bought on date 1. As we have already noted, one can create a synthetic bond by buying $1+r$ units of the state 1 Arrow-Debreu security on date 1 at price $p(1)/(1+r)$ per unit and $1+r$ units of the state 2 Arrow-Debreu security at price $p(2)/(1+r)$ per unit. Since this "portfolio" assures delivery of $1+r$ output units on date 2 regardless of which state occurs, it must have the same date 1 price as a bond paying $1+r$ output units next period (that is, it must cost 1 output unit). Thus,

$$\frac{(1+r)p(1)}{1+r} + \frac{(1+r)p(2)}{1+r} = 1,$$

or equivalently,

$$p(1) + p(2) = 1. \tag{7}$$

[For $S > 2$ states, the obvious generalization of eq. (7) is $\sum_{s=1}^{S} p(s) = 1$.]

We derive the bond Euler equation by adding eqs. (5) over the two states:

$$[p(1) + p(2)]u'(C_1) = (1+r)\{\pi(1)\beta u'[C_2(1)] + \pi(2)\beta u'[C_2(2)]\}.$$

Using the definition of a mathematical expectation and eq. (7), we can write this last equality as the *stochastic Euler equation for riskless bonds*,

$$u'(C_1) = (1+r)\beta \mathrm{E}_1\{u'(C_2)\}, \tag{8}$$

where $\mathrm{E}_t\{\cdot\}$ is the expectation operator conditioned on information known on date $t$. This Euler equation is identical to the stochastic Euler equation (29) in Chapter 2. Its intuitive meaning is the same. Equation (8) can be rewritten as

$$\frac{\beta \mathrm{E}_1\{u'(C_2)\}}{u'(C_1)} = \frac{1}{1+r}.$$

The *expected* marginal rate of substitution of present for future consumption equals $1/(1+r)$, the price of certain future consumption in terms of present consumption.

Another critical implication of eq. (5) is

$$\frac{\pi(1)u'[C_2(1)]}{\pi(2)u'[C_2(2)]} = \frac{p(1)}{p(2)}. \tag{9}$$

That is, the marginal rate of substitution of state 2 for state 1 consumption must equal the relative price of state 1 in terms of state 2 consumption. (This equality is an incarnation of the familiar static optimality condition from consumer theory.) Observe that only when

$$\frac{p(1)}{p(2)} = \frac{\pi(1)}{\pi(2)} \tag{10}$$

does condition (9) imply that $C_2(1) = C_2(2)$, so that it is optimal to equate consumption in different states of nature. We say that Arrow-Debreu security prices are *actuarially fair* when eq. (10) holds. At actuarially fair prices, a country trading in complete asset markets will fully insure against *all* future consumption fluctuations. If prices aren't actuarially fair, however, the country will chose to "tilt" its

consumption across states. Given two equiprobable states, for example, the country will plan for relatively lower consumption in the state for which consumption insurance is relatively expensive. Similarly, if other things are equal, individuals confronted with a higher relative price of auto insurance buy less of it (lower coverage limits, higher deductible, and so on).

### 5.1.5 The Role of Risk Aversion

As you will recall from studying microeconomics, it is the strict concavity of the period utility function $u(C)$ that makes an expected-utility maximizer risk averse and, thus, interested in purchasing insurance. Concavity implies that individuals strictly prefer the expected value of a finite gamble to the gamble itself. The main point we make in this subsection is that, loosely speaking, the degree of concavity of the utility function, which measures the extent of risk aversion, is an inverse measure of the individual's portfolio response to changes in Arrow-Debreu prices.

To see the role of risk aversion in determining the demands for state-contingent consumptions, take natural logarithms of the across-state first-order condition (9), and then totally differentiate it (holding probabilities constant, of course, since they are fixed). The result is

$$\begin{aligned} \mathrm{d}\log\left[\frac{p(1)}{p(2)}\right] &= \frac{u''[C(1)]}{u'[C(1)]}\mathrm{d}C(1) - \frac{u''[C(2)]}{u'[C(2)]}\mathrm{d}C(2) \\ &= \frac{C(1)u''[C(1)]}{u'[C(1)]}\mathrm{d}\log C(1) - \frac{C(2)u''[C(2)]}{u'[C(2)]}\mathrm{d}\log C(2). \end{aligned} \tag{11}$$

The ratio

$$\rho(C) = -\frac{Cu''(C)}{u'(C)} \tag{12}$$

is the celebrated Arrow-Pratt *coefficient of relative risk aversion*. If we assume that $\rho(C)$ is a constant, denoted $\rho$, for all consumption levels, then eq. (11) simplifies to

$$\mathrm{d}\log\left[\frac{C(2)}{C(1)}\right] = \frac{1}{\rho}\mathrm{d}\log\left[\frac{p(1)}{p(2)}\right].$$

This equation shows that the inverse of the coefficient of relative risk aversion is also the elasticity of substitution between state-contingent consumption levels with respect to relative Arrow-Debreu prices. Intuitively, high risk aversion produces an inelastic response of consumption-insurance demands to relative insurance prices.

The *constant relative risk aversion* (CRRA) class of utility functions is given by



$$u(C) = \begin{cases} \dfrac{C^{1-\rho}}{1-\rho} & (\rho > 0,\ \rho \neq 1) \\ \log(C) & (\rho = 1), \end{cases} \tag{13}$$

a description that also fits the familiar isoelastic class if $\sigma$, the intertemporal substitution elasticity, equals $1/\rho$! This example illustrates a shortcoming of the expected utility framework: it does not permit us to vary the consumer's aversions to risk and intertemporal substitution (two very different things) independently of each other. Notwithstanding this drawback, the need for tractability leads us to retain the expected-utility assumption in most of what follows, and we continue to specialize to CRRA (or isoelastic) preferences when it is helpful to do so.[8] (In section 5.1.8 it will prove illuminating to relax the expected-utility assumption briefly.)

A consumer is said to be *risk neutral* when $u''(C) = 0$, implying that $\rho$, as defined previously, is 0. As $\rho \to 0$ and the demand elasticity $1/\rho$ grows without bound, individuals respond by concentrating all of their consumption in states $s$ with $\pi(s) > p(s)$. As we will see in the next section, in a world equilibrium the date 2 output market must clear state by state. Thus the only price vector consistent with general equilibrium as $\rho \to 0$ is $p(s) = \pi(s)$ (in which case risk-neutral agents are indifferent as to the allocation of their consumption across states of nature).

### 5.1.6 Consumption Demands and the Current Account: A Log Utility Example

Thus far we have focused on the first-order conditions characterizing a country's equilibrium. For log utility, it is straightforward to derive closed-form solutions for the current account.

With $u(C) = \log(C)$, the lifetime utility function that the representative individual maximizes subject to lifetime budget constraint (4) is

$$U_1 = \log(C_1) + \pi(1)\beta\log[C_2(1)] + \pi(2)\beta\log[C_2(2)]. \tag{14}$$

Let $W_1$ be the present value of lifetime resources on date 1:

$$W_1 = Y_1 + \frac{p(1)Y_2(1) + p(2)Y_2(2)}{1+r}.$$

From earlier encounters with logarithmic utility we know that a country whose representative resident has the preferences in eq. (14) will spend $W_1/(1+\beta)$,

8. These preferences have the advantage that they are consistent with a steady long-run rate of consumption growth, a factor that will be important in Chapter 7. The reason risk aversion and intertemporal substitutability are indistinguishable with CRRA expected-utility preferences is that utility is additive across states as well as time, with probabilities weighting the period utility function as applied to different states in the same multiplicative fashion that the temporal discount factor weights the value of period utility on different dates.

$\pi(1)\beta W_1/(1+\beta)$, and $\pi(2)\beta W_1/(1+\beta)$ on $C_1$, $C_2(1)$, and $C_2(2)$, respectively.[9] Thus consumption demands are

$$C_1 = \frac{1}{1+\beta}\left[Y_1 + \frac{p(1)Y_2(1) + p(2)Y_2(2)}{1+r}\right], \tag{15}$$

$$\frac{p(s)}{1+r}C_2(s) = \frac{\pi(s)\beta}{1+\beta}\left[Y_1 + \frac{p(1)Y_2(1) + p(2)Y_2(2)}{1+r}\right], \qquad s = 1, 2. \tag{16}$$

The date 1 consumption demand (15) is completely parallel to that for the nonstochastic case with log preferences, but in place of a known $Y_2$, eq. (15) has $p(1)Y_2(1) + p(2)Y_2(2)$, the date 1 value of random date 2 output at world market prices.[10]

Using the solution for $C_1$, one can thus express the date 1 current account balance as

$$CA_1 = Y_1 - C_1 = \frac{\beta}{1+\beta}Y_1 - \frac{1}{1+\beta}\left[\frac{p(1)}{1+r}Y_2(1) + \frac{p(2)}{1+r}Y_2(2)\right]. \tag{17}$$

Again, this expression is parallel to the nonstochastic log case.

In the two-date certainty model of Chapter 1, we showed how a country's current account can be interpreted as depending on comparative advantage in trade across time, by analogy with comparative advantage in trade across different goods at the same point in time in classic international trade theory. In particular, the sign of $CA_1$ depended on the difference between the world and autarky interest rates, $r$ and $r^A$. As we show in appendix 5B, this simple form of comparative advantage does not apply here.[11] The basic difficulty is that there are three goods in this model: consumption on date 1, and consumption on date 2 in each of two states of nature. In standard trade theory, the law of comparative advantage generally holds only in a weaker form when there are more than two goods, and such is the case here.

### 5.1.7 A General Result on Comparative Advantage

Even though comparative advantage does not hold in its simplest and strongest form here, it still holds in a weaker but more general form. As we shall see, the more general form does not require any strong restrictions either on the utility function or on the number of states of nature, $S$.

9. Be sure you can show these conclusions.

10. In general, the world market value in terms of sure date 2 consumption of the country's date 2 output, $p(1)Y_2(1) + p(2)Y_2(2)$, is not the same as its expected output, $E(Y_2) = \pi(1)Y_2(1) + \pi(2)Y_2(2)$. That equality holds only in the actuarially fair case with $p(1) = \pi(1)$ and $p(2) = \pi(2)$.

11. Appendix 5B also derives closed-form solutions for the country's optimal portfolio allocation and shows that gross asset flows can be large even when the net flow (i.e., the current account) is small.



We begin by defining the autarky price $p(s)^A/(1+r^A)$ as the price for the Arrow-Debreu security corresponding to state $s$ that would obtain if the country could not engage in any type of asset trade with the rest of the world. From the first-order Euler condition (5) corresponding to state $s$, we see that for the general case of additive preferences

$$\frac{p(s)^A}{1+r^A} = \frac{\pi(s)\beta u'[Y_2(s)]}{u'(Y_1)},$$

where we have imposed the autarky market-clearing conditions that $C_1^A = Y_1$, $C_2^A(s) = Y_2(s)$.

Now let $C_1$ and $C_2(s)$, $s = 1, 2, \ldots, S$, be the country's consumption choices under free trade. They must satisfy the country's intertemporal budget constraint at free-trade prices, which is the obvious generalization of eq. (4) to $S$ possible states on date 2:

$$C_1 - Y_1 + \sum_{s=1}^{S}\frac{p(s)}{1+r}[C_2(s) - Y_2(s)] = 0. \tag{18}$$

Of course, the country must be made (weakly) better off by the opportunity to trade. Thus its consumption choices under free trade must be (weakly) more valuable than its endowment at autarky prices:

$$C_1 - Y_1 + \sum_{s=1}^{S}\frac{p(s)^A}{1+r^A}[C_2(s) - Y_2(s)] \geq 0. \tag{19}$$

(If the preceding inequality failed, the country would be able to buy its free-trade consumption bundle, and then some, at autarky prices, contradicting the presence of gains from trade.)

Subtracting eq. (18) from eq. (19) and using the second-period budget constraint (3) yields the principle of comparative advantage,

$$\sum_{s=1}^{S}\left[\frac{p(s)^A}{1+r^A} - \frac{p(s)}{1+r}\right][C_2(s) - Y_2(s)] = \sum_{s=1}^{S}\left[\frac{p(s)^A}{1+r^A} - \frac{p(s)}{1+r}\right]B_2(s) \geq 0.$$

This fundamental inequality states that, on average, a country's net imports of date 2, state $s$ consumption tend to be high when the date 1 autarky price of state $s$ consumption is high compared with the world price.

The deterministic case of Chapter 1 corresponds to the standard two-good case of classical trade theory: the two goods are date 1 consumption and (sure) date 2 consumption, so $S = 1$ and the last inequality states merely that countries with low autarky interest rates import date 2 consumption and, by the budget constraint, run current account surpluses on date 1. In general, for $S > 1$ one cannot link net imports of any one specific commodity to the difference between its world and

autarky prices. Comparative advantage applies only in the average sense given by the last inequality.[12]

Because the preceding result requires only the weakest assumptions about preferences (e.g., that more consumption is preferred to less), it is quite general. Indeed, the result holds true for nonexpected- as well as expected-utility preferences.

### *5.1.8 Risk Aversion and Intertemporal Substitution in a General Characterization of the Current Account

The analysis has simplified the problem of choice under uncertainty by treating consumption in each possible date 2 state as a distinct commodity. In this section we draw out further implications of this abstraction, making use of the general analytical approach to multicommodity intertemporal models developed in Chapter 4. A small amount of notational translation permits a straightforward transplant of Chapter 4's framework to complete-markets models of uncertainty. The analogy with Chapter 4 is most transparent if we temporarily relax the assumption of expected utility maximization. Thus this subsection yields as a by-product a quick look at how recent research on choice under uncertainty has attempted to decouple risk aversion and intertemporal substitutability.

We continue to assume that there are $s = 1, 2, \ldots, \mathcal{S}$ possible date 2 states of nature. Suppose, by analogy with section 4.4.1.1, that we can write lifetime utility as

$$U_1 = u(C_1) + \beta u\{\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})]\}, \tag{20}$$

where the consumption index $\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})]$ is homogeneous of degree 1 in $C_2(1), \ldots, C_2(\mathcal{S})$. Let $Z_2$ be total spending on date 2 goods, measured in terms of *sure* (that is, noncontingent) date 2 consumption. Furthermore, define the date 2 consumption-based price index $P$ as the minimal expenditure, measured in terms of sure date 2 consumption, necessary to achieve $\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})] = 1$. Then, since the consumption index is linear homogeneous, we write lifetime utility as

$$U_1 = u(C_1) + \beta u(Z_2/P) \tag{21}$$

and the lifetime budget constraint as

$$C_1 + \frac{Z_2}{1+r} = Y_1 + \frac{1}{1+r}\sum_{s=1}^{\mathcal{S}} p(s)Y_2(s) \tag{22}$$

12. This version of the comparative advantage theorem is offered by Deardorff (1980) and Dixit and Norman (1980). Svensson (1988), who explores a framework similar to the one in this section, notes the application to trade in risky assets. Svensson's model and results cover cases of incomplete asset markets.



[recalling that $1/(1+r)$ is the date 1 price of sure date 2 consumption, the unit in which $Z_2$ is measured]. Given these preliminaries, one can, as in Chapter 4, envision individuals performing a two-stage maximization process:

1. Maximize eq. (21) subject to eq. (22) to find the optimal division of lifetime spending across *dates*, $C_1$ and $Z_2$.

2. Then, to find the optimal division of date 2 spending $Z_2$ across *states*, maximize the consumption index $\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})]$ subject to $\sum_{s=1}^{\mathcal{S}} p(s)C_2(s) = Z_2$.

To highlight the close parallel with section 4.4.1.1, we take $\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})]$ to be of the constant elasticity of substitution (CES) form proposed in eq. (13) of Chapter 4. Section 5.1.5 showed that in the CRRA utility case described by the present chapter's eq. (13), $1/\rho$ is the constant elasticity of substitution between state-contingent consumptions. Our view of state-contingent consumptions as distinct commodities suggests that we replace the intratemporal substitution elasticity $\theta$ in eq. (13) of Chapter 4 by $1/\rho$ and choose

$$\Omega[C_2(1), \ldots, C_2(\mathcal{S}); \pi(1), \ldots, \pi(\mathcal{S})] = \left[\sum_{s=1}^{\mathcal{S}} \pi(s)C_2(s)^{1-\rho}\right]^{\frac{1}{1-\rho}}. \tag{23}$$

Under this choice, date 2 contingent consumption demands are given by the analog of eq. (22) in Chapter 4 [with $\theta$ replaced by $1/\rho$ and the weights of the form $\gamma$ replaced by $\pi(s)^{1/\rho}$],

$$C_2(s) = \left[\frac{p(s)/\pi(s)}{P}\right]^{-1/\rho}\frac{Z_2}{P}, \tag{24}$$

where [the analog of eq. (20) in Chapter 4]

$$P = \left[\sum_{s=1}^{\mathcal{S}} \pi(s)^{\frac{1}{\rho}} p(s)^{\frac{\rho-1}{\rho}}\right]^{\rho/(\rho-1)}. \tag{25}$$

The postulate that $u(C)$ in eq. (20) is itself isoelastic, together with eq. (23), leads to an intertemporal utility function that generalizes both isoelastic and CRRA utility by allowing $\sigma$, the intertemporal substitution elasticity, to differ from $1/\rho$:

$$U_1 = \frac{C_1^{1-1/\sigma}}{1-\frac{1}{\sigma}} + \beta\frac{\left\{\left[\sum_{s=1}^{\mathcal{S}} \pi(s)C_2(s)^{1-\rho}\right]^{\frac{1}{1-\rho}}\right\}^{1-1/\sigma}}{1-\frac{1}{\sigma}}. \tag{26}$$

Notice that the contingent consumption demand (24) confirms that $1/\rho$ is the price elasticity of substitution, so that $\rho$ is the coefficient of relative risk aversion (recall

section 5.1.5). However, $\sigma$, the intertemporal substitution elasticity, is a distinct parameter.[13] Only when $\sigma = 1/\rho$ does eq. (26) reduce to the expected lifetime utility

$$U_1 = \frac{C_1^{1-\rho}}{1-\rho} + \beta \sum_{s=1}^{S} \pi(s) \frac{C_2(s)^{1-\rho}}{1-\rho}.$$

To see the current-account implications of this framework, maximize eq. (21) subject to eq. (22) when $P$ is given by eq. (25). The resulting intertemporal Euler equation for bonds is

$$u'(C_1) = (1+r)\beta \left(\frac{1}{P}\right) u'\left(\frac{Z_2}{P}\right).$$

When $u(C)$ is isoelastic with substitution elasticity $\sigma$, this Euler equation becomes

$$Z_2 = (1+r)^{\sigma} \beta^{\sigma} \left(\frac{1}{P}\right)^{\sigma-1} C_1, \tag{27}$$

which implies that date 1 consumption equals

$$C_1 = \frac{W_1}{1 + \left(\frac{1+r}{P}\right)^{\sigma-1} \beta^{\sigma}}.$$

[As before, $W_1$ is date 1 lifetime resources, the right-hand side of eq. (22).]

Euler equation (27) differs from those we saw in deterministic one-good models only through the presence of the date 2 price index $P$. Absent this factor, expenditure on the two dates, measured in noncontingent output units, would be determined by the same consumption smoothing and tilting factors explained in Chapter 1. [Compare the last equation giving optimal $C_1$ with eq. (26) in Chapter 1.] Indeed, there are some special cases in which it actually is legitimate to extrapolate from the deterministic results in Chapter 1 simply by treating the country's date 2 endowment $\sum p(s)Y_2(s)$ as if it were derived from a deterministic date 2 output level:

1. $\sigma = 1$. In this case the real interest rate effect associated with $P \neq 1$ has exactly offsetting income and substitution effects. The logarithmic expected-utility example of section 5.1.6 has this property, but so do the cases in which $\sigma = 1$ but $\rho \neq 1$.

2. $\pi(s) = p(s)$. For actuarially fair prices, eq. (25) implies that $P = 1$. The current account of a country that has laid off all of its second-period risk in world markets is, quite intuitively, the same as in a deterministic model. More formally, in this case the date 2 consumption index (23) corresponds to noncontingent date 2 consumption.

13. Multiperiod versions of the preferences described by eq. (26) are proposed by Epstein and Zin (1989) and Weil (1989b, 1990). As usual, the special cases with $\rho = 1$ or $\sigma = 1$ are handled by L'Hospital's rule. (Recall Chapter 1, footnote 14.)



What do we make of cases in which $P$ can differ from 1? Notice first that $P$ can only be *below* 1, never above. The reason is that by choosing $C_2(s) = 1$ for all $s$, one can set the consumption index in eq. (23) equal to 1 at an expenditure of $\sum p(s)C_2(s) = 1$, *regardless* of the prices $p(s)$. We have seen that for actuarially fair prices this is the best one can do. For $p(s) \neq \pi(s)$, one therefore does strictly better, implying that $P < 1$.

More intuitively, an economy facing prices that are not actuarially fair *could* finance a noncontingent date 2 consumption level of $Z_2 = (1+r)(Y_1 - C_1) + \sum p(s)Y_2(s)$ by selling date 2 output forward on date 1 and investing the proceeds in bonds that mature on date 2. The date 2 output expenditure required is, of course, just $Z_2$. This strategy is suboptimal, however, when prices aren't actuarially fair, meaning that the consumption index $\Omega$ could have been set to $Z_2$ at a price $PZ_2$ strictly below $Z_2$.

When $P < 1$, the consumption-based real interest rate is above $1+r$. For $\sigma > 1$, date 1 consumption therefore is lower, and $CA_1$ higher, than in a parallel certainty model with $Y_2 = \sum p(s)Y_2(s)$. When $\sigma < 1$ the effect of a real interest rate above $1+r$ is reversed. Thus, even if $\beta(1+r) = 1$ and $Y_1 = \sum p(s)Y_2(s)$, eq. (27) implies that for $P < 1$, the country will have a first-period current account surplus when $\sigma > 1$ and a deficit when $\sigma < 1$.[14]

## 5.2 A Global Model

The last section showed how a small country allocates its consumption across dates and across uncertain states of nature, given world prices of contingent securities. While the small-country case is a useful starting point for thinking about intertemporal trade under uncertainty, several important implications of the complete markets approach becomes clear only in a world general-equilibrium setting.

In this section, we extend the two-period small-country analysis to the global case. Our model highlights the difference between global and country-specific risk and discloses the conditions under which Arrow-Debreu prices are actuarially fair.

14. Equation (27) implies that the "current account autarky" interest rate defined in appendix 5B is given here by

$$1 + r^{CA} = \frac{1}{\beta}\left[\frac{\sum p(s)Y_2(s)}{P^{1-\sigma}Y_1}\right]^{1/\sigma}.$$

As in appendix 5B, if $r > r^{CA}$ the country has a date 1 current account surplus, and it has a date 1 deficit if $r < r^{CA}$.

It also illustrates the very strong restrictions that a complete markets model can place on international consumption-growth comovements.

### 5.2.1 The CRRA Case

As an aid to intuition, we begin by developing a CRRA example. Discussion of more general utility functions is deferred until later in this section.

The world economy consists of two countries, Home and Foreign, with output levels that fluctuate across $S$ states of nature. Foreign's necessary Euler equations are of the same forms as Home's were in section 5.1.4, with due allowance for the possibility that $S > 2$. As usual, Foreign quantities corresponding to Home's are marked with an asterisk ($*$). Home and Foreign consumers have the same degree of risk aversion.

#### 5.2.1.1 Equilibrium Prices

Global general equilibrium requires that supply and demand balance in $S + 1$ markets: the market for date 1 output and those for date 2 output delivered in each of the $S$ states of nature:

$$C_1 + C_1^* = Y_1 + Y_1^*, \tag{28}$$

$$C_2(s) + C_2^*(s) = Y_2(s) + Y_2^*(s), \qquad s = 1, 2, \ldots, S. \tag{29}$$

Equilibrium prices are found by combining these market-clearing conditions with the national representatives' Euler equations. Define $Y^W \equiv Y + Y^*$ as total world output. For a common CRRA period utility function [recall the definition in eq. (13)], Euler equation (5) for state $s$ securities (which holds for any number of states $S$) shows that $C_2(s) = [\pi(s)\beta(1+r)/p(s)]^{1/\rho}C_1$ in Home and $C_2^*(s) = [\pi(s)\beta(1+r)/p(s)]^{1/\rho}C_1^*$ in Foreign. Adding these and applying equilibrium conditions (28) and (29) gives

$$Y_2^W(s) = \left[\frac{\pi(s)\beta(1+r)}{p(s)}\right]^{\frac{1}{\rho}} Y_1^W, \qquad s = 1, 2, \ldots, S,$$

which implies that the date 1 price of the state $s$ contingent security is

$$\frac{p(s)}{1+r} = \pi(s)\beta\left[\frac{Y_2^W(s)}{Y_1^W}\right]^{-\rho}, \qquad s = 1, 2, \ldots, S. \tag{30}$$

Notice the key role of our assumption that Home and Foreign have *the same* risk aversion coefficient, $\rho$. Without this simplification, the model would not necessarily yield a closed-form solution in terms of aggregate output alone.

The preceding results throw light on the conditions under which securities prices will be actuarially fair. Write down eq. (30) for any two states $s$ and $s'$ and conclude



by dividing the two equations that

$$\frac{p(s)}{p(s')} = \left[\frac{Y_2^W(s)}{Y_2^W(s')}\right]^{-\rho} \times \frac{\pi(s)}{\pi(s')}. \tag{31}$$

For $\rho > 0$, contingent claims' prices all will be actuarially fair [that is, satisfy eq. (10)] if and only if total *world* output is the same in all states of nature. The requirement for actuarial fairness, then, is absence of output uncertainty *at the aggregate level*. If there is no aggregate uncertainty, it is feasible for both countries to have state-independent date 2 consumption levels. As a result, equilibrium prices need not provide an incentive for people to "tilt" consumption in favor of states with relatively abundant world output. If, however, world output in state $s'$ exceeds that in state $s$, prices must induce countries to consume relatively more in state $s'$. As eq. (31) shows, state $s$ consumption therefore will command a premium over its actuarially fair price, while state $s'$ consumption will sell at a discount.

With equilibrium Arrow-Debreu prices for date 1 in hand, we can easily find the real interest rate. As an intermediate step, we solve for the date 2 prices $p(s)$. For any state $s'$, the arbitrage condition $\sum p(s) = 1$ and eq. (31) imply

$$p(s') = 1 - \sum_{s \neq s'} p(s) = 1 - p(s') \sum_{s \neq s'} \left[\frac{Y_2^W(s)}{Y_2^W(s')}\right]^{-\rho} \frac{\pi(s)}{\pi(s')},$$

an equation that can be solved for $p(s')$:

$$p(s') = \frac{\pi(s')[Y_2^W(s')]^{-\rho}}{\sum_{s=1}^{S} \pi(s)[Y_2^W(s)]^{-\rho}}. \tag{32}$$

This equation and eq. (30) give the world interest rate:

$$1 + r = \frac{(Y_1^W)^{-\rho}}{\beta \sum_{s=1}^{S} \pi(s)[Y_2^W(s)]^{-\rho}}. \tag{33}$$

Equation (33) is intuitive. Higher world output on date 1 implies a lower real interest rate (raises the price of date 2 consumption relative to date 1 consumption). Similarly, higher future output in any state (that is, higher expected future output) raises the real interest rate.

#### 5.2.1.2 Equilibrium Consumption Levels

The complete-markets model has strong implications concerning correlations in international consumption levels across time and across states of nature. These strong predictions arise because complete markets allow all individuals in Home and

Foreign to equate their marginal rates of substitution between current consumption and state-contingent future consumption to the same state-contingent security prices.

The many-state analogs of eqs. (5) and (9) imply

$$\frac{\pi(s)\beta u'[C_2(s)]}{u'(C_1)} = \frac{p(s)}{(1+r)} = \frac{\pi(s)\beta u'[C_2^*(s)]}{u'(C_1^*)} \tag{34}$$

and

$$\frac{\pi(s)u'[C_2(s)]}{\pi(s')u'[C_2(s')]} = \frac{p(s)}{p(s')} = \frac{\pi(s)u'[C_2^*(s)]}{\pi(s')u'[C_2^*(s')]}$$

for all states $s$ and $s'$. You will recognize these as fundamental necessary conditions for efficient resource allocation: all individuals' marginal rates of substitution in consumption—over time and across states—are equal, so no potential gains from trade remain to be exploited.[15] These equations relate only marginal utilities of consumption, but specific utility functions yield implications for consumption levels.

With CRRA utility, for example, $u'(C) = C^{-\rho}$, so the last two equations, combined with the equations for state-contingent prices (30), imply

$$\frac{C_2(s)}{C_2(s')} = \frac{C_2^*(s)}{C_2^*(s')} = \frac{Y_2^W(s)}{Y_2^W(s')} \tag{35}$$

and

$$\frac{C_2(s)}{C_1} = \frac{C_2^*(s)}{C_1^*} = \frac{Y_2^W(s)}{Y_1^W} \tag{36}$$

for all states. Equation (35) implies the equalities

$$\frac{C_2(s)}{Y_2^W(s)} = \frac{C_2(s')}{Y_2^W(s')}, \qquad \frac{C_2^*(s)}{Y_2^W(s)} = \frac{C_2^*(s')}{Y_2^W(s')},$$

meaning that Home consumption is a constant fraction $\mu$ of world date 2 output regardless of the state. (Foreign's state-invariant share, correspondingly, is $1-\mu$.) Equation (36) says that consumption *growth rates* are the same across countries in every state and are equal to the growth rate of world output.

Equation (36) implies the equalities (for all $s$)

15. As usual, efficiency means *Pareto optimality*: any rise in one country's welfare would have to come at the other country's expense. This property of the equilibrium follows because preferences are quasi-concave, there is complete information, and there are no increasing returns to scale in production (production is exogenous), no nonpecuniary externalities in production or consumption, and no limitations on the kinds of contracts that can be enforced. Furthermore, all consumers can participate in all markets (in contrast to the overlapping generations model of section 5.6).



$$\frac{C_2(s)}{Y_2^W(s)} = \mu = \frac{C_1}{Y_1^W}, \qquad \frac{C_2^*(s)}{Y_2^W(s)} = 1-\mu = \frac{C_1^*}{Y_1^W},$$

meaning that the countries' date 1 consumption shares in world output are the *same* as their date 2 shares. A country's share in world consumption, one can easily show, is the country's share of the world's present discounted output on date 1, evaluated at equilibrium Arrow-Debreu prices.[16]

Note that when date 2 world output is uncertain, *at equilibrium prices* neither Home nor Foreign arranges for constant consumption across states. Each country's consumption is internationally diversified, however, in the sense that any consumption risk it does absorb is entirely due to *systematic* output uncertainty, that is, uncertainty in global output.[17]

Date 2 equilibrium is illustrated in Figure 5.1 for the case $S = 2$. The diagram is a familiar Edgeworth box in which the two goods are state-contingent consumptions and world output is assumed higher in state 1 than in state 2 . At the endowment point **A**, Home has higher output in both states, but it is *relatively* better endowed with state 1 output. Equilibrium point **E** is therefore achieved through Home sales of claims to state 1 output in return for state 2 output claims issued by Foreign. The equilibrium relative price of the two contingent securities, equal to the absolute slope of line **EA**, is $p(1)/p(2)$. Under identical CRRA preferences the fraction of world output each country consumes is constant across states, but this would not necessarily be the case for more general utility functions or with differing national degrees of risk aversion.

16. In equilibrium, Home's budget constraint implies

$$Y_1 + \sum_{s=1}^{S} \frac{p(s)Y_2(s)}{1+r} = C_1 + \sum_{s=1}^{S} \frac{p(s)C_2(s)}{1+r}$$

$$= \mu\left[Y_1^W + \sum_{s=1}^{S} \frac{p(s)Y_2^W(s)}{1+r}\right],$$

an equation readily solved for $\mu$ to verify the text's claim. By substituting equilibrium prices [eq. (30)] for $p(s)/(1+r)$ in the resulting solution, one derives the following reduced-form expression for $\mu$:

$$\mu = \frac{Y_1(Y_1^W)^{-\rho} + \beta\sum_{s=1}^{S}\pi(s)Y_2(s)[Y_2^W(s)]^{-\rho}}{(Y_1^W)^{1-\rho} + \beta\sum_{s=1}^{S}\pi(s)[Y_2^W(s)]^{1-\rho}}.$$

17. The first theorem of welfare economics implies that the equilibrium resource allocation corresponds to the choice of a benevolent social planner who allocates output by command. It is a good exercise to show that here, the equilibrium allocation results if the social welfare function $\kappa U_1 + (1-\kappa)U_1^*$ is maximized subject to resource constraints, where

$$\kappa = \frac{\mu^\rho}{\mu^\rho + (1-\mu)^\rho}.$$

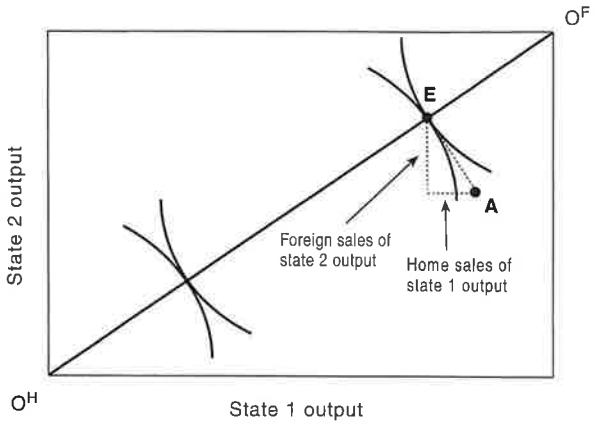



**Figure 5.1**
Efficient risk pooling

### *Application: Comparing International Consumption and Output Correlations*

One of the most powerful predictions of the one-good complete-markets model is that different countries' per capita consumption growth rates should be highly correlated even if growth rates in per capita output are not. We have seen that when two countries have identical CRRA preferences, their shares of world consumption are constant across time, so that their rates of consumption growth must always be equal ex post. However, even for two countries $n$ and $m$ with different constant coefficients of risk aversion, $\rho_n$ and $\rho_m$, and different subjective discount factors, $\beta_n$ and $\beta_m$, the model still yields a very strong empirical prediction. If we use lowercase $c$ and $y$ to denote *per capita* consumption and output, then eq. (34) implies

$$\log\left[\frac{c_2^n(s)}{c_1^n}\right] = \left(\frac{\rho_m}{\rho_n}\right)\log\left[\frac{c_2^m(s)}{c_1^m}\right] + \frac{1}{\rho_n}\log\left(\frac{\beta_n}{\beta_m}\right). \tag{37}$$

This equation shows that any two countries' ex post consumption growth rates, although individually random, are perfectly statistically correlated (they have a correlation coefficient of 1).[18]

Does this perfect-correlation implication match the data? Embarrassingly for the simplest one-good version of the complete markets model, it does not. Table 5.1 presents 1973–92 estimates of correlation coefficients between national per capita consumption growths and world per capita consumption growth. These are compared with correlation coefficients between national per capita output growths and world per capita output growth. [The Penn World Table data used in Table 5.1 allow a meaningful international comparison of real income and consumption levels by correcting for price-level differences among countries (recall Chapter 4).]

For the seven largest industrial countries (the Group of Seven, or G-7), the correlation between domestic and world consumption growth is lower in almost every case than the correlation between domestic and world output growth; the only exception is for Britain where the two correlations are almost equal. When smaller OECD countries are included, the puzzle remains. For developing countries, the average correlation between domestic and world consumption growth is actually slightly negative.[19] In section 5.5 we will consider some possible explanations of

18. The *correlation coefficient* between two random variables $X$ and $Y$, $\text{Corr}\{X, Y\}$, is defined as

$$\text{Corr}\{X, Y\} = \frac{\text{Cov}\{X, Y\}}{\sqrt{\text{Var}\{X\}}\sqrt{\text{Var}\{Y\}}}.$$

If $X$ and $Y$ are linearly related, so that $Y = a_0 + a_1 X$ for some constants $a_0$ and $a_1 > 0$, then these random variables are *perfectly correlated*, by which we mean that $\text{Corr}\{X, Y\} = 1$. Perfect *negative* correlation means that $\text{Corr}\{X, Y\} = -1$, and occurs when $a_1 < 0$. The Cauchy-Schwarz inequality implies that $-1 \leq \text{Corr}\{X, Y\} \leq 1$ always.

19. The observation that consumptions are less correlated internationally than outputs has been made in various ways and for various data samples by Backus, Kehoe, and Kydland (1992), Devereux, Gregory,



**Table 5.1**
Consumption and Output: Correlations between Domestic and World Growth Rates, 1973–92

| Country | $\text{Corr}(\hat{c}, \hat{c}^W)$ | $\text{Corr}(\hat{y}, \hat{y}^W)$ |
|---|---|---|
| Canada | 0.56 | 0.70 |
| France | 0.45 | 0.60 |
| Germany | 0.63 | 0.70 |
| Italy | 0.27 | 0.51 |
| Japan | 0.38 | 0.46 |
| United Kingdom | 0.63 | 0.62 |
| United States | 0.52 | 0.68 |
| OECD average | 0.43 | 0.52 |
| Developing country average | −0.10 | 0.05 |

*Note:* The numbers $\text{Corr}(\hat{c}, \hat{c}^W)$ and $\text{Corr}(\hat{y}, \hat{y}^W)$ are the simple correlation coefficients between the annual change in the natural logarithm of a country's real per capita consumption (or output) and the annual change in the natural logarithm of the rest of the world's real per capita consumption (or output), with the "world" defined as the 35 benchmark countries in the Penn World Table (version 5.6). Average correlations are population-weighted averages of individual country correlations. The OECD average excludes Mexico.

the "paradox" that outputs are more correlated than consumptions, as well as more formal econometric tests. ∎

### 5.2.2 A Rationale for the Representative-Agent Assumption

Throughout much of this book we maintain the convenient fiction that each country is inhabited by a single representative agent. This assumption has allowed us to analyze each country's aggregate per capita consumption as if it were decided by a composite individual endowed with the economy's per capita economic resources. Having introduced an explicit stochastic general equilibrium model, we are now able to present a deeper rationale for the representative-agent approach than the preliminary one we offered in Chapter 1. This subsection shows that if asset markets are complete and agents face the same prices, then, for a broad class of period utility functions, prices and aggregate per capita consumption behave *as if* there were a single representative agent despite substantive differences across individuals.

To take a simple example, consider a complete markets environment in which $I$ agents with possibly different wealth levels have identical discount factors and period utility functions given by the following generalization of the CRRA case,

$$u(c^i) = \frac{(a_0 + a_1 c^i)^{1-\rho}}{1-\rho}, \tag{38}$$

where $a_0$, $a_1$, and $\rho$ all are constants. We know from eq. (5) that individual $i$'s consumption choices for all states $s$ satisfy the Euler equation

$$(a_0 + a_1 c_1^i)^{-\rho} = \frac{\beta(1+r)\pi(s)[a_0 + a_1 c_2^i(s)]^{-\rho}}{p(s)}, \qquad i = 1, 2, \ldots, I.$$

The preceding equation becomes

$$a_0 + a_1 c_1^i = \left[\frac{\beta(1+r)\pi(s)}{p(s)}\right]^{-1/\rho} [a_0 + a_1 c_2^i(s)], \qquad i = 1, 2, \ldots, I,$$

if both sides are raised to the power $-1/\rho$. Sum this equation over all agents, divide by $I$, and then raise both sides of the resulting equality to the power $-\rho$. The result

and Smith (1992), Tesar (1993, 1995), Obstfeld (1994a), and Stockman and Tesar (1995). Time-series structural studies of consumption risk sharing include Obstfeld (1989, 1994a), Canova and Ravn (1994), Bayoumi and MacDonald (1995), and Kollmann (1995). Lewis's (1996) related panel estimates are the subject of an application in section 5.5. Baxter (1995) evaluates the success of simulated dynamic general-equilibrium models with complete markets in matching international consumption and output correlations (as well as other business-cycle regularities). The connection between complete markets and international aggregate consumption correlations was first made by Leme (1984) and Scheinkman (1984).



shows that for all $s$, *per capita* aggregate consumption $c = \sum_i c^i/I$ satisfies the Euler equation

$$(a_0 + a_1 c_1)^{-\rho} = \frac{\beta(1+r)\pi(s)[a_0 + a_1 c_2(s)]^{-\rho}}{p(s)}. \tag{39}$$

Because the individual budget constraints are also linear, of course, one can similarly average these to show that per capita individual consumptions must equal per capita individual resources in value. Equation (39) thus implies the result claimed: the economy's per capita consumption behaves as if chosen by a representative individual who owns the economy's average endowment. This result actually holds for a broader class of utility functions than we have analyzed here.[20] We emphasize, however, that aggregation usually is impossible absent a complete-markets allocation.[21]

Another approach to aggregation is sometimes useful under complete markets even when there are $I$ individuals with *distinct* CRRA utility functions in the restricted class of eq. (13). Now we define the representative consumer to have wealth and consumption equal, respectively, to the geometric averages of all individuals' wealth and consumption levels. The (unweighted) geometric average of individual consumptions is

$$\tilde{c} \equiv \prod_{i=1}^{I} (c^i)^{1/I}.$$

For individual $i$ the Euler equation (5) is equivalent to

20. That class of utility functions is the hyperbolic absolute risk aversion (HARA) class, which includes functions of the form

$u(c) = -a_0 \exp(-c/a_0)$,

$u(c) = \log(a_0 + a_1 c)$,

or

$u(c) = (1-\rho)^{-1}(a_0 + a_1 c)^{1-\rho}$,

see M. Rubinstein (1974). Milne (1979) observes that for utility functions that are additive over commodities (the case here), the HARA class is also the class of *quasi-homothetic* utility functions such that demand functions are linear in lifetime resources.

21. Suppose, for example, that there are two states of nature ($S = 2$) but that the only asset is a risk-free bond. The individual bond Euler equations imply

$$a_0 + a_1 c_1^i = [\beta(1+r)]^{-1/\rho} \left\{\sum_{s=1}^{2} \pi(s)[a_0 + a_1 c_2^i(s)]^{-\rho}\right\}^{-1/\rho},$$

but the right-hand side doesn't aggregate over agents unless their marginal rates of substitution between the two states happen to be the same.

$$c_2^i(s) = \left[\frac{\pi(s)(1+r)\beta}{p(s)}\right]^{1/\rho_i} c_1^i.$$

Taking both sides to the power $1/I$ and multiplying over all agents $i$ gives

$$\tilde{c}_2(s) = \prod_{i=1}^{I}\left[\frac{\pi(s)(1+r)\beta}{p(s)}\right]^{1/I\rho_i} \tilde{c}_1,$$

or

$$\frac{p(s)}{1+r} = \pi(s)\beta\left[\frac{\tilde{c}_2(s)}{\tilde{c}_1}\right]^{-\tilde{\rho}},$$

where $\tilde{\rho}$ is the harmonic mean of the countries' risk-aversion coefficients,

$$\tilde{\rho} \equiv \frac{1}{\frac{1}{I}\sum_{i=1}^{I}\frac{1}{\rho_i}}.$$

The representative's period utility function $u(\tilde{c})$ therefore is of the CRRA class with risk-aversion coefficient equal to the harmonic mean of the possibly distinct individual coefficients.

There are some important situations where perfect aggregation is infeasible, for example, in an overlapping generations framework. (People who haven't yet been born obviously can't participate in markets for future output.) Even there, we may be able to aggregate within generations that have coterminous lifetimes. Nevertheless, our examples show that the representative-agent assumption is a somewhat less restrictive paradigm for analyzing consumption behavior and asset prices than might appear at first glance. It is also apparent that our practice throughout most of this book of assuming a representative agent at the national level but not necessarily at the international level implicitly assumes that national markets are more complete and better integrated than international markets.

### 5.2.3 Dynamic Implications of Complete Markets with an Infinite Horizon

The infinite-horizon case allows a dynamic analysis of the current account's response to unexpected shocks that is impossible in the two-period paradigm, where random events occur only in the second and final period. In this subsection, we contrast how output shocks affect the current account in complete-markets versus bonds-only models with infinite horizons. The discussion is mostly intuitive. Appendix 5C formally extends the two-period Arrow-Debreu model to allow for an infinite horizon. Except for greater notational complexity the extension is straightforward, with results about risk sharing and global consumption correlations closely parallel to those of the two-period model.[22] With regard to the current account, however, the infinite-horizon model yields important new insights.



**Box 5.2**
Are Markets More Complete within than among Countries?

Many of this book's models assume representative national residents in order to focus on country-specific sources of individual heterogeneity. A possible justification for this approach is that markets seem to be better integrated within than between countries. Crucini (1992), for example, finds that for 1971–90 data, the consumption growth rates of different Canadian provinces are more highly correlated than are cross-province output growth rates or different countries' consumption growth rates. Atkeson and Bayoumi (1993), looking at regions within the United States, also find support for the view that risk sharing is higher within countries than among them, as do Bayoumi and Klein (1995) in another study of Canadian provinces. For a discussion of some of the difficulties in making comparisons between regional and international consumption correlations, see Obstfeld (1995).

One major reason to expect higher consumption correlations within than across countries is fiscal federalism. In most countries, federal taxes and transfers are effectively indexed to income, so that government policy plays a substantial role in pooling risks. Sachs and Sala-i-Martin (1992) estimate that in the United States, federal taxes and transfers offset roughly one-third of deviations of regional from national income. Other studies have arrived at somewhat lower estimates of 10 to 25 percent, but as we shall see, even these lower numbers may be large relative to the risk sharing that appears to take place through international capital markets.*

In discussing sovereign risk in Chapter 6, we shall identify another major reason why risk sharing is greater within countries than across them: the relative ease of intranational compared with international contract enforcement. Even if domestic capital markets allow more efficient risk pooling than international markets, they still appear to leave large opportunities for intranational pooling unexploited (see the studies of U.S. microeconomic data by Cochrane, 1991, and Mace, 1991). Indeed, Altonji, Hayashi, and Kotlikoff (1992) find evidence of unexploited risk-sharing opportunities even within U.S. families.

* See Bayoumi and Masson (1994) and the references therein. Bayoumi and Masson estimate that the U.S. federal fiscal system reduces long-term income differentials by 22 cents out of every dollar. They obtain a substantially larger estimate for Canada.

Assume a world of two pure endowment economies, Home and Foreign, identical in every respect on the first date $t$ of economic activity. Not only are they alike in preferences (which are intertemporally additive), but they also have identical outputs $\bar{Y} = \bar{Y}^*$ in all periods except for period $t+1$. On date $t+1$, the countries'

22. This similarity should not be surprising, because one can think of all trade taking place in the initial period of the Arrow-Debreu model, regardless the economy's time horizon (see appendix 5D). As noted by Arrow (1964), however, periodic trading opportunities can lead to efficient risk sharing without the full complement of securities that would be needed were trading restricted to a single date.

outputs are subject to identically and independently distributed mean-zero shocks $\epsilon_{t+1}$ and $\epsilon^*_{t+1}$, so that

$$Y_{t+1} = \bar{Y} + \epsilon_{t+1},$$

$$Y^*_{t+1} = \bar{Y} + \epsilon^*_{t+1},$$

with $E_t\epsilon_{t+1} = E_t\epsilon^*_{t+1} = 0$. Thus the only uncertainty is about period $t+1$ output. Moreover, shocks to $t+1$ output are purely temporary and don't affect outputs in period $t+2$ and beyond.

First consider a world analogous to the infinite-horizon models of Chapter 2, in which there is borrowing and lending in riskless one-period bonds, but no trade in contingent claims. Since the two countries are identical as of period $t$, there are initially no gains from intertemporal trade and the period $t$ current account will be zero. Now suppose that in period $t+1$, the Home output shock is greater than the Foreign output shock, $\epsilon_{t+1} > \epsilon^*_{t+1}$. Since the shock is purely temporary, Home smooths its consumption by lending to Foreign and running a current account surplus. In all future periods, current accounts are zero (the two countries have equal discount rates and, by assumption, there are no further shocks). Home consumption thus remains permanently higher than Foreign consumption.

Now suppose that there are complete markets, so that agents can trade Arrow-Debreu securities on date $t$ for all the possible date $t+1$ states of nature [that is, all possible realizations of the random pair $(\epsilon_{t+1}, \epsilon^*_{t+1})$]. In this case, by the time period $t+1$ arrives both countries will already have pooled their output risks and insured each other against any future country-specific output shocks by effectively trading a 50 percent equity share in the domestic output stream for a 50 percent share in foreign output. If it now turns out that $\epsilon_{t+1} > \epsilon^*_{t+1}$, *absolutely no* current account imbalances result. Realized Home *output* (GDP) $Y_{t+1}$ is higher than realized foreign output $Y^*_{t+1}$, but under complete markets, Home *income* (GNP), equal to GDP plus net dividend payments from abroad, $Y_{t+1} + \frac{1}{2}Y^*_{t+1} - \frac{1}{2}Y_{t+1} = \frac{1}{2}(Y_{t+1} + Y^*_{t+1})$, remains equal to Foreign income. Because Home and Foreign consumption both equal $\frac{1}{2}(Y_{t+1} + Y^*_{t+1})$ as well on date $t+1$, both national current accounts remain at zero. On dates after $t+1$, both Home and Foreign consumption revert to average world output, $\bar{Y}$.

If the date $t+1$ output shocks are *permanent*, the current account remains at zero in the bonds-only case too: Home consumption changes permanently by the change in Home output and Foreign consumption changes permanently by the change in Foreign output. The only difference under complete markets is that Home and Foreign consumption both change by equal amounts (half the change in world output).

While the current account implications of the two asset structures differ when productivity shocks are short-lived, the two models' implications for world interest rates are quite similar. With complete markets, a permanent rise in world output leaves the world interest rate unchanged, regardless of its incidence. This is the same as in the bonds-only model. Similarly, regardless of the asset structure, a temporary rise in world output incipiently raises saving and thus lowers the world interest rate. Thus complete markets remove the relative asymmetric income effects shocks would cause in the absence of risk sharing, but not the relative price effects.[23]



When investment is possible (as in the next section), a country with a persistent positive productivity shock experiences a current account deficit even in an infinite-horizon complete-markets setting.[24] The country's saving increases only partially to match higher domestic investment, and foreign capital flows in to buy up the rest of the expanded productive capacity. Even in this case, however, the current account is merely an accounting device for tracking the international distribution of the new equity claims foreigners must buy to maintain the efficient global pooling of national output risks.

### 5.2.4 Efficient Investment under Uncertainty

Although so far we have considered only endowment economies, it is simple to extend the complete markets model to incorporate investment.[25] All the main issues can be explained within a two-period framework. We continue to take Home's date 1 endowment $Y_1$ as exogenous, but we now assume that the country's date 2 output is produced by a competitive individual or "firm" using the production function

$$Y_2(s) = A(s)F(K_2),$$

where $K_2$ is capital accumulated during period 1 and invested in Home. As in Chapter 1 we abstract from the labor market, and we assume that producers can sell claims to their future outputs. For convenience $K_1 = K^*_1 = 0$ and depreciation is zero, so that date 1 investment equals the date 2 capital stock.

The (representative) Home firm chooses investment $K_2$ in period 1 to maximize the present value of profits, just as in earlier models. Here, however, profits earned in a state $s$ are valued using the Arrow-Debreu security price for output delivered in that state. The present value of the Home firm's state-contingent profits, measured in terms of date 1 output, therefore is

23. This point is emphasized by Stockman (1988a). The basic idea is that asset-market structure affects how shocks' income effects are distributed among countries, but with restrictions on preferences (e.g., homotheticity), international price effects can be unaffected by asset-market structure. This result is less likely to hold when investment is possible (see the following discussion). International differences in time preference or deterministic fluctuations in output growth can generate perfectly anticipated current-account imbalances even under complete markets, but not unexpected imbalances.

24. See, for example, Stockman and Svensson (1987) and Cantor and Mark (1988).

25. The text's discussion abstracts from some interesting and important issues, such as nonconvexities, that would considerably complicate the analysis. For an interesting partial-equilibrium analysis (and references), see Abel, Dixit, Eberly, and Pindyck (1995).

$$\sum_{s=1}^{S} \frac{p(s)}{1+r}[A(s)F(K_2) + K_2] - K_2,$$

and it is maximized (with respect to net investment $K_2$) when

$$\sum_{s=1}^{S} \frac{p(s)}{1+r}[A(s)F'(K_2) + 1] = 1. \tag{40}$$

Since the analogous condition holds in Foreign, the date 2 marginal value product of investment is the same everywhere in the world and equal to the world interest rate,

$$\sum_{s=1}^{S} p(s)A(s)F'(K_2) = \sum_{s=1}^{S} p(s)A^*(s)F^{*\prime}(K_2^*) = r.$$

Equations (5) and (40) imply that profit-maximizing investment decisions satisfy

$$u'(C_1) = \sum_{s=1}^{S} \pi(s)\beta u'[C_2(s)][A(s)F'(K_2) + 1]$$

$$= \sum_{s=1}^{S} \pi(s)\beta u'[C_2(s)][A^*(s)F^{*\prime}(K_2^*) + 1], \tag{41}$$

which states that the marginal utility of current Home consumption equals the expected Home marginal utility due to investing anywhere in the world. Because eq. (5) holds with $C$ replaced by $C^*$, the preceding relationship also holds for $C$ replaced by $C^*$. These two investment Euler equations are necessary conditions for efficient resource allocation.

#### 5.2.4.1 An Example of a Two-Country Global Equilibrium with Investment

An example of a global equilibrium is based on the model of section 5.2.1, with $\rho = 1$ (log preferences) and $F(K_2) = K_2$. When some resources are invested, the market-clearing conditions become

$$Y_1 + Y_1^* = C_1 + C_1^* + K_2 + K_2^*,$$

$$Y_2(s) + Y_2^*(s) + K_2 + K_2^* = C_2(s) + C_2^*(s), \qquad s = 1, 2, \ldots, S.$$

As a result, the price equation (30) for state $s$ is replaced by

$$\frac{p(s)}{1+r} = \frac{\pi(s)\beta(Y_1^W - K_2 - K_2^*)}{Y_2^W(s) + K_2 + K_2^*} = \frac{\pi(s)\beta(Y_1^W - K_2 - K_2^*)}{A(s)F(K_2) + A^*(s)F^*(K_2^*) + K_2 + K_2^*}.$$

Using these relationships, we eliminate $p(s)/(1+r)$ from the firms' profit-maximization condition (40) and find [for our linear production function with $F'(K_2) \equiv 1$] that the Home and Foreign firms' profit-maximization conditions are

$$\sum_{s=1}^{S} \left[\frac{\pi(s)\beta(Y_1^W - K_2 - K_2^*)}{A(s)K_2 + A^*(s)K_2^* + K_2 + K_2^*}\right][A(s) + 1] = 1,$$

$$\sum_{s=1}^{S} \left[\frac{\pi(s)\beta(Y_1^W - K_2 - K_2^*)}{A(s)K_2 + A^*(s)K_2^* + K_2 + K_2^*}\right][A^*(s) + 1] = 1.$$

These two equations in the unknowns $K_2$ and $K_2^*$ can be solved for equilibrium investment levels. We can solve for total world investment, $K_2 + K_2^*$, by multiplying the first of the immediately preceding equations by $K_2$, the second by $K_2^*$, and adding. The result is

$$K_2 + K_2^*$$

$$= \sum_{s=1}^{S} \left[\frac{\pi(s)\beta(Y_1^W - K_2 - K_2^*)}{A(s)K_2 + A^*(s)K_2^* + K_2 + K_2^*}\right]\{[A(s) + 1]K_2 + [A^*(s) + 1]K_2^*\}$$

$$= \sum_{s=1}^{S} \pi(s)\beta[Y_1^W - (K_2 + K_2^*)] = \beta Y_1^W - \beta(K_2 + K_2^*),$$

from which $K_2 + K_2^* = \beta Y_1^W/(1+\beta)$ follows. As a result of risk sharing, each country's consumption now is a date- and state-invariant share of world output *net* of world investment.

#### 5.2.4.2 Reprise: The Independence of Investment from Saving for a Small Country

We close with two further remarks about investment. First, eq. (40) shows that, just as in Chapter 1, a small country's investment is independent of its domestic consumption behavior. Given the production function and the statistical distribution of its productivity shock, world securities prices determine a country's equilibrium investment level. Equation (41) may make it appear as if a small country's consumption preferences affect investment decisions. But that interpretation of eq. (41) would be misleading. With complete markets, all country-specific output risk has been diversified away, which is why eq. (41) also holds with Foreign in place of Home consumption. The relation between consumption and a country's investment productivity in eq. (41) is really a relationship that involves *world* consumption.

Second, we've said nothing about the *ownership* of firms. In a complete-markets world, the nationality of a firm's owners clearly has no bearing on optimal investment decisions. But what determines how people actually do allocate their wealth among firms in various countries? We now turn to a model that can answer that question, and we will explain in detail its relationship to the complete-markets model of this section.

## 5.3 International Portfolio Diversification

By studying a world economy that trades a complete set of Arrow-Debreu securities, we have been able to understand the basic determinants of international trade across states of nature and the requirements for efficient economic allocation under uncertainty. As mentioned earlier, however, real-world international asset trade typically consists of exchanges of much more familiar securities, such as equity shares and risky bonds. How does a world that trades only these familiar assets compare to a complete-markets Arrow-Debreu world?

In this section we explore a world economy in which the only risky assets traded are claims to countries' uncertain outputs—essentially, shares of stock in national economies. We ask how the predictions of this more realistic model compare to those of the Arrow-Debreu model, and we examine various aspects of international portfolio diversification. Lucas (1982) explores a two-country, two-good version of this model. His solution approach, however, is valid only when the equilibrium yields asset prices such that initial wealth is equal across countries. The analysis that follows relaxes this restriction and shows how to derive the international distribution of wealth endogenously.

### 5.3.1 A Model of Portfolio Allocation

There are two dates (1 and 2), $N$ countries in the world, and $S$ states of nature on date 2. People throughout the world have the same preferences.

Let $V_1^n$ be the date 1 market value of country $n$'s uncertain date 2 output—that is, $V_1^n$ is the price of an asset that pays its owner $Y_2^n(s)$ in state $s$. One can think of $V_1^n$ as the value of a mutual fund that owns the totality of country $n$'s productive units. Obviously no single investor is likely to be rich enough to own an entire country's future output, but residents of different countries can exchange fractional shares. Indeed, we assume that the $N$ ownership claims on national outputs, along with a risk-free bond offering a net real interest rate of $r$, are the *only* assets traded on date 1.

A representative resident of country $n$ must divide date 1 income $Y_1^n$ between consumption $C_1^n$ and saving. Given the menu of assets assumed, country $n$'s saving takes the form of net bond purchases $B_2^n$ and net purchases of fractional shares $x_m^n$ in country $m$'s future output, $m = 1, 2, \ldots, N$. [Country $n$ net sales to foreigners of shares in future domestic output, $Y_2^n$, equal $(1 - x_n^n)V_1^n$, that is, the market value of country $n$'s future output less the share of it that country $n$'s own residents desire to retain in their portfolio, $x_n^n$.] The constraint linking consumption and saving on date 1 therefore takes the form

$$Y_1^n + V_1^n = C_1^n + B_2^n + \sum_{m=1}^{N} x_m^n V_1^m. \tag{42}$$



Country $n$'s date 2 consumption depends on how its assets perform between periods 1 and 2, which depends, in turn, on the realized state of nature $s$:

$$C_2^n(s) = (1 + r)B_2^n + \sum_{m=1}^{N} x_m^n Y_2^m(s). \tag{43}$$

In interpreting the budget constraint (43), it is worth stressing the distinction between the country mutual fund shares people trade here and the Arrow-Debreu forward contracts considered in previous sections. A standard Arrow-Debreu claim entitles the agent to a payoff in only one state of nature. Here, an investor holding $x_m$ shares of country $m$'s mutual fund is entitled to $x_m$ percent of country $m$'s date 2 output in *every* state of nature.

Country $n$'s representative resident maximizes the $S$-state analog of eq. (1) subject to eqs. (42) and (43). Substituting the constraints into the expected utility function gives

$$U_1 = u\left[Y_1^n + V_1^n - B_2^n - \sum_{m=1}^{N} x_m^n V_1^m\right] + \beta \sum_{s=1}^{S} \pi(s) u\left[(1 + r)B_2^n + \sum_{m=1}^{N} x_m^n Y_2^m(s)\right].$$

The necessary conditions for a maximum are, with respect to $B_2^n$,

$$u'(C_1^n) = (1 + r)\beta \sum_{s=1}^{S} \pi(s) u'[C_2^n(s)] = (1 + r)\beta \mathrm{E}_1\{u'(C_2^n)\}$$

[which is the same as the bond Euler equation (8)], and, with respect to portfolio shares $x_m^n$,

$$V_1^m u'(C_1^n) = \beta \sum_{s=1}^{S} \pi(s) u'[C_2^n(s)] Y_2^m(s) = \beta \mathrm{E}_1\{u'(C_2^n) Y_2^m\}, \qquad m = 1, 2, \ldots, N. \tag{44}$$

The intuition underlying eq. (44) is similar to that for other Euler conditions we have encountered. The left-hand side of eq. (44) is the marginal utility cost to a country $n$ resident who purchases country $m$'s risky future output on date 1, and the right-hand side is his expected marginal utility gain. In equilibrium these must be equal.[26]

26. Supplement A to this chapter looks at the individual's problem when the horizon is infinite, and derives consumption functions using dynamic programming.

### 5.3.2 Solving the Model with CRRA Utility

To illustrate the model's implications we assume that people in every country $n$ have identical CRRA period utility functions; see eq. (13). To solve the model we take an educated guess at the equilibrium allocation and find equilibrium portfolios and prices that support it. Our conjecture is that the equilibrium allocation is Pareto efficient (even though there is trade only in bonds and country mutual funds), and that equilibrium allocations therefore take the same general form as for the two-country, CRRA complete-markets example in section 5.2.1.

Defining country $n$'s share of initial world wealth as

$$\mu^n = \frac{Y_1^n + V_1^n}{\sum_{m=1}^{N}(Y_1^m + V_1^m)}, \tag{45}$$

we guess that its share of world consumption in each period and state is also $\mu^n$, just as in section 5.2.1:

$$C_1^n = \mu^n \sum_{m=1}^{N} Y_1^m = \mu^n Y_1^{\mathrm{W}}, \tag{46}$$

$$C_2^n(s) = \mu^n \sum_{m=1}^{N} Y_2^m(s) = \mu^n Y_2^{\mathrm{W}}(s), \qquad s = 1, \ldots, \mathcal{S}. \tag{47}$$

For any country $n$ the date 2 budget constraint (43) is consistent with eq. (47) if the representative national holds $\mu^n$ percent of a global mutual fund that encompasses all countries' second-period outputs,

$$x_m^n = \mu^n, \qquad m = 1, \ldots, N, \tag{48}$$

and if bond holdings are zero, $B_2^n = 0$. These consumption and portfolio plans are globally feasible if chosen simultaneously by every country $n$. To establish that they characterize the economy's equilibrium, we must find prices at which they maximize any country $n$'s intertemporal utility subject to satisfying its date 1 budget constraint. (We already know the date 2 constraint holds.)

For CRRA utility, the bond Euler equation (8) is equivalent to

$$1 + r = \frac{(C_1^n)^{-\rho}}{\beta \sum_{s=1}^{\mathcal{S}} \pi(s) C_2^n(s)^{-\rho}}.$$

It is simple to check that the consumption plans posited in eqs. (46) and (47) satisfy this Euler equation, for all countries $n$, if the equilibrium real interest rate is

$$1 + r = \frac{(Y_1^{\mathrm{W}})^{-\rho}}{\beta \sum_{s=1}^{\mathcal{S}} \pi(s) Y_2^{\mathrm{W}}(s)^{-\rho}}. \tag{49}$$



Similarly, the CRRA case of eq. (44) holds for all countries $n$ when consumption plans follow eqs. (46) and (47) if equilibrium share prices are

$$V_1^m = \sum_{s=1}^{\mathcal{S}} \pi(s)\beta \left[\frac{Y_2^{\mathrm{W}}(s)}{Y_1^{\mathrm{W}}}\right]^{-\rho} Y_2^m(s)$$

$$= \beta \mathrm{E}_1 \left\{ \left(\frac{Y_2^{\mathrm{W}}}{Y_1^{\mathrm{W}}}\right)^{-\rho} Y_2^m \right\}, \qquad m = 1, 2, \ldots, N. \tag{50}$$

It remains to check that every country's date 1 budget constraint (42) holds in the hypothesized equilibrium. But this task is easy. By definition (45), country $n$'s date 1 resources, $Y_1^n + V_1^n$, make up a fraction $\mu^n$ of total world wealth—the sum of current output and capitalized future output. Thus it is certainly consistent with eq. (42) for country $n$ to consume a fraction $\mu^n$ of world output [as in eq. (46)], to purchase the same share $\mu^n$ of the global mutual fund [as in eq. (48)], and to choose $B_2^n = 0$.[27]

### 5.3.3 Efficiency of the Allocation

The close parallel between the equilibrium we have just described and the complete-market equilibrium of section 5.2.1 points to a remarkable result. Even when the assets traded are limited to riskless bonds and shares in each of the $N$ countries' future outputs, the equilibrium allocation is efficient. Indeed, it is *identical* to the equilibrium reached through trade in a full set of $\mathcal{S}$ Arrow-Debreu securities, even when $\mathcal{S}$ is much greater than $N$.

To see what is going on, observe by comparing eqs. (50) and (30) that a country $n$'s stock-market value, $V_1^n$, is also the value of the country's uncertain future output evaluated at the date 1 Arrow-Debreu prices that would prevail were a complete set of contingent claims markets in operation. Thus the consumption share $\mu^n$ defined in eq. (45) is the same date- and state-invariant share of world consumption the country would enjoy with complete markets. (We pointed out in section 5.2.1.2 that a country's consumption share under complete markets is its share of wealth in world wealth when uncertain future outputs are evaluated at Arrow-Debreu prices.) Of course, the equilibrium interest rate is the same in the two economies; compare eq. (49) with eq. (33).

27. When people in different countries have different degrees of risk aversion, they may still hold the same mutual fund of *risky* assets, but those who are more risk-averse will hold some of their wealth in safe assets while those who are less risk-averse will go short in safe assets (take negative positions in bonds). As a result, the risk-sharing condition (37) can hold, with countries' consumption growth rates perfectly correlated but with the less risk-averse countries having a higher variance of consumption growth. For one model along these lines, see the continuous-time analysis of Obstfeld (1994c). For an example with exponential period utility, see exercise 5.

The economy behaves *as if* there were complete markets because all agents choose to hold equities in the same proportions, *regardless of their wealth levels*. The resulting equilibrium makes everyone's date 2 consumption move in proportion to the return on a single global mutual fund of risky assets. This common diversification strategy eliminates the gains from any further mutual consumption insurance between countries. We have demonstrated this result for identical CRRA preferences, but the same result can be obtained for a broader preference class.[28]

With other utility functions, or with international differences in preferences, people in different countries around the world might choose different portfolios of risky assets.[29] In this situation we can be certain that trade in equities is a perfect substitute for trade in a complete set of Arrow-Debreu securities only under much more restrictive assumptions. Specifically, Pareto efficiency can be ensured only when the number of states of nature $S$ is no greater than $N + 1$, the number of countries plus 1. The intuition behind this "spanning condition" (which appendix 5A proves formally) is simple. With one state of nature on date 2 (the case of certainty), trade in riskless bonds is sufficient to achieve an efficient allocation. If one adds a second state of nature, efficient allocation can be achieved provided a single risky asset is traded in addition to the bond: the two primal Arrow-Debreu securities can be constructed as linear combinations of the bond (which, as we have seen, is itself a linear combination of the Arrow-Debreu securities) and the risky asset. But adding more uncertain states without a corresponding addition of independent risky assets may compromise the economy's ability to achieve an efficient allocation of all risks. In general, spanning requires as many independent assets as states of nature.

### *Application: International Portfolio Diversification and the Home Bias Puzzle*

In section 5.3.2's simple model investors throughout the world all hold the same globally diversified portfolio of stocks. Even in richer settings, the logic of diversification suggests that people should hold a substantial fraction of their stock portfolios abroad. In practice, however, residents of most countries seem to hold a very large share of their equity wealth at home. A number of studies (for example, Grauer and Hakansson, 1987) suggest that an individual's gains from international stock-portfolio diversification are large, especially when emerging markets are included.

The contradiction between the obvious benefits of holding a globally dispersed set of equities and the apparent reluctance to do so is known as the *home bias*

28. That class is the HARA class noted in section 5.2.2, footnote 20. See M. Rubinstein (1974) for a proof.

29. But in some special cases they might not (see footnote 27).



**Table 5.2**
Share of Domestic Equities in Total Equity Portfolio, End 1989

| United States | United Kingdom | Japan |
|---|---|---|
| 0.96 | 0.82 | 0.98 |

*Source:* French and Poterba (1991).

*puzzle*. Table 5.2, based on figures reported by French and Poterba (1991), gives the share of domestic equities in the equity holdings of the United States, the United Kingdom, and Japan at the end of 1989. Tesar and Werner (1995) find similar results for 1990 data.[30]

Several considerations cloud the interpretation of Table 5.2, however. First, residents who hold shares of domestic-based multinationals obtain some degree of global portfolio diversification, and taking it into account goes a small way toward resolving the puzzle (see Mitra-Stiff, 1995). A second and closely related point is that Table 5.2 only gives equity holdings and does not give direct foreign investment or bond holdings. (The sharp distinction between corporate stocks and bonds becomes blurred once default and exchange rate risk are taken into account.) Direct foreign investment holdings are difficult to measure because there are no data on the market values of historical capital investments abroad. Tesar and Werner (1995) estimate that in 1990, U.S., Canadian, German, and Japanese direct foreign investment holdings stood at 11, 13, 6, and 6 percent of GNP, respectively. These numbers are substantial but not enough to erase home bias. However, home bias in bond holdings is smaller than that for stocks. The International Monetary Fund estimates that the share of public debt held by nonresidents now exceeds 20 percent for the seven major industrial countries. For Germany and France, nonresidents now hold more than 30 percent of negotiable public debt (Folkerts-Landau and Goldstein, 1994).

A number of possible explanations of the home bias puzzle have been presented; we will consider two of the most important (nontraded goods and small utility gains to diversification) later in this chapter. One obvious explanation is relatively high costs connected with foreign stock acquisition or ownership. Tesar and Werner (1995), noting that the turnover rate on nonresident holdings of equities is greater than the turnover rate for resident owners, argue that trading costs are unlikely to be the answer. Trading cost, however, is only one of several possible costs that might inhibit global diversification.[31]

30. Home bias seems to be present but somewhat less extreme for some smaller economies.

31. Kang and Stulz (1995) find that Japanese investors hold a disproportionate share of their foreign assets in large firms. They argue that this practice can be explained by fixed information costs in learning about individual foreign investment opportunities.

Whatever the cause of the current home bias, a growing share of portfolio investment is being channeled abroad today, so that the puzzle may be considerably reduced 25 years from now. American individual investors and mutual funds, for example, substantially increased their foreign equity holdings in the early 1990s. Bohn and Tesar (1995) estimate that the share of foreign equity in U.S. residents' equity portfolios had risen to 8 percent by the end of 1994. ■

## 5.4 Asset Pricing

When the set of assets traded is sufficient to produce a Pareto optimal equilibrium allocation, the price of any asset can be calculated as a linear combination of the underlying Arrow-Debreu prices. The calculation is valid even if the economy does not literally trade Arrow-Debreu securities, provided it *behaves* as if they did exist. In this section we apply this logic to explore asset pricing and the determinants of relative asset returns. Not surprisingly in light of our analysis so far, the less useful an asset is as a consumption hedge, the higher the rate of return that asset must offer investors in equilibrium.

### 5.4.1 The Consumption-Based Capital Asset Pricing Model

#### 5.4.1.1 Essentials of Asset Pricing

Let us begin within the two-period multicountry framework we used in the last section. Because date 1 Arrow-Debreu prices are the marginal rates of substitution of present consumption for future consumption in various states of nature—recall this interpretation of eq. (6)—they can be used to value the totality of state-contingent payoffs that defines any particular asset. For example, the market price of a country $m$ mutual fund that pays off $Y_2^m(s)$ in date 2 state of nature $s$ is simply

$$V_1^m = \sum_{s=1}^{S} \frac{p(s)Y_2^m(s)}{1+r}.$$

The country $m$ fund's payoffs can be perfectly replicated by a portfolio containing $Y_2^m(s)$ state $s$ Arrow-Debreu securities for every state $s$, so a price other than this one would create an arbitrage opportunity. The same argument can be applied to price all other assets.

When countries face the same Arrow-Debreu prices, we may substitute any country $n$'s marginal rate of substitution, given by eq. (6), for $p(s)/(1+r)$ in the preceding formula for $V_1^m$. The result is

$$V_1^m = \sum_{s=1}^{S} \left\{ \frac{\pi(s)\beta u'[C_2^n(s)]}{u'(C_1^n)} \right\} Y_2^m(s).$$



An alternative way to express this equation is as

$$V_1^m = \mathrm{E}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)} Y_2^m \right\}. \tag{51}$$

In eq. (51) we suppress the country superscript on consumption, $n$, since it does not matter which country's intertemporal marginal rate of substitution is used to price the asset: all are equal in complete markets. This is another sense in which the complete-markets assumption rationalizes a representative-agent perspective. We have seen that when countries have identical CRRA utility functions, $C_1$ and $C_2$ in eq. (51) could be replaced by world consumption levels. [This reasoning inspired our "guess" in section 5.3.2 that eq. (50) was the equilibrium value of $V_1^m$.]

#### 5.4.1.2 Asset Returns

We now derive a formula that explains asset prices in terms of expected payoffs and the riskiness of those payoffs. Notice that eq. (51) is equivalent to

$$V_1^m = \mathrm{E}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)} \right\} \mathrm{E}_1\{Y_2^m\} + \mathrm{Cov}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)}, Y_2^m \right\}.$$

Equation (8) permits the substitution $1/(1+r) = \mathrm{E}_1\{\beta u'(C_2)/u'(C_1)\}$ here. The result is

$$V_1^m = \frac{\mathrm{E}_1\{Y_2^m\}}{1+r} + \mathrm{Cov}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)}, Y_2^m \right\}. \tag{52}$$

According to this equation, the date 1 price of country $m$'s uncertain date 2 output is the sum of two terms. The first is the asset's expected payout, discounted at the riskless rate of interest. The second is the covariance between the relative marginal utility of date 2 consumption and the payout. This covariance reflects the asset's value as consumption insurance. Other things being equal, an asset that tends to pay off unexpectedly well on date 2 when the marginal utility of consumption is unexpectedly high (meaning that consumption itself is unexpectedly low) has value as a consumption hedge and therefore will command a price above its "risk-neutral" or actuarially fair price.

Define the *ex post net real rate of return* to a share in country $m$'s output, $r^m$, by

$$r^m = \frac{Y_2^m - V_1^m}{V_1^m}.$$

Multiplying both sides of eq. (52) by $(1+r)/V_1^m$ yields

$$\begin{aligned} \mathrm{E}_1\{r^m\} - r &= -(1+r)\mathrm{Cov}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)}, 1+r^m \right\} \\ &= -(1+r)\mathrm{Cov}_1\left\{ \frac{\beta u'(C_2)}{u'(C_1)}, r^m - r \right\}. \end{aligned} \tag{53}$$

Equation (53) is called the *consumption-based capital asset pricing model* (or consumption-based CAPM).[32] It says that the expected premium an asset must yield over the riskless rate of return (called the asset's risk premium or excess return) depends negatively on the covariance of the asset's excess return with the rate of growth of the marginal utility of consumption. Put differently, the excess return depends *positively* on the correlation of the asset's return with world consumption growth.[33]

Our discussion of how the covariance on the right-hand side of eq. (53) affects the asset's *price* also explains how that factor influences the asset's equilibrium *expected return*. A positive covariance means that the asset tends to yield unexpectedly high returns in states of nature when the marginal utility of world consumption is unexpectedly high or, equivalently, when the level of world consumption is unexpectedly low. Because the asset therefore provides a hedge against world output fluctuations, it offers an expected rate of return below the riskless rate of interest in equilibrium. Conversely, if an asset is more likely to have high returns when world output is high (that is, when the marginal utility of consumption is low), the covariance term in eq. (53) is negative and the asset's expected return will include a positive risk premium. Because individual national outputs sum to world output, the "typical" country stock's payoff is likely to be high when global output is high (and therefore to covary negatively with the marginal utility of world consumption).

#### 5.4.1.3 Market Incompleteness, Asset Pricing, and Investment

We have derived the consumption-based CAPM under the assumption that sufficiently many assets are traded to replicate complete markets. In fact, however, when riskless bonds are traded, eq. (53) holds for any internationally traded risky asset, *even if asset markets are not complete*. Equation (44), derived in section 5.3 without any assumptions on the completeness of markets, implies that eq. (51) holds true for the intertemporal marginal rate of substitution $\beta u'(C_2^n)/u'(C_1^n)$ of any country $n$ that can freely trade the country $m$ mutual fund.[34] If riskless bonds are traded, expected intertemporal rates of substitution will be equal across countries [see eq. (8)]. Thus, even if ex post intertemporal rates of substitution are not internationally equal (so that risks are not allocated efficiently), the first-order conditions for utility maximization nonetheless ensure equality among the *covariances* of countries' intertemporal marginal rates of substitution with the return on *any* risky asset they all trade.

This point also has implications for investment under incomplete markets. As we saw in eq. (40), under complete markets the investment decision maximizes the market value of the country $m$ firm's future dividends net of date 1 investment,

$$\begin{aligned} V_1^m - K_2^m &= \sum_{s=1}^{S} \frac{p(s)}{1+r}[A^m(s)F^m(K_2^m) + K_2^m] - K_2^m \\ &= \sum_{s=1}^{S} \frac{\pi(s)\beta u'[C_2^n(s)]}{u'(C_1^n)}[A^m(s)F^m(K_2^m) + K_2^m] - K_2^m \\ &= \mathrm{E}_1\left\{\frac{A^m(s)F^m(K_2^m) + K_2^m}{1+r}\right\} \\ &\quad + \mathrm{Cov}_1\left\{\frac{\beta u'[C_2^n(s)]}{u'(C_1^n)}, A^m(s)\right\} F^m(K_2^m) - K_2^m, \end{aligned}$$

where the consumption of any country $n$ gives the same value. If markets are incomplete but people trade bonds and shares in the firm, the covariance term in the last equation must still be the same for all national consumptions $n$, for any $K_2^m$. (Again, this equality is true because the same Euler condition corresponding to the asset holds for individuals in all countries.) Thus all owners of the firm, regardless of nationality, will agree that it should raise its investment slightly whenever $\mathrm{d}(V_1^m - K_2^m)/\mathrm{d}K_2^m > 0$, for the derivative

$$\frac{\mathrm{d}(V_1^m - K_2^m)}{\mathrm{d}K_2^m} =$$

$$\mathrm{E}_1\left\{\frac{A^m(s)F^{m\prime}(K_2^m) + 1}{1+r}\right\} + \mathrm{Cov}_1\left\{\frac{\beta u'[C_2^n(s)]}{u'(C_1^n)}, A^m(s)\right\} F^{m\prime}(K_2^m) - 1$$

does *not* depend on which consumption $n$ is used. As a result, optimal investment will still be governed by the condition $\mathrm{d}(V_1^m - K_2^m)/\mathrm{d}K_2^m = 0$, which is equivalent to eq. (41) and will hold for any shareholder's consumption.[35]

32. In deriving it we have used the fact that for any constant $a_0$,

$$\mathrm{Cov}_1\left\{\frac{\beta u'(C_2)}{u'(C_1)}, r^m + a_0\right\} = \mathrm{Cov}_1\left\{\frac{\beta u'(C_2)}{u'(C_1)}, r^m\right\}.$$

33. It is worth emphasizing that the present model is a general equilibrium model, in contrast to the older partial equilibrium capital asset pricing model (as in Sharpe, 1964, for example). In the older (international) CAPM model, assets whose returns are positively correlated with the world market portfolio must offer a higher expected return, a risk premium. There is a substantial empirical literature devoted to testing whether a global or national portfolio is more suitable for measuring an asset's risk premium. See Dumas (1994) and Lewis (1995) for surveys of partial equilibrium asset-pricing models.

34. This statement must be qualified if short sales are prohibited and some countries do not hold positive amounts of the fund. A deep issue concerns sufficient conditions under which a global equilibrium is guaranteed to *exist* without complete markets. For theoretical discussion, see Magill and Quinzii (1996).

35. Our assumption of multiplicative production uncertainty plainly is critical to the argument. For a general discussion, see Ekern and Wilson (1974).

### 5.4.2 The Equity Premium Puzzle

One of the most striking mysteries in empirical finance is the enormous observed average rate of return differential between stocks and bonds. Over the past seventy years, the annual rate of return on stocks has exceeded that on government bonds by more than 5 percent across most major stock markets of the world (see Siegel, 1995). Plainly it is no puzzle that the expected return on stocks is higher than on safer assets. It is the *size* of the differential that perplexes researchers. For seemingly reasonable parameterizations of a general equilibrium complete markets model, premiums of more than 1 or 2 percent cannot easily be rationalized. This problem is the *equity premium puzzle*, first pointed out by Mehra and Prescott (1985) for annual 1889–1978 United States data. As we shall see, the great riskiness of equities that the equity premium puzzle implies only deepens the home bias puzzle.

To see the basic problem, look at the CRRA case of eq. (53), which gives the excess expected return on an equity share of country $m$'s output:

$$\mathrm{E}_1\{r^m\} - r = -(1+r)\mathrm{Cov}_1\left\{\beta\left(\frac{C_2}{C_1}\right)^{-\rho}, r^m - r\right\}. \tag{54}$$

Because annual consumption growth is fairly stable and the annual variability of major stock market returns relatively moderate, the covariance of consumption growth with returns, while positive, is too low to explain the huge equity premium unless the risk-aversion coefficient $\rho$ is extremely high.

To get a better feel for the magnitudes involved we consider an approximate version of eq. (54). The approximation is based on a second-order Taylor expansion of the function

$$G\left(\frac{C_2}{C_1}, r^m\right) \equiv \beta\left(\frac{C_2}{C_1}\right)^{-\rho}(r^m - \mathrm{E}_1 r^m),$$

the expected value of which equals the covariance entering the equity premium in eq. (54).[36] Near the points $C_2/C_1 = 1$ and $r^m = \mathrm{E}_1 r^m$ the approximation is

36. The second-order approximation to $G(X_1, X_2)$ in the neighborhood of $X_1 = \bar{X}_1$, $X_2 = \bar{X}_2$ is

$$\begin{aligned} G(X_1, X_2) \approx\ & G(\bar{X}_1, \bar{X}_2) + G_{X_1}(\bar{X}_1, \bar{X}_2)(X_1 - \bar{X}_1) + G_{X_2}(\bar{X}_1, \bar{X}_2)(X_2 - \bar{X}_2) \\ & + \tfrac{1}{2}G_{X_1X_1}(\bar{X}_1, \bar{X}_2)(X_1 - \bar{X}_1)^2 + \tfrac{1}{2}G_{X_2X_2}(\bar{X}_1, \bar{X}_2)(X_2 - \bar{X}_2)^2 \\ & + G_{X_1X_2}(\bar{X}_1, \bar{X}_2)(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) \end{aligned}$$

(assuming the listed derivatives exist). In omitting any higher-order Taylor-expansion terms when we equate expectations of this approximation's two sides (as we will do), we are implicitly assuming that higher-order moments of the joint distribution of $X_1$ and $X_2$ are negligible compared to the first and second moments. Ingersoll (1987, pp. 259–262) describes the class of "compact" distributions for which this will be the case if the trading interval is sufficiently small.



$$G\left(\frac{C_2}{C_1}, r^m\right) \approx \beta(r^m - \mathrm{E}_1 r^m) - \beta\rho\left(\frac{C_2}{C_1} - 1\right)(r^m - \mathrm{E}_1 r^m).$$

Taking conditional expectations of both sides above yields

$$\begin{aligned} \mathrm{E}_1 G\left(\frac{C_2}{C_1}, r^m\right) &= \mathrm{Cov}_1\left\{\beta\left(\frac{C_2}{C_1}\right)^{-\rho}, r^m - r\right\} \\ &\approx -\beta\rho\mathrm{Cov}_1\left\{\frac{C_2}{C_1} - 1, r^m - r\right\} \end{aligned}$$

(recall footnote 32). Using this estimate to eliminate the covariance term in eq. (54) yields the approximation we seek,

$$\begin{aligned} \mathrm{E}_1\{r^m\} - r &= (1+r)\beta\rho\mathrm{Cov}_1\left\{\frac{C_2}{C_1} - 1, r^m - r\right\} \\ &= (1+r)\beta\rho\kappa\,\mathrm{Std}_1\left\{\frac{C_2}{C_1} - 1\right\}\mathrm{Std}_1\{r^m - r\}, \end{aligned}$$

where $\kappa \equiv \mathrm{Corr}_1(C_2/C_1, r^m - r)$ is the conditional correlation coefficient between consumption growth and the excess return on equity, and the conditional standard deviation, $\mathrm{Std}_1$, is the square root of the conditional variance.[37]

Mankiw and Zeldes (1991) point out that, taking averages over the entire Mehra-Prescott data set, $\kappa = 0.4$, the standard deviation of per capita consumption growth is 0.036 per year, and that of the excess equity return is 0.167 per year. If we assume that $(1+r)\beta = 1$, then the model fits the excess average annual 1889–1978 equity return of

$$\mathrm{E}\{r^m\} - r = 0.0698 - 0.0080 = 0.0618$$

only if $\rho$ is roughly 26![38] As Mankiw and Zeldes (1991) also observe, the equity premium puzzle is even more severe in postwar U.S. data than in the original

37. When variables are jointly lognormally distributed, one can obtain an exact expression for the equity premium that closely parallels this one. See, for example, Aiyagari (1993).

38. Another way to see the puzzle is to let $mrs_2$ be the ex post marginal rate of intertemporal substitution realized on date 2 and rewrite eq. (51) as

$$\begin{aligned} 1 &= \mathrm{E}_1\{mrs_2(1 + r^m)\} \\ &= \mathrm{Cov}_1\{mrs_2, 1 + r^m\} + \mathrm{E}_1\{mrs_2\}\mathrm{E}_1\{1 + r^m\}, \end{aligned}$$

which implies that

$$1 - \mathrm{E}_1\{mrs_2\}\mathrm{E}_1\{1 + r^m\} = \mathrm{Cov}_1\{mrs_2, 1 + r^m\}.$$

Observe that $\mathrm{Std}_1\{mrs_2\}\mathrm{Std}_1\{1 + r^m\} = \mathrm{Cov}_1\{mrs_2, 1 + r^m\}/\mathrm{Corr}_1\{mrs_2, 1 + r^m\}$, where $\mathrm{Corr}_1\{mrs_2, 1 + r^m\}$, the correlation coefficient, is below 1 in absolute value. Thus,

Mehra-Prescott sample, with values of $\rho$ as high as 100 needed to justify the observed premium on stocks. Most economists feel that such high levels of risk aversion are unrealistic.[39]

If agents are really so risk averse, it is all the more puzzling that they do not take greater advantage of opportunities to diversify their equity portfolios internationally. From this perspective there is an intimate connection between the home bias and equity premium puzzles: if equity risk is so significant, why isn't much more of it laid off on foreign investors?

Several possible solutions of the equity premium puzzle have been proposed. One factor, stressed by Aiyagari and Gertler (1991), is that transaction costs on stock trades may be higher than on bond trades.[40] The cost difference leads to a liquidity premium on bonds that can explain the high equity premium. Transaction costs must be very high, however, to explain a significant portion of it. Aiyagari and Gertler (1991) and Weil (1992) argue that the presence of nondiversifiable labor-income risk also can help to resolve the puzzle. [With nondiversifiable income eq. (54) won't generally hold for aggregate per capita consumption, only for the individual consumptions of those who can trade equities.] Heaton and Lucas (1995) show how the interaction of several market frictions and imperfections can explain asset-pricing puzzles.

Another line of explanation, due to Constantinides (1990), abandons time-additive utility. Consider a period utility function with *habit persistence*,

$$u(C_t, D_t) = \frac{(C_t - D_t)^{1-\rho}}{1-\rho},$$

where $D_t$ is a stock of "consumption experience" that evolves according to

$$D_t = (1-\delta)D_{t-1} + \delta\zeta C_{t-1}, \qquad 0 < \zeta, \delta < 1.$$

(Supplement B to Chapter 2 shows a different way of relaxing time additivity.) With habit persistence, accumulated consumption experience $D_t$ tends to make $C_t - D_t$ a small number if $\zeta$ is close to 1, allowing realistic consumption fluctuations to have a very large effect on the marginal rate of substitution between current and future consumption even when $\rho$ is not large. Since a more volatile ex post intertemporal marginal rate of substitution raises the warranted equity premium [by analogy to what eq. (54) would predict], habit persistence can help resolve the puzzle. But the attendant increase in the volatility of expected intertemporal marginal substitution rates [by analogy to what eq. (8) would predict] makes real interest rates counterfactually volatile.

A related asset-pricing puzzle, raised by Mehra and Prescott (1985) and emphasized by Weil (1989b), is the "low riskless interest rate puzzle." With CRRA utility, eq. (8) can be written

$$1 + r = \frac{1}{\mathrm{E}_t\left\{\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\rho}\right\}}.$$

To illustrate the riskless rate puzzle we assume that $C$ is lognormally distributed.[41] Then the last equation yields the approximation

$$r \approx \log(1+r) = \rho \mathrm{E}_t\left\{\log\left(\frac{C_{t+1}}{C_t}\right)\right\} - \frac{\rho^2}{2}\mathrm{Var}\left\{\log\left(\frac{C_{t+1}}{C_t}\right)\right\} - \log(\beta).$$

In the Mehra-Prescott sample the mean annual per capita consumption growth rate is 0.018 per year, with a variance of 0.0013. Thus for a reasonable $\rho$ value such as 2, even with $\beta = 1$, the model predicts a yearly riskless rate of 3.34 percent, more than four times the 0.80 percent that Mehra and Prescott estimated.

Some of the same factors that help explain the equity premium puzzle—for example, transaction costs and uninsured labor income—also imply a low riskless rate. In the early 1980s real interest rates rose globally, largely erasing the riskless rate conundrum from recent data. As the next application shows, several other historical epochs have been characterized by quite high real interest rates.

---

$$\mathrm{Std}_t\{mrs_2\}\mathrm{Std}_t\{1+r^m\} \geq 1 - \mathrm{E}_t\{mrs_2\}\mathrm{E}_t\{1+r^m\},$$

or

$$\mathrm{Std}_t\{mrs_2\} \geq \frac{1 - \mathrm{E}_t\{mrs_2\}\mathrm{E}_t\{1+r^m\}}{\mathrm{Std}_t\{1+r^m\}}.$$

This inequality, due to Hansen and Jagannathan (1991), uses only observable data on returns to define combinations of the mean and standard deviation of $mrs_2$ (which is not directly observable) that are consistent with an Euler equation like eq. (51). The rationale for this approach, which does not rely on estimating the marginal rate of substitution in consumption, is that the econometrician may be unsure of the underlying utility function or else the consumption data may be measured with error. For $mrs_2 = \beta(C_2/C_1)^{-\rho}$ and low values of $\rho$, U.S. equity returns violate the Hansen-Jagannathan bound.

39. Kandel and Stambaugh (1991) argue that risk aversion as high as $\rho = 30$ should not be ruled out, but even that high value would be insufficient to match the model to postwar data. The prewar-postwar difference may be due in part to spurious volatility in prewar U.S. national income data; see C. Romer (1989).

40. In this spirit, He and Modest (1995) show that even for low degrees of risk aversion, transaction costs and trading constraints can make aggregate consumption data appear consistent with modified Hansen-Jagannathan bounds.

41. If $X$ is a normally distributed random variable with mean $\mu_X$ and variance $\sigma_X^2$, then $\exp X$ is lognormal with mean

$$\mathrm{E}\{\exp X\} = \exp\left(\mu_X + \tfrac{1}{2}\sigma_X^2\right).$$

Notice that $\mathrm{E}\{\exp X\} > \exp \mu_X$ is consistent with Jensen's inequality [the function $\exp(\cdot)$ is strictly convex].

***Application: The Equity Premium Puzzle over the Very Long Run***

Mehra and Prescott (1985) calculated that the average annual real return on a market portfolio of U.S. stocks exceeded that on U.S. Treasury securities by 6.18 percent over the period 1889–1978.[42] Table 5.3 reports data on real equity and bond returns over subperiods of the years 1802–1992, a span of nearly two centuries.

Despite its year-to-year variability, the real return on the U.S. stock market over long periods has been remarkably stable. Real bond returns, in contrast, were much lower after 1925 than earlier. (A major reason for these low average returns was that bonds yielded negative real returns during the high-inflation 1970s.) Thus there is some sense in which the equity premium puzzle becomes less severe when one looks at longer time periods. Still, even the period 1926–92 is a very long one.

One might conjecture that the high ex post long-term returns on U.S. equities simply reflect the fact the United States has been one of the world's great economic success stories. The equity premium phenomenon, however, while not universal, certainly has characterized several other industrial-country financial markets. Over the 1926–92 period, annual real yields of 5.71 percent on British stocks and of 5.43 percent on German stocks far exceeded bond returns in those countries. (British bond yields averaged below comparable U.S. yields while German bonds suffered the ravages of postwar inflation.) Twentieth-century Britain is hardly an example of unalloyed economic dynamism. German stocks performed well despite the nearly total destruction or dismantling of the country's physical capital stock during and just after World War II. Thus the equity premium puzzle seems to have arisen in several countries despite disparate economic circumstances. This finding is not totally surprising: to the extent that international capital markets have been integrated since 1926, relative average asset returns over long periods should reflect global rather than country-specific factors.

Some researchers suggest that the equity premium is simply a historical anomaly of the post-Depression period. Blanchard (1993) develops ex ante measures of the equity premium for a number of OECD countries (the numbers in Table 5.3 are based on ex post returns) and contends that the equity premium has become much smaller since 1980. Brown, Goetzman, and Ross (1995) argue for an international investing perspective that accounts for survival bias, which appears to be more important when one looks at very long time periods. They point out that in 1792, investors could buy stocks in markets in Holland, France, Austria, and Germany

42. Over 1889–1919 they used a commercial paper rate, as the first Treasury certificates were issued only in 1920.



**Table 5.3**
Real U.S. Stock and Government Bond Returns (annual geometrically compounded percent rate of return)

| Period | Stocks | Short Bonds | Long Bonds |
|---|---|---|---|
| 1802–1992 | 6.7 | 2.9 | 3.4 |
| 1871–1992 | 6.6 | 1.7 | 2.6 |
| 1802–70 | 7.0 | 5.1 | 4.8 |
| 1871–1925 | 6.6 | 3.2 | 3.7 |
| 1926–92 | 6.6 | 0.5 | 1.7 |
| 1946–92 | 6.6 | 0.4 | 0.4 |

*Source:* Siegel (1995).

in addition to the United States and Britain. But only the U.S. and British markets have operated continuously since then. When market casualties are taken into account, Brown, Goetzman, and Ross argue, the equity premium becomes much smaller. ■

### 5.4.3 The Infinite-Horizon Consumption-Based CAPM

The model we have been exploring in this section and section 5.3 is easily extended to encompass an infinite horizon. The representative resident of country $n$ now maximizes the expected utility function

$$U_t = \mathrm{E}_t \left\{ \sum_{s=t}^{\infty} \beta^{s-t} u(C_s^n) \right\} \tag{55}$$

on any date $t$, where the conditional expectation operator averages over all possible future contingency plans for consumption. The country $n$ individual's period-by-period finance constraint is

$$B_{s+1}^n + \sum_{m=1}^{N} x_{m,s+1}^n V_s^m = (1 + r_s) B_s^n + \sum_{m=1}^{N} x_{m,s}^n (Y_s^m + V_s^m) - C_s^n, \tag{56}$$

where $V_s^m$ now denotes the *ex dividend* date $s$ market price of a claim to the entire income stream of country $m$ in *all* future periods, $x_{m,s}^n$ denotes the fractional share of country $m$'s security purchased by agent $n$ in period $s - 1$, and $r_s$ is the one-period real interest rate between periods $s - 1$ and $s$. Maximizing the utility function (55) subject to the constraints (56) shows that on every date $s$ the first-order conditions

$$u'(C_s^n) V_s^m = \beta \mathrm{E}_s \{ u'(C_{s+1}^n)(Y_{s+1}^m + V_{s+1}^m) \}, \tag{57}$$

$$u'(C_s^n) = (1 + r_{s+1})\beta \mathrm{E}_s\{u'(C_{s+1}^n)\} \tag{58}$$

will hold.[43]

To derive asset prices take the $s = t + 1$ Euler condition (57) and use it to eliminate $V_{t+1}^m$ from the $s = t$ condition. The result is

$$V_t^m = \mathrm{E}_t\left\{\frac{\beta u'(C_{t+1})}{u'(C_t)} Y_{t+1}^m\right\} + \mathrm{E}_t\left\{\frac{\beta u'(C_{t+1})}{u'(C_t)} \mathrm{E}_{t+1}\left\{\frac{\beta u'(C_{t+2})}{u'(C_{t+1})}(Y_{t+2}^m + V_{t+2}^m)\right\}\right\}$$

(where we have dropped the country superscript $n$ on consumption because the equilibrium asset price does not depend on which nation's consumption we use to value future payoffs). Because $\beta u'(C_{t+1})/u'(C_t)$ is in the set of information upon which date $t + 1$ expectations are conditioned, we may pass that marginal rate of substitution through the expectations operator $\mathrm{E}_{t+1}\{\cdot\}$ and express the second summand on the right-hand side of the preceding equation as

$$\mathrm{E}_t\left\{\frac{\beta u'(C_{t+1})}{u'(C_t)} \mathrm{E}_{t+1}\left\{\frac{\beta u'(C_{t+2})}{u'(C_{t+1})}(Y_{t+2}^m + V_{t+2}^m)\right\}\right\}$$

$$= \mathrm{E}_t\left\{\mathrm{E}_{t+1}\left\{\frac{\beta^2 u'(C_{t+2})}{u'(C_t)}(Y_{t+2}^m + V_{t+2}^m)\right\}\right\}.$$

By the law of iterated conditional expectations, $\mathrm{E}_t\{\mathrm{E}_{t+1}\{X\}\} = \mathrm{E}_t\{X\}$ for any random variable $X$, so we obtain the simplified formula for $V_t^m$:

$$V_t^m = \mathrm{E}_t\left\{\frac{\beta u'(C_{t+1})}{u'(C_t)} Y_{t+1}^m\right\} + \mathrm{E}_t\left\{\frac{\beta^2 u'(C_{t+2})}{u'(C_t)}(Y_{t+2}^m + V_{t+2}^m)\right\}.$$

The usual process of iterative forward substitution, coupled with the condition

$$\lim_{T\to\infty} \mathrm{E}_t\left\{\beta^T [u'(C_{t+T})/u'(C_t)] V_{t+T}^m\right\} = 0$$

to exclude speculative bubbles, shows the date $t$ price of the country $m$ equity to be

$$V_t^m = \mathrm{E}_t\left\{\sum_{s=t+1}^{\infty} \frac{\beta^{s-t} u'(C_s)}{u'(C_t)} Y_s^m\right\}$$

$$= \sum_{s=t+1}^{\infty} R_{t,s} \mathrm{E}_t\{Y_s^m\} + \sum_{s=t+1}^{\infty} \mathrm{Cov}_t\left\{\frac{\beta^{s-t} u'(C_s)}{u'(C_t)}, Y_s^m\right\}. \tag{59}$$

43. Just solve eq. (56) for $C_s^n$, substitute the result into eq. (55), and differentiate with respect to $x_{m,s+1}^n$ and $B_{s+1}^n$. Alternatively, Supplement A to this chapter gives a dynamic programming approach and also shows how to derive the optimal consumption function in special cases.



The second equality is derived in the same fashion as eq. (52), where $R_{t,s}$ (as usual) is the date $t$ market discount factor for noncontingent date $s > t$ consumption,

$$R_{t,s} = \mathrm{E}_t\left\{\frac{\beta^{s-t} u'(C_s)}{u'(C_t)}\right\}, \tag{60}$$

that is, the inverse of the gross interest rate on a riskless $s - t$ period discount bond.[44] Equation (59) is a very natural extension of the two-period pricing equation (52). It states that an asset's price is the expected present value of payouts (discounted at riskless rates) plus a sum of risk adjustments, each of which reflects the asset's contribution to consumption insurance on a different future date.

All the results on the efficiency of equity trade in a two-period model carry over to an infinite horizon. For CRRA utility the results of section 5.3.2 generalize immediately. In particular, each country holds a fixed share of the world equity portfolio, each country's consumption is a constant fraction of world output $Y_t^{\mathrm{W}}$ on every date $t$, and the multiperiod analog of eq. (50) holds:

$$V_t^m = \mathrm{E}_t\left\{\sum_{s=t+1}^{\infty} \beta^{s-t}\left(\frac{Y_s^{\mathrm{W}}}{Y_t^{\mathrm{W}}}\right)^{-\rho} Y_s^m\right\}, \qquad m = 1, 2, \ldots, N. \tag{61}$$

***Application: GDP-Linked Securities and Estimates of $V^m$***

If claims to countries' entire future outputs were traded, how big would $V^m$ be in practice? Shiller (1993) has advocated creation of such securities on the grounds that they would facilitate hedging against fluctuations in currently nontradable income components, notably much of labor income. (Labor income is largely uninsurable because it depends on the worker's effort and thus is subject to moral hazard absent some harsh enforcement mechanism such as slavery.) Shiller estimates the market value of a perpetual claim to a county's entire GDP, along with the standard deviation of the return on that asset. His measures of a country's worth far exceed those produced by standard financial wealth calculations, in part because roughly two-thirds of output in most countries goes to labor, and payment streams earned by labor are not securitized. Positing a common, constant market discount rate of 6.8 percent per annum (which is assumed to incorporate an appropriate risk correction), Shiller calculates the dollar market value of a claim to all future U.S.

44. The same reasoning behind Euler equations (8) or (58) shows that when there is a riskless long-term discount bond that pays 1 output unit after $s - t$ periods in return for $R_{t,s}$ output units invested today, individual consumption optimality requires

$$R_{t,s} u'(C_t) = \beta^{s-t} \mathrm{E}_t\{u'(C_s)\}.$$

Equation (60) for $R_{t,s}$ follows.

**Table 5.4**
Measures of $V^m$, the Securitized Value of a Claim to a Country's Entire Future GDP, 1992 (billions of U.S. dollars)

| Country | $V^m$ | Std($r^m$) | Country | $V^m$ | Std($r^m$) |
|---|---|---|---|---|---|
| Argentina[a] | 2,460 | 9.86 | Nigeria | 2,019 | 10.06 |
| Australia | 4,340 | 3.88 | Pakistan | 2,894 | 2.45 |
| Brazil | 10,032 | 8.88 | Philippines[a] | 1,602 | 3.68 |
| Canada | 7,663 | 4.22 | South Africa | 1,722 | 8.98 |
| France | 12,901 | 5.38 | Spain | 6,721 | 6.30 |
| Germany (West) | 16,796 | 4.47 | Sweden | 1,972 | 5.70 |
| India | 20,378 | 4.32 | Switzerland | 1,911 | 5.30 |
| Italy | 11,540 | 4.68 | Thailand | 4,007 | 3.99 |
| Japan | 31,762 | 8.41 | Turkey | 3,868 | 3.38 |
| Kenya | 418 | 4.34 | United Kingdom | 13,495 | 1.46 |
| Mexico | 9,583 | 5.33 | United States | 82,075 | 2.03 |
| Netherlands | 3,607 | 4.68 | Venezuela | 2,501 | 6.87 |

*Source:* Methodology is based on Shiller (1993, ch. 4). Underlying annual real GDP data are from Penn World Table, version 5.6. Standard deviations are on annual return (income plus appreciation) of a perpetual claim to GDP.
[a] 1990 value based on 1950–90 data.

output at $81 trillion as of 1990, or more than 14 times that year's GDP. By contrast, standard measures of wealth (real estate, stocks and bonds, and so on) arrive at a 1990 figure of $18 trillion, just over three times GDP.

In Table 5.4 we update Shiller's $V^m$ calculations to 1992; also presented are estimates of the standard deviations of annual net returns $r^m$ on the perpetual GDP claims, defined by

$$r_t^m = \frac{Y_t^m}{V_{t-1}^m} + \frac{V_t^m - V_{t-1}^m}{V_{t-1}^m},$$

the sum of the GDP "dividend" and the capital gain, per dollar invested. (In the two-period model of this section, we could omit a term containing the future asset price in defining returns because assets had no resale value after paying their date 2 dividends.) The $V^m$ estimates are constructed using simple univariate time series processes for each country's GDP, estimated over 1950–92. The parameter estimates are then used to construct the forecasts of future GDP growth that enter into the present value formula for $V^m$; see Shiller (1993, ch. 4) for details.

Though the market value of Japanese financial wealth (including stocks, bonds, and land) frequently exceeds that of the United States, a perpetual claim to Japan's entire output, including labor's product, is estimated to be worth less than half as much. The standard deviation of the yearly return on a claim to U.S. output is only 2.03 percent; the perpetual Japanese claim's annual return volatility, at 8.41 percent, is more than four times higher. Note that because the claims are present



values of current and all expected future GDPs, routine rate-of-return fluctuations can imply very large international wealth redistributions.

These present-value calculations can be questioned on a number of grounds: they depend on the discount factor chosen and on the assumption that it is constant, and they can be sensitive to the time series processes used to represent expectations of future GDP. Moreover, the discount factor for future outputs is not explicitly linked to a risk premium, either of the consumption-based CAPM variety or a covariance with financial-market returns as in Sharpe (1964). But Shiller's basic point that existing asset markets are not nearly complete enough to substitute for perpetual claims on national GDPs probably is robust.

Shiller goes on to argue that markets in perpetual GDPs are feasible and that policymakers should be actively working to facilitate them. A number of practical obstacles would have to be surmounted, however, before this step could be taken. One is moral hazard. If the claims are indexed to government reports on GDP, and if a country is short in its own stock (as it would be in the models considered thus far), one can envision large incentives for governments to underreport output. Another problem, enforceability, will be central in Chapter 6. How would other countries enforce a trillion-dollar claim against Japan, much less against the United States? Individuals would still be subject to idiosyncratic nontradable risks. Finally, as section 5.5 will discuss in detail, there are limits to a country's ability to insure against fluctuations in the nontraded component of its GDP, even when problems of contract enforceability or moral hazard are absent. Despite these causes for caution, Shiller's plan raises interesting and important questions about the possibility of more efficient global risk sharing. ■

## 5.5 The Role of Nontradables

Chapter 4 pointed out that many goods, especially service-intensive goods, are never traded across national borders. Since nontraded goods constitute a large share of total output in most countries, it is important to ask how their presence modifies our analysis of international diversification and portfolio choice. We shall see that even when there are no restrictions on trade in assets, the existence of nontraded goods can lead to lower international consumption correlations and to a home preference in asset holdings.

### 5.5.1 Trade in Contingent Securities

The effects of nontradable goods can be illustrated by introducing them into a two-period, $S$ state complete markets model. Assume again $N$ countries, with a representative country $n$ resident maximizing a function of tradable and nontradable consumption levels,

$$U_1 = u(C^n_{\mathrm{T},1}, C^n_{\mathrm{N},1}) + \beta \sum_{s=1}^{S} \pi(s) u[C^n_{\mathrm{T},2}(s), C^n_{\mathrm{N},2}(s)]. \tag{62}$$

For simplicity we consider the case of a pure endowment economy, in which country $n$'s per capita endowments of tradable and nontradable goods are denoted by $Y^n_{\mathrm{T}}$ and $Y^n_{\mathrm{N}}$, respectively. We assume it is costless to ship tradables.

What types of securities can be traded if some goods are not tradable? People from any country can still promise to deliver specified amounts of *tradable* output in the various states of nature. But securities promising the delivery of *nontradable* output can be traded only among residents of the same country. Nonetheless, international trade is possible in claims that are *indexed* to random nontradable endowments but *payable* in traded goods.

Now $p(s)/(1+r)$ stands for the price, in terms of date 1 tradables, of a unit of tradables delivered on date 2 if and only if the state is $s$. Under free trade, countries face common prices $p(s)/(1+r)$. The relative price of nontradables in terms of tradables, however, is country specific (because international arbitrage is not possible for nontradables). We denote by $p^n_{\mathrm{N},1}$ the country $n$ price of nontradables in terms of tradables on date 1, and by $p^n_{\mathrm{N},2}(s)$ the same relative price on date 2 *in state s*. Thus, $p^n_{\mathrm{N},2}(s)p(s)/(1+r)$ is the price of date 2, state $s$, *non*tradables in terms of date 1 tradables. Given these prices, a typical resident of country $n$ faces the budget constraint

$$\begin{aligned} C^n_{\mathrm{T},1} + p^n_{\mathrm{N},1} C^n_{\mathrm{N},1} + \sum_{s=1}^{S} \frac{p(s)C^n_{\mathrm{T},2}(s) + p^n_{\mathrm{N},2}(s)p(s)C^n_{\mathrm{N},2}(s)}{1+r} \\ = Y^n_{\mathrm{T},1} + p^n_{\mathrm{N},1} Y^n_{\mathrm{N},1} + \sum_{s=1}^{S} \frac{p(s)Y^n_{\mathrm{T},2}(s) + p^n_{\mathrm{N},2}(s)p(s)Y^n_{\mathrm{N},2}(s)}{1+r}. \end{aligned} \tag{63}$$

Necessary first-order conditions for maximization of the lifetime utility function (62) subject to the constraint (63) are derived in the usual way. These conditions show that on date 1 and for each state $s$ on date 2, the marginal rate of substitution between traded and nontraded goods is equal to their relative price

$$\frac{\partial u(C^n_{\mathrm{T}}, C^n_{\mathrm{N}})/\partial C^n_{\mathrm{N}}}{\partial u(C^n_{\mathrm{T}}, C^n_{\mathrm{N}})/\partial C^n_{\mathrm{T}}} = p^n_{\mathrm{N}}, \tag{64}$$

where the relevant date subscripts and state indexes are understood. Similarly, Euler equations analogous to eq. (5) also follow for every $s = 1, \ldots, S$,

$$\begin{aligned} \frac{p(s)}{1+r} \cdot \frac{\partial u(C^n_{\mathrm{T},1}, C^n_{\mathrm{N},1})}{\partial C^n_{\mathrm{T}}} &= \pi(s)\beta \frac{\partial u[C^n_{\mathrm{T},2}(s), C^n_{\mathrm{N},2}(s)]}{\partial C^n_{\mathrm{T}}}, \\ \frac{1}{p^n_{\mathrm{N},1}} \cdot \frac{p^n_{\mathrm{N},2}(s)p(s)}{1+r} \cdot \frac{\partial u(C^n_{\mathrm{T},1}, C^n_{\mathrm{N},1})}{\partial C^n_{\mathrm{N}}} &= \pi(s)\beta \frac{\partial u[C^n_{\mathrm{T},2}(s), C^n_{\mathrm{N},2}(s)]}{\partial C^n_{\mathrm{N}}}. \end{aligned} \tag{65}$$



The second, less familiar, Euler condition states that at an optimum, the date 1 marginal utility cost of a unit of date 2, state $s$ nontradables must equal the expected date 2 marginal utility benefit.

Nontraded goods have to be consumed domestically, so equilibrium in nontradables within country $n$ requires that on date 1 and in every date 2 state,

$$C^n_{\mathrm{N}} = Y^n_{\mathrm{N}}. \tag{66}$$

Since agents throughout the world face a common set of Arrow-Debreu prices for state-contingent payments of tradables, Euler equation (65) and the market-clearing condition (66) imply that for any two countries $m$ and $n$,

$$\frac{\pi(s)\beta\partial u[C^m_{\mathrm{T},2}(s), Y^m_{\mathrm{N},2}(s)]/\partial C^m_{\mathrm{T}}}{\partial u(C^m_{\mathrm{T},1}, Y^m_{\mathrm{N},1})/\partial C^m_{\mathrm{T}}} = \frac{\pi(s)\beta\partial u[C^n_{\mathrm{T},2}(s), Y^n_{\mathrm{N},2}(s)]/\partial C^n_{\mathrm{T}}}{\partial u(C^n_{\mathrm{T},1}, Y^n_{\mathrm{N},1})/\partial C^n_{\mathrm{T}}}. \tag{67}$$

This condition generalizes eq. (34); it implies that ex post growth rates in the marginal utility of tradables are equal when countries are able to transact in a complete set of state-contingent claims on future tradable goods.

For national consumptions of nontradables there is no simple condition on marginal utilities analogous to eq. (67). Because forward contracts for nontradables can be traded only domestically, no common relative prices force equality across countries in marginal rates of intertemporal substitution for nontradables. Condition (67) implies that further international exchanges of tradables across time and across states of nature yield no efficiency gains, but further exchanges of nontradables could—if only nontradables could be traded! As a result, the allocation of world resources is only *constrained* Pareto efficient, not Pareto efficient *tout court*.

It is important to avoid a semantic trap here. Is the allocation that the market produces inefficient in any meaningful sense? To answer, one must determine if a utilitarian planner could improve on it. In this case the answer is *no*, unless the planner somehow has the technological capacity to move goods across national borders when the market will not. If, for example, goods are nontraded because they are prohibitively costly to transport, then a planner cannot usefully ship them across borders either, and has no advantage over the market. If, on the other hand, a good is nontraded in part because of an artificial government restriction, such as a licensing requirement that allows only French electricians to wire houses in France, then a planner able to shift electrician services around internationally can improve on the market. We say that the market equilibrium is *constrained* Pareto efficient if a benevolent economic planner facing the same technological and information constraints as the market cannot Pareto-improve on the market outcome (that is, cannot make one country better off without making at least one other country worse off). In practice, there are nontraded goods in both categories, those that are literally impossible to transport and those that would be traded except for government restrictions. For purposes of clarity, the reader should assume in the following analysis

that nontraded goods are internationally immobile for technological reasons. Because most of the analysis is positive rather than normative, however, which way we choose to interpret nontradability is not essential.

In the presence of nontraded goods, it is generally no longer possible to derive neat relationships such as eq. (37) between countries' total real per capita consumption growth rates as conventionally measured. However, in the special case where utility within a period is additive in tradeds and nontradeds and of the form

$$u(C_{\mathrm{T}}, C_{\mathrm{N}}) = \frac{C_{\mathrm{T}}^{1-\rho}}{1-\rho} + v(C_{\mathrm{N}}),$$

eq. (67) does imply that national growth rates of *tradables* consumption are perfectly correlated with each other. In the more general case of nonadditive period utility, nontradable consumption affects marginal rates of substitution for tradable consumption and prevents any simple general relationship analogous to eq. (37) from holding even for tradable consumption.

It is nonetheless straightforward to extend the consumption-based CAPM, even when individuals' marginal utility of tradables is not independent of their nontradable consumption. Claims to a country's nontradable endowment are priced in the same way as claims to its tradable endowment. However, nontradable output must be converted into tradables locally, and at the local relative price, before foreign shareholders can be paid. For a claim to country $m$'s tradable date 2 output,

$$V_{\mathrm{T},1}^m = \sum_{s=1}^{S} \frac{p(s)Y_{\mathrm{T},2}^m(s)}{1+r} = \sum_{s=1}^{S} \frac{\pi(s)\beta\partial u[C_{\mathrm{T},2}^n(s), Y_{\mathrm{N},2}^n(s)]/\partial C_{\mathrm{T}}^n}{\partial u(C_{\mathrm{T},1}^n, Y_{\mathrm{N},1}^n)/\partial C_{\mathrm{T}}^n} Y_{\mathrm{T},2}^m(s), \tag{68}$$

where $1/(1+r)$ is the date 1 tradables price of sure date 2 tradables. By the marginal equality (67) implied by complete markets for future tradables, $V_{\mathrm{T},1}^m$ is independent of the country $n$ whose residents' consumptions are used to evaluate that price. For a claim to country $m$'s nontradable date 2 output, again independently of $n$,

$$V_{\mathrm{N},1}^m = \sum_{s=1}^{S} \frac{p(s)p_{\mathrm{N},2}^m(s)Y_{\mathrm{N},2}^m(s)}{1+r}$$

$$= \sum_{s=1}^{S} \frac{\pi(s)\beta\partial u[C_{\mathrm{T},2}^n(s), Y_{\mathrm{N},2}^n(s)]/\partial C_{\mathrm{T}}^n}{\partial u(C_{\mathrm{T},1}^n, Y_{\mathrm{N},1}^n)/\partial C_{\mathrm{T}}^n} p_{\mathrm{N},2}^m(s)Y_{\mathrm{N},2}^m(s). \tag{69}$$

### 5.5.2 Preference Shocks

Throughout the chapter, we have assumed that $u(C)$ is stable across states of nature. What if there are unanticipated shocks to national health, weather patterns, birth rates, and so on, that affect a country's marginal rates of substitution across various market goods? A model with preference shocks can be interpreted as a special case of the nontraded goods model.



Let us assume that all goods are traded internationally and all countries $n$ have utility functions of the form

$$u(C_1^n) + \beta \sum_{s=1}^{S} \pi(s)u[C_2^n(s); \varepsilon^n(s)], \tag{70}$$

where $\varepsilon^n(s)$ is a country-specific, state-specific preference shock in period 2. Since preference shocks don't affect endowments, the budget constraint is the same as in the traded-goods-only case. The intertemporal state-specific Euler equations for representatives from countries $m$ and $n$ imply an efficiency condition completely analogous to eq. (67) for the model with nontradables,

$$\frac{\pi(s)\beta u'[C_2^m(s); \varepsilon^m(s)]}{u'(C_1^m)} = \frac{\pi(s)\beta u'[C_2^n(s); \varepsilon^n(s)]}{u'(C_1^n)}. \tag{71}$$

Just as in the case of nontraded goods, the presence of preference shocks rules out any exact relationship between consumption growth rates across countries. In particular, equations as simple as eq. (37) generally won't hold. By putting additional structure on the stochastic processes governing differentials in shocks across countries, one can sometimes derive predictions about consumption correlations. In any case, complete Arrow-Debreu contracts still yield a Pareto efficient allocation provided preference shocks are fully observable. (An omnipotent benevolent planner could not improve upon the market outcome.) When preference shocks aren't observable, the question of efficiency becomes more subtle; see Atkeson and Lucas (1992) and Lucas (1992).

Preference shocks that are large and erratic present a problem for empirical work, since it may be hard to control for such shocks using observable variables (see the next application). Nevertheless, many of our results can be generalized to allow for preference shocks (though we do not treat the issue in detail here). The remaining theoretical development in this section focuses on nontraded goods rather than preference shocks, but which factor is more important in practice ultimately is an empirical question.

---

***Application: Nontradability and International Consumption Correlations***

The earlier application on international consumption correlations (p. 290) implicitly assumed that all output is tradable. But if a significant share of output cannot be traded, a country's consumption growth could be more highly correlated with domestic output growth than with world consumption growth, despite complete markets in claims on future tradables, simply because it is technologically

infeasible for countries to pool nontraded consumption risks. If the period utility function is additive over tradables and nontradables, then efficiency condition (67) implies for internationally identical CRRA preferences that consumption growth rates for *tradables* are equalized in a constrained-efficient equilibrium. But without intraperiod additivity, even that simple prediction of the model doesn't follow. Nevertheless, Lewis (1996) shows that one can still test the complete-market model's predictions about international consumption correlations by controlling for nontradable consumption levels.

To illustrate, assume that the period utility function takes the CES-CRRA form

$$u(C_{\mathrm{T}}, C_{\mathrm{N}}) = \frac{\left\{\left[\gamma^{\frac{1}{\theta}} C_{\mathrm{T}}^{\frac{\theta-1}{\theta}} + (1-\gamma)^{\frac{1}{\theta}} C_{\mathrm{N}}^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}\right\}^{1-\rho}}{1-\rho}, \tag{72}$$

where $\rho > 0$ continues to denote the risk aversion coefficient and $\theta > 0$ is the intratemporal substitution elasticity between traded and nontraded goods. Efficiency condition (67) implies that on any date $t$, the ex post intertemporal marginal rate of substitution for tradables is a constant $\lambda_t$ that is the same for any country $n$,

$$\frac{\partial u(C^n_{\mathrm{T},t}, Y^n_{\mathrm{N},t})/\partial C^n_{\mathrm{T}}}{\partial u(C^n_{\mathrm{T},t-1}, Y^n_{\mathrm{N},t-1})/\partial C^n_{\mathrm{T}}} = \lambda_t.$$

Using the functional form (72), let's logarithmically differentiate with respect to the date $t$ variables, starting at their date $t-1$ values. With "hats" denoting log derivatives (infinitesimal percent changes), we arrive at the equation

$$\widehat{C}^n_{\mathrm{T}} = \frac{-\theta}{1-[\phi(1-\theta\rho)]}\widehat{\lambda} + \frac{(1-\phi)(1-\theta\rho)}{1-[\phi(1-\theta\rho)]}\widehat{Y}^n_{\mathrm{N}}, \tag{73}$$

where

$$\phi \equiv \frac{\gamma^{\frac{1}{\theta}}(C^n_{\mathrm{T}})^{\frac{\theta-1}{\theta}}}{\gamma^{\frac{1}{\theta}}(C^n_{\mathrm{T}})^{\frac{\theta-1}{\theta}} + (1-\gamma)^{\frac{1}{\theta}}(C^n_{\mathrm{N}})^{\frac{\theta-1}{\theta}}} < 1.$$

Equation (73) shows that in a constrained-efficient equilibrium, the evolution of country $n$'s traded-goods consumption depends on both the change in the "global" factor $\lambda_t$ (which is higher the scarcer world tradable resources are) and the change in the domestic endowment of nontradeds. Only when $\theta\rho = 1$ is the function (72) additive, so that growth rates of traded-goods consumption are perfectly correlated across countries.

Lewis (1996) tests eq. (73) econometrically by estimating an equation of the form



$$\Delta \log(C^n_{\mathrm{T},t} - Y^n_{\mathrm{D},t})$$
$$= \upsilon_t + \psi_1 \Delta \log Y^n_{\mathrm{N},t} + \psi_2 \Delta \log Y^n_{\mathrm{D},t} + \psi_3 \Delta \log(Y^n_{\mathrm{T},t} - Y^n_{\mathrm{D},t}) + \epsilon^n_t$$

for a panel of countries, where $Y^n_{\mathrm{D},t}$ is a proxy for the services of durables, $\upsilon_t$ is a global consumption shock common to all countries on date $t$, $Y^n_{\mathrm{N},t}$ is nontradable output as conventionally measured, $Y^n_{\mathrm{T},t}$ is tradable output, and the composite error $\epsilon^n_t$ incorporates measurement error as well as random preference shocks.[45] Under constrained-efficient risk sharing, the coefficient $\psi_3 = 0$: tradable consumption growth can depend on domestic nontraded output growth but not on domestic traded output growth, as shocks to the latter have been internationally pooled and influence $\Delta \log(C^n_{\mathrm{T},t} - Y^n_{\mathrm{D},t})$ only through their effect on the common time-varying intercept $\upsilon_t$.

Lewis finds in her full sample of countries that $\psi_3$ is significantly positive, so that idiosyncratic nondurable tradable output growth has a detectable positive correlation with nondurable tradable consumption growth even after possible correlations with nontradable supply shocks have been controlled for. However, a split of her sample into countries with severe official foreign exchange restrictions and others reveals that $\psi_3$ is significant only for the former group. Lewis concludes that after corrections for nontradability and durability are made, constrained-efficient risk sharing cannot be rejected for the group of countries with relatively low government barriers to asset trade.

Lewis's results are based on a number of maintained hypotheses and data compromises, and therefore warrant cautious interpretation. Further, her tests do not examine risks other than output risks. Nonetheless, the Lewis study suggests that the low international correlations between aggregate national consumptions may in fact be compatible with substantial use of feasible opportunities for pooling output risks by the more developed economies. The nature of the assets countries trade to achieve such risk pooling is an open question, since even trade in noncontingent bonds can facilitate partial sharing of risks.[46] ■

### 5.5.3 International Diversification with Nontraded Goods

As noted earlier, the significant shares of nontradables in most countries' consumption baskets may help explain the puzzlingly low international diversification of

45. Lewis argues that the services consumer durables provide are likely to be nontradable, despite the tradability of the durable goods themselves. Her paper provides a detailed discussion of the data, which are drawn from 1970–85 studies underlying the Penn Word Table.

46. Baxter and Crucini's (1995) simulations of a two-country model economy suggest that, for industrial-country levels of uncertainty, consumption in a complete-markets world behaves much like it does in a world with only bond trade when productivity shocks are transitory. When a shock is permanent, however, the country in which it occurs adjusts its consumption more or less fully in a bonds-only regime, in sharp contrast to what would happen if the shock's effects had been shared ex ante.

private equity portfolios (see Table 5.2). To illustrate, we continue to assume a global endowment economy with the preferences of a country $n$ resident described by eq. (62).

Assume now, however, that people can trade only shares of future national outputs (and bonds) as in section 5.3 rather than the Arrow-Debreu securities of the last subsection. We continue to denote by $V^m_{\mathrm{T},1}$ the price in terms of date 1 tradables of the claim to country $m$'s random output of tradables on date 2. People also can trade claims on any country $m$'s output of nontradables, but as we have seen, because nontradables cannot cross national borders, a *foreign* owner of such a claim must be paid in tradables. Again, $V^m_{\mathrm{N},1}$ denotes the price (in terms of date 1 tradables) of the claim to $p^m_{\mathrm{N},2}(s)Y^m_{\mathrm{N},2}(s)$ units of the tradable in state $s$ on date 2.

Section 5.3.2 showed that for an important class of period utility functions, international trade in output equity claims can bring about an efficient resource allocation when all output is tradable. When some output is nontradable, however, the conditions leading to efficiency are more stringent. To illustrate, we start by following Stockman and Dellas (1989) in considering a period utility function that is *additive* in $C_{\mathrm{T}}$ and $C_{\mathrm{N}}$. More specifically, suppose any country's expected utility function takes the form

$$U_1 = \frac{C_{\mathrm{T},1}^{1-\rho}}{1-\rho} + v(C_{\mathrm{N},1}) + \beta \sum_{s=1}^{S} \pi(s) \left\{ \frac{C_{\mathrm{T},2}(s)^{1-\rho}}{1-\rho} + v[C_{\mathrm{N},2}(s)] \right\}.$$

This case is relatively simple because the additivity of period utility allows the results of section 5.3 to carry through with minimal modification.

Following the same steps as in section 5.3.2, one easily shows that the equilibrium for tradables here parallels the one described there. All agents hold shares in a single global mutual fund of tradable output processes: for any country $n$, share holdings $x^n_{\mathrm{T},m}$ in country $m$ tradables and all consumption shares $C^n_{\mathrm{T}}/Y^{\mathrm{W}}_{\mathrm{T}}$ equal $n$'s initial equilibrium share in world tradable wealth. Each country's net bond holdings are zero. The reader can confirm that the traded-goods equilibrium is characterized by equations identical to eqs. (45)–(50), except that consumption, output, portfolio-share, and share-price variables all have T subscripts. [Equation (49), so modified, gives the world own interest rate on tradables.] Equation (67) holds and implies ($\forall m, n$)

$$\frac{C^m_{\mathrm{T},2}(s)}{C^m_{\mathrm{T},1}} = \frac{C^n_{\mathrm{T},2}(s)}{C^n_{\mathrm{T},1}} = \frac{Y^{\mathrm{W}}_{\mathrm{T},2}(s)}{Y^{\mathrm{W}}_{\mathrm{T},1}}, \tag{74}$$

so that the equilibrium is constrained-efficient.[47]

47. As usual, this efficiency result generalizes to utility-of-tradables functions in the HARA class. (Recall footnote 28.)



What about investors' holdings of shares in foreign *nontradable* output streams? The answer is surprisingly simple. It is an equilibrium for all claims to a country $n$'s risky *nontraded* output to be held *only* by residents of country $n$: for all $n$,

$$x^n_{\mathrm{N},m} = \begin{cases} 1 & (m = n) \\ 0 & (m \neq n). \end{cases}$$

Thus overall stock portfolios can exhibit a home bias, containing all of the claims to domestic nontradables industries despite being diversified among all the world's tradable industries. The larger the total share of nontradable output in world output, the larger the home bias.

How can we be sure that investors will not avail themselves of the opportunity to hold claims indexed to foreign countries' nontraded endowments? There is no benefit from doing so: with an additive period utility function, diversification among the world's tradable endowments exhausts all available gains from trade. To see this point formally, note the implication of eqs. (69) and (74) that the price in terms of tradables of a claim indexed to country $m$'s nontraded goods is valued at

$$V^m_{\mathrm{N},1} = \sum_{s=1}^{S} \pi(s)\beta \left[ \frac{Y^{\mathrm{W}}_{\mathrm{T},2}(s)}{Y^{\mathrm{W}}_{\mathrm{T},1}} \right]^{-\rho} p^m_{\mathrm{N},2}(s)Y^m_{\mathrm{N},2}(s),$$

in every country $n$, independently of its residents' portfolio of nontradables. This convergence of valuations tells us that once people have fully diversified their portfolios of claims to tradables, they cannot gain further by diversifying their holdings of nontradables.[48]

Unfortunately, this explanation at best goes only partway toward explaining the facts. One reason is empirical. The portfolio bias observed through the mid-1990s was too extreme to be due entirely to nontradables. For a small country, even if half of all output is nontradable, the model still implies that roughly half of agents' portfolios should be held abroad. Nothing near this proportion was observed.

Furthermore, a *nonadditive* intraperiod utility function can imply that individuals should hold shares in foreign industries that produce nontradables. This case is difficult to analyze because an equilibrium with trade in bonds and equity only (rather than Arrow-Debreu contracts) need not be constrained-efficient. Nonetheless, provided an equilibrium exists, we can develop some intuition about it. Suppose, for example, that the period utility function is of the nonadditive CES-CRRA form in eq. (72). Since the nontradables equilibrium condition (66) must hold in each country, date 2 shocks to nontradables output, like preference shocks, cause the marginal utility of tradables consumption to move idiosyncratically across

48. Recall the meaning of the Euler condition that this pricing equation satisfies in equilibrium. It implies that when $x^n_{\mathrm{N},n} = 1$, $x^n_{\mathrm{N},m} = 0$ ($m \neq n$), a country $n$ resident (for any $n$) has no positive incentive to alter his portfolio of risky claims to nontradable outputs.

countries ex post even when all countries' portfolios are fully diversified among tradables industries. Differentiation of the period utility function shows that the cross-partial derivative

$$\frac{\partial^2 u(C_T, C_N)}{\partial C_N \partial C_T}$$

has the same sign as $1 - \theta\rho$. Thus when $\theta > 1/\rho$, a higher date 2 endowment of nontradables *lowers* the marginal utility of date 2 tradables consumption. If a rise in nontradables output raises industry revenue measured in tradables, $p_{N,2}\, Y_{N,2}$ (this requires $\theta > 1$), then when $\theta > 1/\rho$ all countries may be able simultaneously to achieve a better allocation of risks by swapping ownership claims to their nontradables industries. Afterward, any country with unexpectedly high nontradables output will ship additional dividends, payable in tradables, abroad, reducing the incipient discrepancy between the domestic and foreign ex post marginal rates of intertemporal substitution of tradables. Thus trade in nontraded industry shares can enable all countries to hedge against fluctuations in the marginal utility of tradables consumption caused by output shocks in nontradables. This conclusion also follows when $\theta < 1/\rho$, but in that case investors may hold more than 100 percent of domestic nontraded industries and assume a negative position (that is, "go short") in foreign nontraded industries. (Someone with a short position in an asset pays rather than receives its ex post return. In the aggregate the market must hold the entire existing supply of any asset, but some individuals can take negative positions if others demand more than the total supply.)[49]

There is another problem with the view that nontradables explain home portfolio bias. Even though individuals are content to concentrate their nontradables portfolios domestically in the case of additive intraperiod utility, they might have no positive reason to do so. One special but interesting case occurs when the period utility function for the representative agent (of any country) is given by

$$u(C_T, C_N) = \gamma \log C_T + (1 - \gamma) \log C_N,$$

making the intratemporal elasticity of substitution between tradables and nontradables 1. In this case, payoffs on all countries' nontradable equities are perfectly

49. Eldor, Pines, and Schwartz (1988), Tesar (1993), Feeney and Jones (1994), Pesenti and van Wincoop (1994), and Baxter, Jermann, and King (1995) all stress the interplay between $\rho$ and $\theta$ in determining international diversification patterns under nonadditive period utility. Tesar and Pesenti-van Wincoop explore the possibility of a home bias in portfolios of traded-goods industries when shares in nontraded-goods industries cannot be traded. Baxter, Jermann, and King study a static model of risk allocation with infinitesimal uncertainty and trade in claims to nontradable outputs. They show that the complete-markets allocation may be attainable through trade in shares, and argue that it remains optimal for investors to hold the global mutual fund of traded-goods industries. Their conclusions do not hold generally for noninfinitesimal uncertainty, however. Stulz (1981) develops a very general dynamic analysis of portfolio choice and asset pricing when residents of different countries face different consumption opportunity sets. He does not, however, model the source of international discrepancies in consumption opportunities.



correlated with the mutual fund of world tradables and, as a result, are perfectly correlated with each other. (The reason stochastic variation in nontradable outputs plays no role is that a rise in nontradable output, for example, causes a proportional fall in price, leaving the nontradable endowment's value in terms of tradables unchanged.) But perfect correlation in payoffs makes any nontradable equity a *perfect substitute* for the world tradable portfolio, a fact that renders countries' portfolios *indeterminate*. As a result, we can no longer make predictions about home bias purely on the basis of some part of output being nontradable.

Cole (1988), Golub (1994), Brainard and Tobin (1992), Baxter and Jermann (1993), and Ghosh and Pesenti (1994) discuss models in which all goods are traded, but in which, realistically, there are forward markets only in income earned by capital, not in income earned by labor. If nontradable labor income is positively correlated with the returns on domestic equities, then individuals should go short in the domestic stock market to hedge labor-income risk. In this case, recognizing that labor income is nontraded would seem to make the home bias puzzle even deeper. Bottazzi, Pesenti, and van Wincoop (1996) argue that empirically, domestic labor and capital income actually tend to be negatively correlated in most OECD countries; that is, the relative share of labor and capital income in output is quite volatile. They suggest that domestic rather than foreign equity may provide the better hedge against redistributive shocks, although the quantitative significance of this effect is unlcear. In most countries, of course, fixed transaction costs and borrowing constraints result in equity holdings being concentrated disproportionately among a relatively small number of better-off individuals, many of whom derive a substantial fraction of their total income from financial wealth (see, for example, Mankiw and Zeldes, 1991). For such individuals, the correlation between aggregate labor and capital income might be a comparatively minor factor in portfolio decisions.

---

***Application: How Large Are the Gains from International Risk Sharing?***

International financial markets aid in allocating risks more efficiently among countries, but are the welfare benefits they confer in theory substantial in practice? One school of thought holds that the benefits are actually quite minimal—a view that could throw light on the home bias diversification puzzle, since it implies that small transaction costs could go far to discourage trade in risky assets.

An influential estimate by Lucas (1987) of the welfare cost of variability in United States consumption conveys intuitively the grounds for believing that the gains from risk sharing are small. Lucas considers a representative individual whose lifetime utility is

$$U_t = \mathrm{E}_t \left\{ \sum_{s=t}^{\infty} \beta^{s-t} \frac{C_s^{1-\rho}}{1-\rho} \right\},$$

where $C_s = (1+g)^{s-t}\bar{C}\exp[\epsilon_s - \frac{1}{2}\mathrm{Var}(\epsilon_s)]$ and $\epsilon_s$ is a normal, i.i.d., mean-zero shock. By direct calculation,

$$\mathrm{E}_t\{C_s^{1-\rho}\} = (1+g)^{(1-\rho)(s-t)}\bar{C}^{1-\rho}\exp\left[-\tfrac{1}{2}(1-\rho)\rho\mathrm{Var}(\epsilon)\right]$$

(where we have used the fact that $\exp[1-\rho]\epsilon_s$ is lognormally distributed with mean $\exp\left\{\frac{1}{2}[1-\rho]^2\mathrm{Var}[\epsilon]\right\}$). Thus, prior to observing $\epsilon_t$,

$$U_t = \frac{\bar{C}^{1-\rho}}{1-\rho}\left[\frac{1}{1-\beta(1+g)^{1-\rho}}\right]\exp\left[-\tfrac{1}{2}(1-\rho)\rho\mathrm{Var}(\epsilon)\right]$$

[assuming $\beta(1+g)^{1-\rho} < 1$].

Lucas asks us to imagine that all consumption uncertainty could be eliminated, resulting in the expected-value consumption path $\bar{C}_s \equiv \mathrm{E}_t C_s = (1+g)^{s-t}\bar{C}$. The associated lifetime utility is

$$\bar{U}_t = \frac{\bar{C}^{1-\rho}}{1-\rho}\left[\frac{1}{1-\beta(1+g)^{1-\rho}}\right].$$

What percent increase $\tau$ in annual consumption has the same positive utility effect as the total elimination of consumption uncertainty? This equivalent variation is given by

$$\frac{[(1+\tau)\bar{C}]^{1-\rho}}{1-\rho}\exp\left[-\tfrac{1}{2}(1-\rho)\rho\ \mathrm{Var}(\epsilon)\right] = \frac{\bar{C}^{1-\rho}}{1-\rho},$$

which can be solved to yield

$$\tau = \left\{\exp\left[\tfrac{1}{2}(1-\rho)\rho\ \mathrm{Var}(\epsilon)\right]\right\}^{1/(1-\rho)} - 1.$$

A first-order Taylor approximation in the neighborhood of $\mathrm{Var}(\epsilon) = 0$ yields

$$\tau \approx \tfrac{1}{2}\rho\ \mathrm{Var}(\epsilon). \tag{75}$$

Thus we end up with a remarkably simple approximation to the cost of consumption variability, expressed as a perpetual percent tax on consumption.

For annual 1950-90 U.S. data on total per capita consumption, $\mathrm{Var}(\epsilon) = 0.000708$, which corresponds to a standard deviation slightly below 2.7 percent per year. Thus, even if $\rho = 10$, the total elimination of consumption variability would be worth only about a third of a percent of consumption per year to a representative U.S. consumer, a surprisingly small amount. (The reader may note the strong parallels with our earlier discussion of the equity risk premium. Here, too, the low measured variability of U.S. aggregate consumption makes it difficult to assign a large role to consumption risk.)



One might conclude that if the gain from *eliminating* unpredictable consumption variability altogether is so small, then the gain from perfect international pooling of risks, which still leaves people facing systematic global consumption risk, must be even smaller. However, a closer look at Lucas's reasoning suggests that caution is warranted:

1. Lucas's assumption that consumption fluctuates randomly around a fixed time trend has been questioned by many economists. That setup makes all consumption fluctuations temporary; but if permanent shocks to consumption sometimes occur, the welfare cost of incomplete insurance is potentially much more severe.

2. The United States is atypical in the relative stability of its aggregate per capita consumption and output. Some industrial countries and many developing countries have substantially greater variability. The numbers in Table 5.4 are suggestive of the disparity across different countries. Since the measure in eq. (75) depends on the square of variability, it rises sharply as variability rises.

3. Lucas's calculations do not allow for individual heterogeneity—that is, they abstract altogether from uninsurable idiosyncratic risks and look only at the cost of systematic risk. Thus Lucas's numbers are relevant at best to the *marginal* benefit that global risk pooling may yield once domestic risk pooling possibilities have been exhausted.

Cole and Obstfeld (1991) assessed the gains from international risk sharing in a version of the Lucas (1982) model with representative national agents and permanent as well as transitory output shocks. Assuming output innovations comparable in variance to those of the United States, they found small gains from perfect pooling of output risks, on the order of a fifth of a percent of output per year even for $\rho = 10$. Backus, Kehoe, and Kydland (1992) find that essentially for this reason, even small transactions costs can seriously limit international asset trade. Introducing fairly minor trade restrictions into their calibrated dynamic general equilibrium model leads to an equilibrium close to autarky.

Mendoza (1995) and Tesar (1995) look at calibrated world-economy models with nontraded goods, endogenous labor supply, and investment, allowing for different degrees of persistence in (statistically stationary) productivity shocks but no nondiversifiable risks. Their general conclusion is that the gains from risk pooling are of the order of those found by Cole and Obstfeld (or smaller), in part because domestic investment possibilities offer greater scope for self-insurance through intertemporal domestic reallocations. Obstfeld (1995) argues that for many developing countries, the gains from international risk sharing are nonetheless likely to be large (a point we reexamine in the next chapter).

When output shocks are persistent, with effects that propagate over time, utility functions that make risk aversion high only when intertemporal substitutability is low may understate the cost of uncertainty (see Obstfeld, 1994b). Van Wincoop (1994) evaluates gains from risk pooling among the OECD economies in a model with a random walk assumed for log per capita consumption, nonexpected-utility preferences, and an idiosyncratic component of consumption risk that cannot be diversified away domestically or internationally. He finds much larger average benefits than did Cole and Obstfeld, averaging as high as 5.6 percent of GDP per year for the OECD when risk aversion $\rho = 3$ and the intertemporal substitution elasticity $\sigma = 1$. The wide divergence in estimates shows how sensitive answers can be to seemingly minor differences in assumptions.

In reality, uninsurable risks seem to be potentially important determinants of what countries gain from pooling insurable risks globally. Similarly, research on the equity premium puzzle suggests that uninsurable risks greatly increase the price people will pay to lay off insurable risks. By implication, uninsurable risks raise the welfare benefit of feasible risk reduction.

More detailed and explicit modeling of consumer heterogeneity would shed light on both the effects of international financial integration and on the domestic distribution of risk-sharing gains. The noninsurability of much labor-income risk plainly is related to another puzzle: the contrast between the often slim apparent gains from pooling aggregate national consumption risks, and the finding in the finance literature that global diversification of equity portfolios can yield large benefits to consumers whose incomes come entirely from financial wealth (for example, Grauer and Hakansson, 1987).

Even if the gains from risk pooling are small in conventional dynamic models with representative national individuals, they may be much larger in models where investment rises in response to expanded diversification opportunities and where higher investment endogenously generates a higher rate of long-run economic growth. We will return to this idea in Chapter 7. ■

## *5.6 A Model of Intragenerational Risk Sharing

Models of incomplete risk sharing have aggregate implications that can differ from those of models with complete markets. In this section we explore a model in which an overlapping generations structure prevents all people alive on a given date from pooling endowment risks for that date. The reason complete risk sharing fails is not the adverse incentives full insurance might create. (Such models are the topic of the next chapter.) Here markets fail to pool all consumption risks efficiently simply because those who have not yet been born cannot sign contracts. Interpreted broadly, the model shows how aggregate consumption can behave when some, but not all, individuals have access to insurance markets.



The world consists of two countries, Home and Foreign, whose residents consume a single consumption good on each date. The world economy has an infinite horizon but people do not: every period a new generation with a two-period lifetime is born in each country. A Home resident born on date $t$ maximizes lifetime expected utility,

$$U_t = \log c_t^{\mathrm{Y}} + \beta \mathrm{E}_t \log c_{t+1}^{\mathrm{O}}.$$

In Foreign, expected utility is the same function of $c_t^{\mathrm{Y}*}$ and $c_{t+1}^{\mathrm{O}*}$. The size of a generation (in either country) is constant and normalized to 1.

The only uncertainty concerns the values of individuals' exogenous endowments. Everyone in Home, whether young or old, receives the same endowment $y_t(s)$ on date $t$, where $s$ is one of $\mathcal{S}$ possible states of nature. Similarly, everyone in Foreign receives $y_t^*(s)$. The probability distribution of the states $s = 1, \ldots, \mathcal{S}$ occurring on a date $t$ is described by the probability density function denoted $\pi_{t-1}(s)$.

In this model, someone born on date $t$ cannot insure date $t$ consumption because the initial-period endowment is revealed before there is a chance to transact in asset markets. (If populations formed immortal dynasties, parents could transact on behalf of their offspring, but that is not the case here.) After the date $t$ state is revealed, however, those born on date $t$ can trade on a complete set of markets for contingent claims to date $t + 1$ consumption. Only *intra*generational asset trade is possible: the young do not sign contracts with the old because the old will not be around tomorrow to fulfill commitments made today. (Nor is there any long-lived asset in the model, such as capital or government debt, for the old to sell to the young, though even this would not generally result in an efficient allocation.) Clearly the young of a given date $t$ could attain higher unconditional expected utility (expected utility conditional on no information about endowments) if they could trade claims contingent on the date $t$ state *before* it became known.

To see the model's implications for international consumption covariation and the distribution of wealth, consider how a Home resident, young on date $t$, chooses current and future consumption. This person maximizes

$$U_t = \log c_t^{\mathrm{Y}} + \beta \sum_{s=1}^{\mathcal{S}} \pi_t(s) \log c_{t+1}^{\mathrm{O}}(s)$$

subject to the constraint

$$c_t^{\mathrm{Y}} + \frac{1}{1 + r_{t+1}} \sum_{s=1}^{\mathcal{S}} p_t(s) c_{t+1}^{\mathrm{O}}(s) = y_t + \frac{1}{1 + r_{t+1}} \sum_{s=1}^{\mathcal{S}} p_t(s) y_{t+1}(s),$$

where $p_t(s)/(1 + r_{t+1})$ is the date $t$ price of the Arrow-Debreu security for state $s$ occurring on date $t + 1$. This is the same type of problem analyzed in section 5.2.1. Now retrace the steps taken there, remembering that all trade occurs within age

cohorts. You can confirm that *equilibrium* consumption levels for the young on date $t$ are

$$c_t^Y = \mu_t(y_t + y_t^*), \qquad c_t^{Y*} = (1 - \mu_t)(y_t + y_t^*),$$

and, for this same cohort when old on date $t + 1$, they are

$$c_{t+1}^O(s) = \mu_t[y_{t+1}(s) + y_{t+1}^*(s)], \qquad c_{t+1}^{O*}(s) = (1 - \mu_t)[y_{t+1}(s) + y_{t+1}^*(s)],$$

where (set $\rho = 1$ in footnote 16, section 5.2.1.2)

$$\mu_t = \frac{1}{1+\beta}\left[\frac{y_t}{y_t + y_t^*} + \beta \sum_{s=1}^{S} \pi_t(s) \frac{y_{t+1}(s)}{y_{t+1}(s) + y_{t+1}^*(s)}\right]$$

$$= \frac{1}{1+\beta}\left[\frac{y_t}{y_t + y_t^*} + \beta \mathrm{E}_t\left\{\frac{y_{t+1}}{y_{t+1} + y_{t+1}^*}\right\}\right].$$

For example, if Home's output is exactly half of world output today and expected to be half tomorrow, the current Home young will consume exactly half of their generation's world output [that is, $\frac{1}{2}(y + y^*)$] in both periods of life, regardless of the state of nature when they are old. As in section 5.2.1, the equilibrium prices associated with the preceding consumption allocation are

$$1 + r_{t+1} = \frac{\dfrac{1}{y_t + y_t^*}}{\beta \mathrm{E}_t\left\{\dfrac{1}{y_{t+1} + y_{t+1}^*}\right\}}$$

and, for all states $s$,

$$p_t(s) = \frac{\dfrac{\pi_t(s)}{y_{t+1}(s) + y_{t+1}^*(s)}}{\mathrm{E}_t\left\{\dfrac{1}{y_{t+1} + y_{t+1}^*}\right\}}.$$

Turn next to the model's implications for *aggregate* national consumption levels. Aggregate Home per capita consumption, $c_t$, is

$$c_t = \tfrac{1}{2}(c_t^Y + c_t^O) = \tfrac{1}{2}(\mu_t + \mu_{t-1})(y_t + y_t^*)$$

$$= \frac{y_t + y_t^*}{2(1+\beta)}\left[\frac{y_t}{y_t + y_t^*} + \beta \mathrm{E}_t\left\{\frac{y_{t+1}}{y_{t+1} + y_{t+1}^*}\right\}\right.$$

$$\left. + \frac{y_{t-1}}{y_{t-1} + y_{t-1}^*} + \beta \mathrm{E}_{t-1}\left\{\frac{y_t}{y_t + y_t^*}\right\}\right]. \qquad (76)$$



**Box 5.3**
A Test of Complete Markets Based on Consumption Divergence within Age Cohorts

The model of this section suggests that if the only major departure from complete markets were the inability of future generations to write contracts today, the consumption levels of individuals within the *same* cohort would still be highly correlated. What does the evidence suggest?

Deaton and Paxson (1994) analyze cohort survey data on consumer income and expenditure for three countries—the United States, Britain, and Taiwan. They use the data to track income and consumption dispersion among individuals born in a specific year (an age cohort) within a given country. (They do not compare consumptions across countries.) While Deaton and Paxson's data do not permit them actually to track the exact same group of individuals for the entire sample period, they are able to make comparisons across time by using random sampling techniques. The results, Deaton and Paxson argue, provide a compelling rejection of the complete-markets model. In all three countries, consumption inequality within a cohort tends to rise sharply over time.

On the other hand, the divergence of cohort consumptions over time is consistent with the bonds-only model of Chapter 2, which allows individual shares in the economy's total consumption to drift arbitrarily far apart. Evidence such as Deaton and Paxson's is one compelling reason not to assume automatically that the complete-markets model of this chapter is necessarily a closer approximation to reality than the models of Chapters 2 and 3. However, it would be much more satisfying from a theoretical standpoint to justify market incompleteness rigorously rather than simply assuming it. Deepening our understanding of the reasons for incompleteness in international financial markets will be the main goal of the next chapter.

A parallel formula holds for Foreign per capita consumption. With complete markets and a representative infinitely-lived consumer with log utility in each country, the ratio of national to world per capita consumption would be constant through time. In the present model, however, consumption aggregates do not behave as if chosen by infinitely-lived representative consumers, despite the access of all agents to state-contingent claims on second-period-of-life output.

Equation (76) shows that here, instead, the distribution of world consumption changes over time, even with time-invariant probability distributions for outputs, because of the dependence of domestic consumption on current and lagged idiosyncratic components of domestic output. This conclusion is reminiscent of models in which countries trade only noncontingent claims to future outputs.

## Appendix 5A Spanning and Completeness

In the $N$-country diversification model of section 5.3, we saw that trade in country-fund equities may lead to the same resource allocation that trade in a complete set of Arrow-Debreu claims would produce. That remarkable result holds only under a restricted class of utility

functions. This appendix makes precise and derives a result that is weaker but does not depend on utility functions: given a sufficient number of assets with independently varying returns, investors can synthesize a complete set of derivative Arrow-Debreu securities.

Recall (see p. 307) that in a two-period model the state $s$ net real rate of return on shares in country $m$, $r^m(s)$, is output in state $s$ divided by the date 1 price of a claim to the output, less 1:

$$r^m(s) = \frac{Y_2^m(s) - V_1^m}{V_1^m}.$$

Next define the $\mathcal{S} \times (N+1)$ matrix $\mathbf{R}$ of gross ex post returns:

$$\mathbf{R} \equiv \begin{bmatrix} 1+r & 1+r^1(1) & \dots & 1+r^N(1) \\ 1+r & 1+r^1(2) & \dots & 1+r^N(2) \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 1+r & 1+r^1(\mathcal{S}) & \dots & 1+r^N(\mathcal{S}) \end{bmatrix}.$$

Each column of $\mathbf{R}$ is a state-by-state list of returns on a different asset, with the first column corresponding to the riskless bond.

If $\mathcal{S} \leq N+1$, the rank of the return matrix $\mathbf{R}$ is at most $\mathcal{S}$. The *spanning condition* states that

$$\text{Rank}(\mathbf{R}) = \mathcal{S}. \tag{77}$$

Condition (77) means that $\mathbf{R}$ contains a set of $\mathcal{S}$ linearly independent columns; that is, there is a set of $\mathcal{S}$ assets such that no member's state-by-state return vector can be replicated by a linear combination (or *portfolio*) of the $\mathcal{S}-1$ other assets. Notice that eq. (77) couldn't possibly hold were $N+1 < \mathcal{S}$, because then the rank of $\mathbf{R}$ could be at most $N+1$.

The spanning condition allows international diversification to lead to the same Pareto-optimal resource allocation that would occur with complete Arrow-Debreu securities markets, regardless of what the utility functions are. The existence of $\mathcal{S}$ assets with linearly independent return vectors provides market risk-sharing opportunities as rich as those provided by $\mathcal{S}$ Arrow-Debreu securities with linearly independent return vectors.

In showing why the Arrow-Debreu equilibrium results if eq. (77) holds, we restrict attention to the special case of $N+1 = \mathcal{S}$ assets, in which $\mathbf{R}$ is a square matrix. (More assets can never reduce insurance possibilities.) To see why condition (77) allows us to form a portfolio paying 1 unit of output in exactly one state $s$ and 0 in all the others, let $a_{0s}$ be the value of the safe asset in the hypothesized portfolio and $a_{ns}$ (for $n = 1, \dots, N$) the value of the country $n$ output claim. Thus, the portfolio contains a share $a_{ns}/V_1^n$ of claim $n$. If

$$\mathbf{a}_s = [\, a_{0s} \quad a_{1s} \quad \dots \quad a_{Ns} \,]^{\mathcal{T}}$$

(where $\mathcal{T}$ denotes matrix transposition) and $\mathbf{1}_s$ is the $\mathcal{S} \times 1$ vector with 1 as its $s$th row entry and 0 elsewhere, then by choosing $\mathbf{a}_s = \mathbf{R}^{-1}\mathbf{1}_s$, we form a portfolio with the state-by-state return vector $\mathbf{R}\mathbf{a}_s = \mathbf{1}_s$. [The spanning condition (77) tells us that the inverse $\mathbf{R}^{-1}$ exists.] But $\mathbf{1}_s$ is simply the payoff vector for a state $s$ Arrow-Debreu security. Thus the $\mathcal{S}$ portfolios described by the vectors $\mathbf{a}_1$ through $\mathbf{a}_{\mathcal{S}}$ are indistinguishable from the primal Arrow-Debreu securities.



The circumstances under which the spanning condition is met might seem implausible, but the issue is more subtle in richer dynamic settings. In models with continuous trading, dynamic hedging strategies (strategies involving continual rebalancing of portfolios) can, under certain conditions, lead to a complete markets outcome even when the number of securities available is small relative to the number of states of nature. Interested readers should consult Duffie and Huang (1985).

## Appendix 5B Comparative Advantage, the Current Account, and Gross Asset Purchases: A Simple Example

In section 5.1.6, we derived the current account under log period utility in a two-period, two-state, small-country model with complete markets. The current account is given by eq. (17),

$$CA_1 = Y_1 - C_1 = \frac{\beta}{1+\beta}Y_1 - \frac{1}{1+\beta}\left[\frac{p(1)Y_2(1) + p(2)Y_2(2)}{1+r}\right].$$

In this appendix, we explore further the role of comparative advantage in determining the current account. The extended example illustrates why the strong results on comparative advantage in Chapter 1 extend only weakly to the stochastic case. In addition to studying net capital flows, we also derive closed-form solutions for gross capital flows in the log case.

### 5B.1 The Autarky Interest Rate and the Current Account

To interpret the current account balance in terms of the relationship between relative autarky prices and world prices, let us begin by temporarily assuming that $\beta(1+r) = 1$. Equation (17) then shows that if the value of output is the same on both dates—that is, if $Y_1 = p(1)Y_2(1) + p(2)Y_2(2)$—then the current account balance $CA_1$ will be zero, just as in Chapter 1's model. With $Y_1 < p(1)Y_2(1) + p(2)Y_2(2)$, $CA_1$ is negative, and it is positive in the opposite case.

The intuition behind this result is easy to grasp. Observe that in the log case, the Euler equation for the state $s$ security, eq. (5), reads

$$\frac{p(s)}{(1+r)C_1} = \frac{\pi(s)\beta}{C_2(s)}.$$

Multiply both sides by $(1+r)C_1C_2(s)$ and sum over states $s$. The result is

$$p(1)C_2(1) + p(2)C_2(2) = \beta(1+r)C_1 = C_1,$$

where the last equality follows from our assumption that $\beta(1+r) = 1$. People seek a level (across time) expenditure path, and if their first-period income is below (above) the constant expenditure level consistent with the intertemporal budget constraint, they will shift purchasing power to the present (future) through a current account deficit (surplus). In short, the current account is determined just as in the certainty case, but with the date 2 endowment's *value* at world Arrow-Debreu prices in place of the nonstochastic date 2 endowment. (Section 5.1.8 of the text showed the very limited extent to which this result generalizes beyond the log case.)

If the country were somehow forbidden from altering the timing of its overall spending through the current account, but could still trade date 2 risks at world prices $p(s)$, it would

have no choice but to consume $C_1 = Y_1$, and it would then set $C_2(s) = \pi(s)[p(1)Y_2(1) + p(2)Y_2(2)]/p(s)$. The (gross) domestic real interest rate in the resulting circumstance of "current account autarky" would be

$$1 + r^{CA} = \frac{u'(Y_1)}{\beta\left(\sum_{s=1}^{2}\pi(s)u'\{\pi(s)[p(1)Y_2(1) + p(2)Y_2(2)]/p(s)\}\right)}$$

$$= \frac{1}{\beta Y_1}[p(1)Y_2(1) + p(2)Y_2(2)], \tag{78}$$

which is simply the interest rate consistent with the intertemporal Euler condition (8). The autarky interest rate when the country has access to *no* financial markets at all is, however,

$$1 + r^{A} = \frac{u'(Y_1)}{\beta\sum_{s=1}^{2}\pi(s)u'[Y_2(s)]} = \frac{1}{\beta Y_1}\left[\frac{\pi(1)}{Y_2(1)} + \frac{\pi(2)}{Y_2(2)}\right]^{-1} \tag{79}$$

[because consumption levels necessarily are $C_1 = Y_1$ and $C_2(s) = Y_2(s)$ in complete autarky]. Plainly, $r^{CA}$ and $r^{A}$ coincide only in very special circumstances, for example, when $Y_2(1) = Y_2(2)$, so that the country has no output uncertainty.[50] But since $\beta = 1/(1 + r)$ in the present example, eq. (78) and the discussion in the two paragraphs preceding it show that $CA_1$ has the same sign as $r - r^{CA}$, not $r - r^{A}$.[51] This is a good example of how the textbook $2 \times 2$ comparative advantage theorem is inapplicable with more than two goods.

Now relax the assumption that $\beta(1 + r) = 1$. An alternative way to characterize the current account in the logarithmic case is in terms of the autarky date 1 prices of Arrow-Debreu securities. Using condition (5) with log utility, we read off the equilibrium autarky price ratios

$$\frac{p(s)^{A}}{1 + r^{A}} = \frac{\pi(s)\beta Y_1}{Y_2(s)}, \qquad s = 1, 2, \tag{80}$$

If one rewrites the current account equation (17) as

$$CA_1 = \frac{1}{1+\beta}\left[\pi(1)\beta Y_1 - \frac{p(1)Y_2(1)}{1+r} + \pi(2)\beta Y_1 - \frac{p(2)Y_2(2)}{1+r}\right],$$

then the formula for autarky prices, eq. (80), shows that

50. When $Y_2(1) = Y_2(2) = Y_2$, $r^{CA} = r^{A}$ because

$$p(1)Y_2 + p(2)Y_2 = Y_2 = \frac{Y_2}{\pi(1) + \pi(2)} = \left[\frac{\pi(1)}{Y_2} + \frac{\pi(2)}{Y_2}\right]^{-1}.$$

The date 1 current account is simply

$$CA_1 = \frac{Y_2}{1+\beta}\left(\frac{1}{1+r^{A}} - \frac{1}{1+r}\right).$$

Even in this case, the country opts for different consumption levels in states 1 and 2 if world prices $p(s)$ aren't actuarially fair.

51. The latter result holds even when $\beta(1 + r) \neq 1$, as eqs. (17) and (78) show.



$$CA_1 = \frac{Y_2(1)}{1+\beta}\left[\frac{p(1)^{A}}{1+r^{A}} - \frac{p(1)}{1+r}\right] + \frac{Y_2(2)}{1+\beta}\left[\frac{p(2)^{A}}{1+r^{A}} - \frac{p(2)}{1+r}\right]. \tag{81}$$

According to eq. (81) the current account balance depends positively on the difference between the autarky and world market prices of Arrow-Debreu securities. If other things are equal, the higher are the autarky prices of assets relative to world prices on date 1, the greater will be the country's date 1 net asset purchases from abroad, the sum of which equals the date 1 current account surplus. (This is reminiscent of—but distinct from—the result in standard two-good trade theory, that a country imports the good with an autarky price above the world price.) An alternative (and equivalent) interpretation of eq. (81) is that countries with relatively high autarky prices for future state-contingent consumption will tend to import future consumption, and export current consumption, through a date 1 current account surplus.[52]

## 5B.2 Understanding Gross Capital Flows

Equation (81) explains the economy's *net* asset purchases or sales from the rest of the world. In special cases the simple form of the comparative advantage principle can be useful in understanding *gross* imports or exports of Arrow-Debreu securities, which may be quite large even when the current account is small.

By eq. (80), the autarky relative price of state 1 in terms of state 2 consumption is

$$\frac{p(1)^{A}}{p(2)^{A}} = \frac{\pi(1)/Y_2(1)}{\pi(2)/Y_2(2)}.$$

Let us suppose that the country desires a balanced current account, so that [by eqs. (17) and (78)] its interest rate in "current account autarky" equals the world rate, $r^{CA} = r$. In that case the country will spend a total of $p(1)Y_2(1) + p(2)Y_2(2)$ on date 2, choosing state-contingent consumption levels by $C_2(s) = \pi(s)[p(1)Y_2(1) + p(2)Y_2(2)]/p(s)$, as we have seen. The date 2 budget constraint (3) now gives the country's demand for state 1 securities as

$$B_2(1) = C_2(1) - Y_2(1)$$

$$= \frac{\pi(1)[p(1)Y_2(1) + p(2)Y_2(2)]}{p(1)} - Y_2(1)$$

$$= \frac{p(2)\pi(1)}{p(1)}Y_2(2) - \pi(2)Y_2(1)$$

$$= \frac{\pi(2)p(2)Y_2(1)}{p(1)}\left[\frac{\pi(1)/Y_2(1)}{\pi(2)/Y_2(2)} - \frac{p(1)}{p(2)}\right]$$

$$= \frac{p(2)}{p(1)}\pi(2)Y_2(1)\left[\frac{p(1)^{A}}{p(2)^{A}} - \frac{p(1)}{p(2)}\right].$$

52. By combining eqs. (79) and (80) you can compute the autarky prices or Arrow-Debreu securities in terms of date 2 output. Equation (81) implies that when $p^{A}(s) = p(s)$ for $s = 1, 2$, we have another case in which $CA_1$ depends only on the difference between $r$ and $r^{A}$. [International equality of absolute state-contingent prices, $p^{A}(s) = p(s)$, follows from equality of relative prices because the prices sum to 1.]

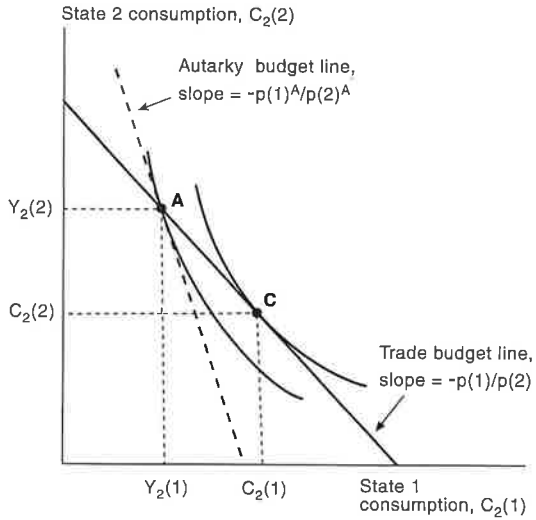



**Figure 5.2**
Trade across states of nature

The assumption of a zero current account implies that the demand for state 2 securities must be

$$B_2(2) = -\pi(2)Y_2(1)\left[\frac{p(1)^A}{p(2)^A} - \frac{p(1)}{p(2)}\right].$$

When the current account is zero, so that the country's international trade consists entirely of trade across date 2 states of nature, it imports the asset with the relatively high autarky price and exports the one with the relatively low autarky price. These opposite but equal (in value) gross capital flows allow the country to use output from the relatively prosperous state to hedge against the low consumption level that would prevail in the other state in the absence of asset trade. Figure 5.2 illustrates how this pure trade across states of nature maximizes expected second-period utility, $\pi(1)\log[C_2(1)] + \pi(2)\log[C_2(2)]$, given total spending of $p(1)Y_2(1) + p(2)Y_2(2)$ on date 2 consumption.

The preceding asset-trade pattern is consistent with the simplest version of the comparative advantage principle. If the country had an unbalanced current account, however, it might import or export both assets irrespective of their relative autarky prices. End-of-chapter exercise 1 considers that more general case.

## Appendix 5C An Infinite-Horizon Complete-Markets Model

In this appendix, we extend to an infinite horizon the chapter's two-period Arrow-Debreu model. The extension, discussed informally in section 5.2.3, is quite straightforward with the help of some additional notation.



### 5C.1 Dynamic Uncertainty

Our two-period model involved uncertainty on one future date only. Thus we were able to summarize the future state of the economy by the outcome of only one set of random events, those occurring "tomorrow." With many periods, random events occur not only tomorrow, but also the day after tomorrow, the day after that one, and so on. The *history* of these random events up to and including a date $t$ determines the state of nature the economy occupies on date $t$.

More formally, denote by $\mathbf{h}_t$ the history of the world economy on date $t$, that is, the set whose elements are a list of how events that were not predictable with certainty before they occurred actually turned out. The set of positive-probability random events whose outcomes become known on date $t$ itself is denoted by $\mathcal{S}(\mathbf{h}_{t-1})$: this set is a function of past history $\mathbf{h}_{t-1}$ because the history of the economy through date $t-1$ determines the set of events that can occur on date $t$. (For example, U.S. President Kennedy's death in 1963 precluded his reelection in 1964.) If event $s_t \in \mathcal{S}(\mathbf{h}_{t-1})$ occurs on date $t$, then $\mathbf{h}_t$ is given recursively by $\mathbf{h}_t = \{s_t\} \cup \mathbf{h}_{t-1}$. More generally, we will say that $\mathbf{h}_{t'}$, $t' > t$, is a *continuation of* $\mathbf{h}_t$ *through date* $t'$ if there are events $s_{t+1} \in \mathcal{S}(\mathbf{h}_t)$, $s_{t+2} \in \mathcal{S}(\{s_{t+1}\} \cup \mathbf{h}_t), \ldots, s_{t'} \in \mathcal{S}(\{s_{t'-1}, \ldots, s_{t+1}\} \cup \mathbf{h}_t)$ such that $\mathbf{h}_{t'} = \{s_{t'}, \ldots, s_{t+1}\} \cup \mathbf{h}_t$.

In a dynamic model, it is natural for contingent contracts to specify payoffs for a date as a function of the economy's realized *future* history through that date, which is not known with certainty when contracts are signed. We therefore assume that on an initial date 1 (and only then) agents can trade Arrow-Debreu securities of the following form: a history $\mathbf{h}_t$ security pays its owner one unit of output on date $t > 1$ if the economy's history through that date turns out to be $\mathbf{h}_t$, but it pays zero otherwise. We denote by $p(\mathbf{h}_t \mid \mathbf{h}_1)$ the date 1 price of this security, measured in terms of sure date $t$ output. Correspondingly, $\pi(\mathbf{h}_t \mid \mathbf{h}_1)$ is the conditional probability that the economy's history on date $t$ turns out to be $\mathbf{h}_t$, given what has happened through date 1. The set $\mathbf{H}_t(\mathbf{h}_1)$ is defined to consist of all positive-probability continuations of $\mathbf{h}_1$ through date $t$.

### 5C.2 Individual Optimality

Given these notational conventions, the analysis of a representative country $n$ individual's behavior goes through much as in section 5.1. The individual from country $n$ maximizes expected utility,

$$U_1 = u(C_1^n) + \sum_{t=2}^{\infty} \beta^{t-1} \left\{ \sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_1)} \pi(\mathbf{h}_t \mid \mathbf{h}_1) u[C^n(\mathbf{h}_t)] \right\}, \tag{82}$$

subject to

$$C_1^n + \sum_{t=2}^{\infty} R_{1,t} \left[ \sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_1)} p(\mathbf{h}_t \mid \mathbf{h}_1) C^n(\mathbf{h}_t) \right]$$

$$= Y_1^n + \sum_{t=2}^{\infty} R_{1,t} \left[ \sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_1)} p(\mathbf{h}_t \mid \mathbf{h}_1) Y^n(\mathbf{h}_t) \right], \tag{83}$$

where $R_{1,t}$ is the multiperiod discount factor for sure date $t$ consumption defined in eq. (60), section 5.4.3. Constraint (83) generalizes constraint (4) from the two-period, two-state model of the text. You can easily check that the necessary conditions for individual optimality imply

$$R_{1,t}\,p(\mathbf{h}_t \mid \mathbf{h}_1)u'(C_1^n) = \pi(\mathbf{h}_t \mid \mathbf{h}_1)\beta^{t-1}u'[C^n(\mathbf{h}_t)] \tag{84}$$

for all dates $t$ and histories $\mathbf{h}_t$. [Compare this with eq. (5) for the two-period model.] From this equation, we derive the analog of eq. (9) for any two date $t$ histories $\mathbf{h}_t^1$ and $\mathbf{h}_t^2$,

$$\frac{\pi(\mathbf{h}_t^1 \mid \mathbf{h}_1)u'[C^n(\mathbf{h}_t^1)]}{\pi(\mathbf{h}_t^2 \mid \mathbf{h}_1)u'[C^n(\mathbf{h}_t^2)]} = \frac{p(\mathbf{h}_t^1 \mid \mathbf{h}_1)}{p(\mathbf{h}_t^2 \mid \mathbf{h}_1)},$$

along with Euler equations for noncontingent bonds that generalize eq. (8),

$$u'(C_1^n) = \frac{\beta^{t-1}}{R_{1,t}}\mathrm{E}_1\{u'(C_t^n)\} = \frac{\beta^{t-1}}{R_{1,t}}\mathrm{E}[u'[C^n(\mathbf{h}_t)] \mid \mathbf{h}_1\}. \tag{85}$$

Equation (85) follows from the observation that for any date $t$,

$$\sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_1)} p(\mathbf{h}_t \mid \mathbf{h}_1) = 1$$

[a generalization of the no-arbitrage condition (7)]. Notice that the conditional expectation $\mathrm{E}_1\{\cdot\}$ in eq. (85) is a function of the conditioning information $\mathbf{h}_1$, consistent with our notation for conditional probabilities and securities prices.

## 5C.3 Equilibrium in the CRRA Case

With CRRA utility, $u(C) = C^{1-\rho}/(1-\rho)$, it is easy to build on these relationships to obtain closed-form solutions for asset prices and interest rates in a global equilibrium. Prices are determined so that aggregate output demands and supplies are equal on every date, for every history. In equilibrium, as in the two-period case, country $n$'s consumption on any date, and for any history, will be an unchanging fraction $\mu^n$ of world output,

$$C^n(\mathbf{h}_t) = \mu^n Y^{\mathrm{W}}(\mathbf{h}_t),$$

where $\mu^n$ equals country $n$'s share in the date 1 present value of current and future world output. By eq. (85), the equilibrium long-term interest rate is given by

$$R_{1,t} = \frac{\beta^{t-1}\sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_1)} \pi(\mathbf{h}_t \mid \mathbf{h}_1)Y^{\mathrm{W}}(\mathbf{h}_t)^{-\rho}}{(Y_1^{\mathrm{W}})^{-\rho}}.$$

The history-contingent securities prices $p(\mathbf{h}_t \mid \mathbf{h}_1)$ are similarly derived (we leave this as an exercise).

Notice why the current account always is zero after period 1 in this infinite-horizon economy. World saving is zero, and since countries differ only in their scale (measured by $\mu^n$), each country's saving is zero as well. With investment, countries' saving rates relative to income still would always be identical, but not necessarily their investment rates. Stochastic investment opportunities localized in individual countries would thus open up the possibility of random current-account imbalances.



The formalism we have developed in this appendix can be applied to the bonds-only economy studied in Chapter 2. Naturally the present approach leads to identical results.

## Appendix 5D Ongoing Securities Trade and Dynamic Consistency

In the many-period Arrow-Debreu model as presented in appendix 5C, agents commit on date 1 to a sequence of contingent consumption plans for all dates $t > 1$. They do this by trading contingent claims to future outputs in a market that meets on date 1 *only*. After date 1 there is no further asset trade: as the economy evolves stochastically over time, individuals have only to execute the sequence of history-dependent output transfers they contracted to make on date 1.

There is no reason, however, not to consider *continuing* asset trade over time. Indeed, this is how asset markets work in real economies, and such trade would arise endogenously if it offered opportunities for mutual gain. On date 2, for example, after execution of contracts written contingent on $\mathbf{h}_2$, people will still be holding securities for contingencies occurring on dates 3, 4, and so on. The arrival of new information makes some of these securities more valuable than when purchased on date 1, some less. If the remaining contingent securities could be bought and sold again on date 2, won't individuals sometimes wish to recontract and thus alter the contingent consumption levels they selected on date 1?

The answer, it turns out, is no. We will prove this result by showing that after we allow for continuing security-market activity, the solution to maximizing eq. (82) subject to eq. (83) is *dynamically consistent*: contingency plans that appear optimal on date 1 remain so as the dates and states in which they are supposed to be implemented arrive.

In section 2.1.4, we raised the potential problem of dynamic inconsistency in individual intertemporal plans. The question is a bit more subtle in models with uncertainty because as time passes, individuals update the probabilities entering the expected utility function (82) in light of new information. This updating appears to make preferences change over time, but in reality preferences are stable here. Because this stability may not be completely obvious to you, we include a formal proof of dynamic consistency.

To prove dynamic consistency in an economy with ongoing securities trade, we first make a useful notational economy. Let $\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1)$ denote the price of the Arrow-Debreu security for history $\mathbf{h}_t$ *in terms of date 1 output*; that is, $\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1) \equiv p(\mathbf{h}_t \mid \mathbf{h}_1)R_{1,t}$. Similarly, $\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2)$ denotes the price of the same security on date 2 measured in date 2 output units. Using this convention we can, for example, rewrite eq. (84) as

$$\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1)u'(C_1) = \pi(\mathbf{h}_t \mid \mathbf{h}_1)\beta^{t-1}u'[C(\mathbf{h}_t)], \tag{86}$$

where we drop the country superscripts, which are superfluous for current purposes.

To begin, consider the position on date 2 of an individual who learns $s_2$ (and therefore $\mathbf{h}_2$) after having maximized eq. (82) subject to eq. (83) on date 1 and planned for history-contingent consumption levels $C(\mathbf{h}_t)$, $\forall t > 1$. In contrast to appendix 5C, securities markets reopen on date 2 so that people can recontract, if they wish, on the basis of their new information about the economy's state. Individuals are legally bound to make the payments they contracted for on date 1, but they are not bound to stick with their original consumption plans for dates $t \geq 2$. To establish dynamic consistency, we have to show that the consumption contingency plans chosen on date 1 remain feasible and optimal on date 2.

Let's take feasibility first. Given a realized history $\mathbf{h}_2$ on date 2, the consumption levels contracted on date 1 determine the individual's endowment. So he faces the date 2 budget constraint:

$$C_2' + \sum_{t=3}^{\infty}\left[\sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_2)} \widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2) C(\mathbf{h}_t)'\right]$$

$$= C(\mathbf{h}_2) + \sum_{t=3}^{\infty}\left[\sum_{\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_2)} \widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2) C(\mathbf{h}_t)\right],$$

where primes indicate new consumption levels that could be picked on date 2. Clearly the original consumption plans $C(\mathbf{h}_t)$ are still feasible given this constraint.

But are they optimal? They are provided they satisfy the date 2 versions of the first-order conditions in eq. (86),

$$\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2) u'[C(\mathbf{h}_2)] = \pi(\mathbf{h}_t \mid \mathbf{h}_2)\beta^{t-2} u'[C(\mathbf{h}_t)], \tag{87}$$

for all histories $\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_2)$ [recall that in eq. (87), $\widetilde{p}$ is a date 2 present value].

Consumption was planned on date 1 to satisfy eq. (84) for all $t > 1$; in particular,

$$\widetilde{p}(\mathbf{h}_2 \mid \mathbf{h}_1) u'(C_1) = \pi(\mathbf{h}_2 \mid \mathbf{h}_1)\beta u'[C(\mathbf{h}_2)].$$

Combining this with eq. (86) to eliminate $u'(C_1)$, we find that the date 1 plan sets

$$\frac{\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1)}{\widetilde{p}(\mathbf{h}_2 \mid \mathbf{h}_1)} u'[C(\mathbf{h}_2)] = \left[\frac{\pi(\mathbf{h}_t \mid \mathbf{h}_1)}{\pi(\mathbf{h}_2 \mid \mathbf{h}_1)}\right]\beta^{t-2} u'[C(\mathbf{h}_t)] \tag{88}$$

By the properties of conditional expectations (Bayes's rule),[53]

$$\frac{\pi(\mathbf{h}_t \mid \mathbf{h}_1)}{\pi(\mathbf{h}_2 \mid \mathbf{h}_1)} = \pi(\mathbf{h}_t \mid \mathbf{h}_2).$$

Furthermore, because history $\mathbf{h}_t \in \mathbf{H}_t(\mathbf{h}_2)$ can only occur if $\mathbf{h}_2$ does, a simple arbitrage argument shows that

$$\frac{\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1)}{\widetilde{p}(\mathbf{h}_2 \mid \mathbf{h}_1)} = \widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2).$$

On date 1 one can buy a unit of date $t$ output contingent on $\mathbf{h}_t$ in two ways: pay $\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_1)$ directly, or buy $\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2)$ units of $\mathbf{h}_2$-contingent date 2 output at date 1 cost $\widetilde{p}(\mathbf{h}_2 \mid \mathbf{h}_1)\widetilde{p}(\mathbf{h}_t \mid \mathbf{h}_2)$, using the proceeds (if $\mathbf{h}_2$ is realized) to buy a unit of $\mathbf{h}_t$-contingent date $t$ output on date 2. These two strategies must entail the same sacrifice of date 1 output.

The last two equalities show, however, that eq. (88) reduces to eq. (87), completing our proof that to continue the date 1 optimal plan always remains optimal on date 2.

53. In general, for all $s \geq t+1$,

$$\pi(\mathbf{h}_s \mid \mathbf{h}_{t+1}) = \pi(\mathbf{h}_s \mid \mathbf{h}_t)/\pi(\mathbf{h}_{t+1} \mid \mathbf{h}_t).$$

**Proof:** Let $\mathbf{h}_{t+1} = \{s_{t+1}\} \cup \mathbf{h}_t$ and $\mathbf{h}_s \in \mathbf{H}_s(\mathbf{h}_{t+1})$. Then by the definition of conditional expectation,

$$\begin{aligned}\pi(\mathbf{h}_s \mid \mathbf{h}_{t+1}) &= \pi(\mathbf{h}_s \mid s_{t+1} \text{ and } \mathbf{h}_t)\\ &= \pi(\mathbf{h}_s \text{ and } s_{t+1} \mid \mathbf{h}_t)/\pi(s_{t+1} \mid \mathbf{h}_t)\\ &= \pi(\mathbf{h}_s \mid \mathbf{h}_t)/\pi(\mathbf{h}_{t+1} \mid \mathbf{h}_t).\end{aligned}$$



## Exercises

1. *More on the logarithmic small-country example.* Consider the small-country Arrow-Debreu model with log preferences of section 5.1.6 and appendix 5B. Show that in general, when the current account need not be zero, the country's *gross* purchases of the individual Arrow-Debreu security purchases, $B_2(1)$ and $B_2(2)$ satisfy

$$\frac{p(1)}{1+r}B_2(1) = \frac{p(2)\pi(2)Y_2(1)}{1+r}\left[\frac{p(1)^{\wedge}}{p(2)^{\wedge}} - \frac{p(1)}{p(2)}\right] + \pi(1)CA_1,$$

$$\frac{p(2)}{1+r}B_2(2) = \frac{-p(2)\pi(2)Y_2(1)}{1+r}\left[\frac{p(1)^{\wedge}}{p(2)^{\wedge}} - \frac{p(1)}{p(2)}\right] + \pi(2)CA_1.$$

Provide an intuitive interpretation. [Hint: Any date 2 expenditure $(1+r)CA_1$ above (below) $p(1)Y_2(1) + p_2(2)Y_2(2)$ goes to increase (reduce) state $s$ consumption in the proportion $\pi(s)/p(s)$ with log preferences. The country then does any additional portfolio rebalancing it desires by a pure swap between state 1 and state 2 securities, as in the text.]

2. *An example with risk neutrality.* Consider the two-country, two-period, two-state Arrow-Debreu endowment economy in section 5.2.1. Suppose the utility function at home and abroad is given by

$$U_1 = \log(C_1) + \pi(1)\beta C_2(1) + \pi(2)\beta C_2(2),$$

so that countries are risk-neutral with respect to second-period consumption. Determine equilibrium Arrow-Debreu prices $p(1)$ and $p(2)$ for this world economy. Are they actuarially fair? Also describe equilibrium Home and Foreign consumption levels, and determine the interest rate $1+r$.

3. *Comparing optimal consumption with complete and incomplete markets.* Consider a two-period small open endowment economy facing the world interest rate $r$ for riskless loans. Date 1 output is $Y_1$. There are $S$ states of nature on date 2 that differ according to the associated output realizations $Y_2(s)$ and have probabilities $\pi(s)$ of occurring. The representative domestic consumer maximizes the expected lifetime utility function

$$U_1 = C_1 - \frac{a_0}{2}(C_1)^2 + (1+r)^{-1}\mathrm{E}_1\left\{C_2 - \frac{a_0}{2}(C_2)^2\right\}, \qquad a_0 > 0,$$

in which period utility is quadratic. The relevant budget constraints when markets are *incomplete* can be written as

$$B_2 = (1+r)B_1 + Y_1 - C_1,$$

$$C_2(s) = (1+r)B_2 + Y_2(s), \qquad s = 1, 2, \ldots, S,$$

where $B_1$ is given. The last constraint is equivalent to the $S$ constraints: for all states $s$,

$$C_1 + \frac{C_2(s)}{1+r} = (1+r)B_1 + Y_1 + \frac{Y_2(s)}{1+r}.$$

(You may assume that all output levels are small enough that the marginal utility of consumption $1 - a_0 C$ is safely positive.)

(a) Start by temporarily ignoring the nonnegativity constraints $C_2(s) \geq 0$ on date 2 consumption. Compute optimal date 1 consumption $C_1$. What are the implied values of $C_2(s)$? What do you think your answer for $C_1$ would be with an infinite horizon and output uncertainty in each future period? [Hint: Recall Chapter 2.]

(b) Now let's worry about the nonnegativity constraint on $C_2$. Renumber the date 2 states of nature (if necessary) so that $Y_2(1) = \min_s\{Y_2(s)\}$. Show that if

$$(1+r)B_1 + Y_1 + \frac{2+r}{1+r}Y_2(1) \geq \mathrm{E}_1 Y_2,$$

then the $C_1$ computed in part a (for the two-period case) is still valid. What is the intuition? Suppose the preceding inequality does not hold. Show that the optimal date 1 consumption is lower (a precautionary saving effect) and equals

$$C_t = (1+r)B_1 + Y_1 + \frac{Y_2(1)}{1+r}.$$

[Hint: Apply the Kuhn-Tucker theorem.] Explain the preceding answer. Does the bond Euler equation hold in this case?

(c) Now assume the consumer faces *complete* global asset markets with $p(s)$, the state $s$ Arrow-Debreu security price, equal to $\pi(s)$. Find the optimal values of $C_1$ and $C_2(s)$ now. Why can nonnegativity constraints be disregarded in the complete-markets case?

4. *An alternative solution of the diversification model.* This problem illustrates an alternative approach to solving the model in section 5.3.1 when period utility is logarithmic. Note that eq. (42) implies that a representative individual from country $n$ has a lifetime income equal to $Y_1^n + V_1^n$, the sum of current output and the market value on date 1 of uncertain future output. Given the agent's log utility function, a reasonable guess is that optimal date 1 consumption will be

$$C_1^n = \frac{1}{1+\beta}(Y_1^n + V_1^n). \tag{89}$$

Let us further guess that the agent invests savings $\frac{\beta}{1+\beta}(Y_1^n + V_1^n)$ in the global mutual fund that gives the agent an equal share in every country's output:

$$C_2^n(s) = \frac{\beta}{1+\beta}(Y_1^n + V_1^n)\frac{\sum_{m=1}^N Y_2^m(s)}{\sum_{m=1}^N V_1^m}. \tag{90}$$

Use these guesses to find asset prices consistent with individual agents' first-order conditions and with global equilibrium.

5. *Consequences of exponential period utility.* Suppose we have the two-country, two-period, $\mathcal{S}$ state endowment setup of section 5.2 and of the $N = 2$ case from section 5.3. Now, however, in both Home and Foreign, agents have the exponential period utility function $u(C) = -\exp(-\gamma C)/\gamma$, $\gamma > 0$, rather than CRRA period utility.



[The parameter $\gamma = -u''(C)/u'(C)$ is called the coefficient of *absolute* risk aversion.]

(a) For the case of complete markets (paralleling section 5.2), calculate equilibrium prices and consumption levels.

(b) Suppose that instead of complete markets, people are restricted to trading riskless bonds and shares in Home and Foreign period 2 outputs. Show that the resulting allocation is still efficient (paralleling section 5.3), and that Home and Foreign consumptions on both dates are given by $C = \frac{1}{2}Y^{\mathrm{W}} - \mu$, $C^* = \frac{1}{2}Y^{\mathrm{W}} + \mu$, where $Y^{\mathrm{W}}$ is world output and $\mu$ is a time-invariant constant. Show that to support this equilibrium, both countries purchase equal shares in the risky world mutual fund on date 1 and one country makes riskless loans to the other.

(c) How does your answer to part b change when Home and Foreign have distinct coefficients of absolute risk aversion, $\gamma \neq \gamma^*$?

6. *The Lucas (1982) two-good model.* There are two countries, Home and Foreign, with exogenous stochastic endowments of *distinct* goods, the outputs of which we denote by $X$ and $Y$. Residents of the two countries have identical tastes, such that for the representative Home resident, say,

$$U_t = \mathrm{E}_t\left\{\sum_{s=t}^{\infty}\beta^{s-t}u(C_{\mathrm{X},s}, C_{\mathrm{Y},s})\right\}.$$

We also assume that the countries start out endowed with equal and perfectly-pooled portfolios of risky claims, such that each country owns exactly half the Home output process and half the Foreign output process. Thus, either country's initial endowment is $\frac{1}{2}(X + pY)$, where $p$ denotes the price of good $Y$ (Foreign's output) in terms of good $X$ (Home's output). As usual, countries are free to trade away from these initial endowments.

(a) Paralleling section 5.3, let $V_{\mathrm{X},t}$ be the ex dividend price on date $t$ of the claim to Home's entire future output process (where asset prices are measured in terms of current units of good $X$). Correspondingly, let $V_{\mathrm{Y},t}$ be the ex dividend date $t$ price (in units of good $X$) of the claim to Foreign's future output process. Write down the finance constraints corresponding to eq. (56) in section 5.4.3, but with two risky assets. Derive a representative consumer's first-order Euler conditions corresponding to the two risky assets and the riskless bond.

(b) Show that the equilibrium consumption allocation is $C_{\mathrm{X},t} = C^*_{\mathrm{X},t} = \frac{1}{2}X_t$, $C_{\mathrm{Y},t} = C^*_{\mathrm{Y},t} = \frac{1}{2}Y_t$, in every period. (People find it optimal to hold initial endowments.) In particular, the distribution of world wealth is constant. Show that in equilibrium,

$$p_t = p(X_t, Y_t) = \frac{\partial u\left(\frac{1}{2}X_t, \frac{1}{2}Y_t\right)/\partial Y}{\partial u\left(\frac{1}{2}X_t, \frac{1}{2}Y_t\right)/\partial X}.$$

(c) Calculate the equilibrium levels of $V_{\mathrm{X},t}$ and $V_{\mathrm{Y},t}$ as expected present values.

(d) What is the date $t$ price of a unit of $X$ to be delivered with certainty on date $t+1$? Similarly, what is the own-rate of interest on good $Y$?